

EIGENFUNCTIONS, FREE BOUNDARIES, AND TIME-FREQUENCY LOCALIZATION

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ABSTRACT. We study Gaussian time–frequency localization operators through their Bargmann–Fock representation, developing an inverse and free-boundary approach to their spectral theory. Our perturbative inverse theory shows that functions sufficiently close to the ground-state Hermite function can be realized as eigenfunctions of localization operators; as a byproduct we recover the Abreu–Dörfler disk characterization in the simply connected setting, and we prove that the stability estimate of Gómez–Guerra, Ramos and Tilli is sharp.

We then apply these inverse results to Faber–Krahn-type problems for Gaussian time–frequency concentration: local maximizers of the first localization eigenvalue under a measure constraint are necessarily disks, and global maximizers exist by a concentration–compactness argument in the Bargmann–Fock space, recovering the Nicola–Tilli theorem from this perspective.

Keywords: Gaussian STFT, localization operators, Bargmann–Fock space, inverse spectral problem, free boundary, quadrature domains, Faber–Krahn inequality, time-frequency concentration.

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1. INTRODUCTION

How does a function concentrate in phase space? This is the founding question of time–frequency analysis, dating back to Wigner, Gabor, and Daubechies, and it admits a precise formulation in terms of the *short-time Fourier transform*

$$V_\varphi f(x, \omega) = \int_{\mathbb{R}} f(t) \overline{\varphi(t-x)} e^{-2\pi i t \omega} dt, \quad f, \varphi \in L^2(\mathbb{R}), \quad (1.1)$$

whose squared modulus is the canonical phase-space energy density of f . Identifying which signals achieve maximal concentration on a prescribed set $\Omega \subset \mathbb{R}^2$ leads to the spectral analysis of the *time–frequency localization operator* $\mathcal{L}_\Omega$ introduced by Daubechies [15]; see [22, 37, 18, 16, 42, 43, 14, 12, 7, 31, 41, 44] for the broader theory and [34, 2] for the underlying uncertainty and large-sieve phenomena.

The Gaussian window $\varphi(x) = 2^{1/4} e^{-\pi x^2}$ is exceptional. Through the Bargmann transform [4], the operator $\mathcal{L}_\Omega$ becomes a Toeplitz-type integral operator on the Bargmann–Fock space $\mathcal{F}^2(\mathbb{C})$ of entire functions [8, 9, 10, 11, 51, 22, 18], and time–frequency questions translate into weighted complex analysis. Two foundational rigidity theorems shape the modern picture of the Gaussian model:

- (NT) (Nicola–Tilli [40]; see also [41, 5, 6, 45]) *Geometric extremality*. Among all measurable sets of a given measure, disks maximize the first localization eigenvalue λ_1 , sharpening the Faber–Krahn paradigm to phase space.
- (AD) (Abreu–Dörfler [1]) *Spectral rigidity*. If a Hermite function is an eigenfunction of $\mathcal{L}_\Omega$ on a simply connected bounded domain, then Ω is a disk.

The two theorems articulate, from opposite directions, the same heuristic: *the special role of the Gaussian among signals must be matched by the special role of the disk among phase-space sets*. Yet the geometric and the spectral side have so far been developed as parallel, but largely independent, viewpoints. The variational theory leading to (NT) and its quantitative refinement [20] is essentially based on rearrangement and integral-geometric inequalities; the rigidity (AD) is proved by overdetermined boundary conditions and unique-continuation arguments. Neither perspective, on its own, makes the connection between eigenfunctions and domains constructive.

Eigenfunctions as geometric data. The aim of this paper is to construct that connection. The unifying thesis we develop is that, for Gaussian time–frequency localization, an eigenfunction must be regarded as *geometric data*: it determines a localization domain, and prescribing it has strong consequences for the variational theory. This thesis is realized in three coupled steps.

(i) *An inverse construction*. Near the Gaussian, the eigenvalue equation $\mathcal{L}_U f_0 = \lambda f_0$ admits a perturbative solution *in the unknown domain* U for arbitrarily prescribed near-Gaussian f_0 . We solve this inverse problem by a free-boundary moment system on $\mathcal{F}^2(\mathbb{C})$ and an analytic implicit-function theorem at the Daubechies disk.

(ii) *Constructive consequences of rigidity*. Combining the inverse construction with eigenfunction rigidity yields, on the one hand, a short conceptual proof of (AD) that does *not* require any boundary regularity, and, on the other hand, the optimal exponent in the quantitative stability inequality of [20].

(iii) *A new road to (NT) through local rigidity*. Local maximizers of λ_1 , suitably defined in a weighted symmetric-difference topology, are themselves rigid: each is a disk up to translation. Combined with a Fock-space concentration–compactness profile decomposition, this local rigidity yields a new proof of (NT) that is independent of rearrangement.

The technical core of the paper is the rewriting of the eigenvalue equation, after Bargmann transform, as an infinite weighted moment system on the Bargmann–Fock space:

$$\int_U F(z) \overline{e_k(z)} e^{-\pi|z|^2} dA(z) = \lambda \langle F, e_k \rangle_{\mathcal{F}^2}, \quad k \geq 0, \quad (1.2)$$

where $\{e_k\}$ is the standard monomial basis. This is the λ -dependent *weighted quadrature identity* of the title. Classical quadrature-domain theory studies analogous identities for harmonic test functions against Lebesgue measure [23, 48, 47, 3, 29, 13]; here the Gaussian weight and the spectral parameter λ make the system genuinely new and place it at the intersection of phase-space analysis, free-boundary problems, and inverse potential theory [28]. The remainder of the introduction states the four main results; the methods, technical inputs, and connections with the literature are gathered in §1.2–§1.3.

1.1. Main results. The first main result is the inverse construction at the Daubechies disk. It says that any sufficiently small Hermite-polynomial perturbation of the Gaussian arises as an eigenfunction of *some* localization operator at any prescribed eigenvalue.

Theorem 1 (Inverse construction near the Gaussian). *Let $\lambda \in (0, 1)$ and $N \in \mathbb{N}$. There exists $\varepsilon_0 = \varepsilon_0(\lambda, N) > 0$ such that for any*

$$f_0 = h_0 + \sum_{k=1}^N a_k h_k \quad \text{with} \quad \max_k |a_k| < \varepsilon_0,$$

there exists a domain $U_\lambda \subset \mathbb{R}^2$, a real-analytic perturbation of a disk, with

$$\mathcal{L}_{U_\lambda} f_0 = \lambda f_0.$$

This appears to be the first general inverse construction of localization domains beyond the highly symmetric examples (radial multiplets, scaled Gaussians) accessible via direct symmetry arguments. The threshold ε_0 depends explicitly on λ and on the spectral gap of the disk localization operator restricted to the Hermite subspace of degree $\leq N$, with quantitative bounds extracted from the proof. The construction is perturbative in nature, and the local nature of the conclusion is essential: the analogue at fixed scale, away from the Gaussian, remains an open problem.

The same machinery yields a self-contained, methodologically transparent recovery of (AD) — and, in fact, a strengthening of it.

Theorem 2 (Spectral rigidity, regularity-free). *Let $U \subset \mathbb{C}$ be a simply connected bounded domain. If a Hermite function is an eigenfunction of $\mathcal{L}_U$, then U is a disk.*

We give two proofs (§4). The first, in the spirit of overdetermined free-boundary problems, transports the eigenvalue equation into the Bargmann–Fock space, applies a Paley–Wiener argument and an Ehrenpreis–Malgrange-type divisibility lemma, and forces the logarithmic potential of $\mathbf{1}_U$ to be radial along ∂U . The second proof eliminates *all* hypotheses on the regularity of ∂U , replacing them with a holomorphic auxiliary function attached to the associated Poisson problem; it strengthens (AD) to the form stated above (Theorem 7), in which ∂U is allowed to be merely a closed measurable set with no perimeter or differentiability hypothesis. The version is sharp at the level of currently available statements, and we view it as a clean instance of the regularity-bypass principle made possible by the inverse machinery.

The third theorem deploys Theorem 1 to settle the question of optimal exponents in the quantitative stability estimate of [20]. For a measurable set $U \subset \mathbb{R}^2$ of finite positive measure, write $\lambda_1(U)$ for the first eigenvalue of $\mathcal{L}_U$, f_U for an L^2 -normalized first eigenfunction, φ_{z_0} for a Gaussian optimizer centred at $z_0 \in \mathbb{C}$, and $\mathcal{A}(U)$ for the Fraenkel asymmetry of U . Corollary 1.4 of [20] provides

$$\min_{z_0 \in \mathbb{C}, |c|=1} \|f_U - c \varphi_{z_0}\|_{L^2(\mathbb{R})} \leq C e^{|U|/2} \left(1 - \frac{\lambda_1(U)}{1 - e^{-|U|}}\right)^{1/2}, \quad \mathcal{A}(U) \leq K(|U|) \left(1 - \frac{\lambda_1(U)}{1 - e^{-|U|}}\right)^{1/2}. \quad (1.3)$$

Theorem 3 (Sharpness of the GGRT stability inequality). *The exponent $1/2$ in both inequalities of (1.3) is optimal: it cannot be replaced by any $\beta > 1/2$.*

The argument is direct. Theorem 1 applied to second-order Hermite perturbations of h_0 produces a one-parameter family U_ε of perturbed disks whose boundary deformation is governed by quadratic harmonics. A first-variation analysis based on the Reynolds transport identity — and a sharp explicit lower bound on $|U_\varepsilon \triangle B(z_0, r)|$ valid uniformly over (z_0, r) — shows that these harmonics cannot be removed by translations of the disk, and that the symmetric-difference deficit is exactly of the order predicted by the upper bounds in (1.3).

The fourth main result concerns local extremizers of the Faber–Krahn problem in the Gaussian STFT setting and is a key input to the new proof of (NT) below.

Theorem 4 (Local-extremizer rigidity). *Any local maximizer of $\lambda_{1,\varphi}$ under a measure constraint, in the weighted symmetric-difference topology of §6, is a disk up to translation.*

The proof identifies any local maximizer with a superlevel set of the Bargmann–Fock density of its first eigenfunction (via a Lieb–Loss bathtub argument), uses a Hadamard shape-derivative computation to derive the Euler–Lagrange overdetermined system, and exploits simplicity of λ_1 to force the eigenfunction into Gaussian form, hence the domain into a disk. Since the local maximizer is shown to be open with real-analytic boundary as a *consequence* of the variational equations, no boundary regularity is assumed.

Combining Theorem 4 with a Fock-space concentration–compactness profile decomposition gives a new proof of (NT). Existence of a global maximizer is established by ruling out vanishing and dichotomy via strict subadditivity of the localization functional $M(s) := \sup\{\lambda_1(U) : |U| = s\}$; uniqueness up to translation then follows because every global maximizer is in particular a local one.

Theorem 5 (New proof of Nicola–Tilli). *Among all measurable sets $\Omega \subset \mathbb{R}^2$ of fixed finite measure, the global maximizers of $\lambda_{1,\varphi}$ are exactly the translates of a single disk.*

New results vs. new proofs. Theorems 1, 3, and 4 are new. Theorem 2 is a new proof, and a regularity-free strengthening, of the Abreu–Dörfler rigidity theorem; Theorem 5 is a new proof of the Nicola–Tilli theorem. We make no claim of priority on the statements of the latter two; their interest here is the placement of (NT) and (AD) inside a single inverse/free-boundary framework that simultaneously yields the new results.

1.2. Methods and central identity. The central object of the paper is the moment identity (1.2). Three groups of techniques act on it.

(a) *Free-boundary methods on weighted Hardy spaces.* The boundary of the unknown domain is parametrized as an analytic radial graph over the disk in a Wiener-type Banach algebra of holomorphic functions on a strip. The moment system (1.2) is recast as a nonlinear map between such spaces; its Fréchet derivative at the disk is computed explicitly as a Fourier multiplier diagonal in modes, with explicit eigenvalues and an isometric inverse on the orthogonal complement of the translation kernel. An analytic implicit-function theorem then produces the perturbed domain U_λ . This step is in the spirit of Isakov’s perturbative free-boundary framework [28] and of the inverse Newtonian-potential results of Cherednichenko [13], but the function

spaces and the linearization are specifically tailored to the Gaussian weight and the spectral parameter, and the implicit-function step is reproved in the form needed.

(b) *Complex analysis and divisibility.* A coefficientwise conjugate differential operator $P^\#$, defined explicitly via Wirtinger derivatives, intertwines the eigenvalue equation with a moment-to-Cauchy-transform identity. A Paley–Wiener–Schwartz argument applied to a compactly supported distribution on $\mathbb{R}^2$ encodes the eigenfunction’s structure as the vanishing of an entire function on the complex variety $\xi^2 + \eta^2 = 0$. An Ehrenpreis–Malgrange-type divisibility lemma — which we state and prove self-containedly in the form needed (Lemma 2.6) — then forces the underlying logarithmic potential to be locally radial.

(c) *Variational and geometric measure theory.* Shape derivatives are computed via Reynolds transport for C^1 deformation fields. The bathtub principle of Lieb–Loss [34] converts the measure-constrained variational problem into a superlevel-set problem. Simplicity of the principal eigenvalue at a local maximizer is established by a positivity argument together with Berezin estimates. Finally, a profile-decomposition theorem in $\mathcal{F}^2(\mathbb{C})$ in the spirit of P.-L. Lions’s concentration–compactness, adapted to the lack of translation compactness in the Fock space, drives the existence half of the proof of (NT).

The detailed list of external theorems we use — with explicit indication of which are quoted, which are adapted, and which are reproved as lemmas — appears as §2.1 of the preliminaries.

1.3. Relation to the literature. *Quadrature domains and inverse potential theory.* A classical quadrature domain $D \subset \mathbb{R}^2$ is characterized by the fact that integration of harmonic test functions against Lebesgue measure on D reduces to a finite combination of point evaluations and their derivatives; see Gustafsson–Shahgholian [48] for the obstacle-problem characterization, the Gustafsson lectures [23], and Shahgholian [47]. Aharonov–Schiffer–Zalcman [3], Karp [29], and Cherednichenko [13] address the inverse potential problem in adjacent settings. Our moment system (1.2) is structurally distinct from a classical quadrature identity in two ways: the test space is the Bargmann–Fock space of entire functions rather than harmonic functions; and the eigenvalue λ enters as an explicit spectral parameter. We expect the resulting notion of λ -dependent weighted quadrature domain to be of independent interest.

Toeplitz inverse-spectral problems on Fock and Bergman spaces. Symbol-recovery and inverse-spectral problems for Toeplitz operators have been pursued, in various directions, by Berger–Coburn [9, 10, 11], Berezin [8], and Zhu [51], and more recently by Bastianoni–Cordero–Nicola–Teofanov [5, 6], Fulsche–Hagger [19], and Kisil [31]; see also [2] for related polyanalytic and large-sieve phenomena. In all of these the symbol typically belongs to a function class on phase space, and the inverse problem is to read off properties of the symbol from the spectrum. By contrast, the inverse problem we solve here is geometric: the symbol is the indicator $\mathbf{1}_U$ of an unknown domain, and the spectral datum is a single eigenfunction.

Concentration–compactness in time-frequency analysis. The profile decomposition we use in §7 extends and complements the time-frequency adaptations of P.-L. Lions’s method developed by Nicola–Tilli [40, 41] and Nicola–Romero–Trapasso [39], with the principal subtlety being the

lack of translation compactness in $\mathcal{F}^2(\mathbb{C})$. Closely related concentration phenomena and profile decompositions in adjacent time-frequency settings appear in [38, 46, 19].

Hermite structures in the Bargmann model. The role of Gaussian decay and Hermite-type rigidity is also visible in nearby phase-space problems [33, 21, 50]; the inverse-spectral perspective adopted here is, however, of a different character, in that it produces the domain rather than constraining it.

1.4. Organization. Section 2 collects the analytic preliminaries on the Bargmann–Fock space, the main technical inputs from the external literature, and the bridge from time–frequency analysis to free-boundary problems via the divisibility lemma. Section 3 is devoted to the proof of Theorem 1, including the self-contained disk inverse step. Section 4 contains the two proofs of Theorem 2; the second yields the regularity-free strengthening Theorem 7. Section 5 deploys Theorem 1 to prove Theorem 3. Section 6 is devoted to local maximizers and Theorem 4. Section 7 establishes the profile decomposition in $\mathcal{F}^2(\mathbb{C})$, rules out vanishing and dichotomy via strict subadditivity, and combines existence with Theorem 4 to give the new proof of (NT).

2. PRELIMINARIES

2.1. Main technical inputs. For the convenience of the reader, and to make the boundaries between quoted material, adapted material, and original arguments transparent, we collect here the principal external theorems used in this paper, together with an indication of the form in which each is used.

- *Isakov’s perturbative free-boundary theorem.* In Section 3 we use the implicit-function-theorem framework for inverse free-boundary and inverse-source problems developed in Isakov [28]. The general theorem is *adapted with proof*: the underlying perturbative implicit-function strategy is Isakov’s, but the function spaces, the linearized moment map at the disk, and its invertibility on the relevant subspace are reformulated and reproved here in a form tailored to the Gaussian Bargmann–Fock setting.
- *Cherednichenko’s inverse Newtonian-potential result.* The recovery of a domain from its exterior logarithmic potential, used as a conceptual guide in Sections 3 and 4, is taken from Cherednichenko [13]; see also [29, 3] for the closely related classical statements. We *quote* these results in the form needed and do not reprove them.
- *Bathtub principle.* The Lieb–Loss bathtub principle is used in Sections 6 and 7 to identify maximizers of $\int_{\Omega} u_F dA$ under a measure constraint with superlevel sets of u_F . We quote it *verbatim* from [34]; see also Lemma 2.5(i) above.
- *Concentration–compactness profile decomposition.* In Section 7 we use a profile-decomposition statement in $\mathcal{F}^2(\mathbb{C})$ in the spirit of P.-L. Lions’s concentration–compactness method and of its time–frequency adaptations by Nicola and Tilli [40, 41] and by Nicola, Romero, and Trapasso [39]. The form we use is *adapted with proof*: the abstract concentration mechanism is classical, while the precise profile decomposition for maximizing sequences of $I_F(s)$ in the Fock space, including the treatment of the lack of translation compactness, is reproved here.

- *Ehrenpreis–Malgrange divisibility.* The divisibility of entire functions on $\mathbb{C}^2$ vanishing on a complex hypersurface, which underlies Lemma 2.6 above, is the classical Ehrenpreis–Malgrange divisibility theorem. The version we need is *proved as a self-contained lemma* (Lemma 2.6), using only the Paley–Wiener–Schwartz theorem and elementary two-variable polynomial divisibility, in order to keep track of explicit support and regularity information.
- *Paley–Wiener and Paley–Wiener–Schwartz theorems.* The classical Paley–Wiener theorem (and its Schwartz extension to compactly supported distributions) is used several times to translate vanishing of a Fourier transform along complex lines into vanishing of a function on the support side. We *quote* these results from [25, 18] in the standard form.

The remainder of Section 2 is devoted to the analytic preliminaries on the Bargmann–Fock space and the divisibility lemma proper.

2.2. Properties of the Bargmann–Fock space. We briefly recall the connection between the Gaussian short-time Fourier transform (STFT) and the Bargmann–Fock space, which will be used systematically in what follows. Excellent references for the material in this subsection are [4, 18, 22, 51]; see also [8, 10, 15].

Let $\varphi(x) = 2^{1/4}e^{-\pi x^2}$ be the normalized Gaussian. For $f \in L^2(\mathbb{R})$, its Gaussian STFT is

$$V_\varphi f(x, \omega) = \int_{\mathbb{R}} f(t) \varphi(x - t) e^{-2\pi i t \omega} dt. \quad (2.1)$$

The Bargmann transform $\mathcal{B} : L^2(\mathbb{R}) \rightarrow \mathcal{F}^2(\mathbb{C})$ is defined by

$$(\mathcal{B}f)(z) = 2^{1/4} \int_{\mathbb{R}} f(t) e^{2\pi i t z - \pi t^2 - \frac{\pi}{2} z^2} dt, \quad z \in \mathbb{C}. \quad (2.2)$$

It is well known that $\mathcal{B}$ is a unitary isomorphism onto the Bargmann–Fock space $\mathcal{F}^2(\mathbb{C})$, endowed with the norm

$$\|F\|_{\mathcal{F}^2}^2 = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2} dA(z).$$

A direct computation shows that, for $z = x + i\omega$,

$$V_\varphi f(x, -\omega) = e^{\pi i x \omega} (\mathcal{B}f)(z) e^{-\frac{\pi}{2}|z|^2}. \quad (2.3)$$

In particular, setting $F = \mathcal{B}f$, one obtains

$$\int_{\mathbb{R}^2} |V_\varphi f(x, \omega)|^2 dx d\omega = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2} dA(z) = \|F\|_{\mathcal{F}^2}^2. \quad (2.4)$$

Weyl operators and invariance. For $a \in \mathbb{C}$, define the operator

$$(T_a F)(z) := e^{\pi \bar{a} z - \frac{\pi}{2}|a|^2} F(z - a). \quad (2.5)$$

Then T_a is unitary on $\mathcal{F}^2(\mathbb{C})$. Indeed,

$$\|T_a F\|_{\mathcal{F}^2}^2 = \int_{\mathbb{C}} e^{2\pi \Re(\bar{a} z) - \pi|a|^2} |F(z - a)|^2 e^{-\pi|z|^2} dA(z)$$

$$= \int_{\mathbb{C}} |F(z-a)|^2 e^{-\pi|z-a|^2} dA(z) = \|F\|_{\mathcal{F}^2}^2,$$

because

$$2\Re(\bar{a}z) - |a|^2 - |z|^2 = -|z-a|^2.$$

Moreover,

$$u_{T_a F}(z) = u_F(z-a). \quad (2.6)$$

Indeed, since $2\Re(\bar{a}z) - |a|^2 - |z|^2 = -|z-a|^2$,

$$u_{T_a F}(z) = |T_a F(z)|^2 e^{-\pi|z|^2} = e^{2\pi\Re(\bar{a}z) - \pi|a|^2} |F(z-a)|^2 e^{-\pi|z|^2} = |F(z-a)|^2 e^{-\pi|z-a|^2} = u_F(z-a). \quad (2.7)$$

In particular, the functionals I_F defined by, for $s > 0$,

$$I_F(s) := \sup_{|\Omega|=s} \int_{\Omega} u_F dA,$$

satisfy, by the translation-invariance of the Lebesgue measure on $\mathbb{C}$,

$$I_{T_a F}(s) = I_F(s), \quad a \in \mathbb{C}, \quad (2.8)$$

reflecting the invariance of the Gaussian STFT under time–frequency shifts; see [15, 22, 33, 21].

Connection with localization. If $F = \mathcal{B}f$, then for every measurable $\Omega \subset \mathbb{C}$,

$$\int_{\Omega} u_F dA = \int_{\tilde{\Omega}} |V_{\varphi} f(x, \omega)|^2 dx d\omega,$$

where $\tilde{\Omega} = \{(x, \omega) : (x, -\omega) \in \Omega\}$. Thus, concentration properties of the STFT are naturally encoded by the function

$$u_F(z) := |F(z)|^2 e^{-\pi|z|^2}, \quad z \in \mathbb{C}, \quad (2.9)$$

which is the main object of interest in the Bargmann–Fock setting, in terms of the quantities $I_F(s)$.

We next collect a few elementary facts that will be used repeatedly.

Lemma 2.1. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) *The space $\mathcal{F}^2(\mathbb{C})$ is a reproducing kernel Hilbert space with kernel*

$$K_w(z) = e^{\pi z \bar{w}}, \quad z, w \in \mathbb{C}.$$

(ii) *(Sharp pointwise bound) For every $z \in \mathbb{C}$,*

$$u_F(z) \leq \|F\|_{\mathcal{F}^2}^2,$$

and equality holds at some point $w \in \mathbb{C}$ if and only if F is proportional to the reproducing kernel K_w .

(iii) *$u_F(z) \rightarrow 0$ as $|z| \rightarrow \infty$.*

(iv) *One has the estimate*

$$\|u_F - u_G\|_{L^1(\mathbb{C})} \leq (\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}) \|F - G\|_{\mathcal{F}^2}.$$

Proof. (i) For $w \in \mathbb{C}$, the function $K_w(z) = e^{\pi z \bar{w}}$ is entire and satisfies, after completing the square via $2\Re(z\bar{w}) - |z|^2 = -|z - w|^2 + |w|^2$,

$$\|K_w\|_{\mathcal{F}^2}^2 = \int_{\mathbb{C}} e^{2\pi\Re(z\bar{w}) - \pi|z|^2} dA(z) = e^{\pi|w|^2} \int_{\mathbb{C}} e^{-\pi|z-w|^2} dA(z) = e^{\pi|w|^2},$$

since the Gaussian $e^{-\pi|\cdot|^2}$ integrates to 1 over $\mathbb{C}$ (with the Lebesgue measure dA on $\mathbb{C} \simeq \mathbb{R}^2$). In particular $K_w \in \mathcal{F}^2(\mathbb{C})$. Now, expanding $F \in \mathcal{F}^2(\mathbb{C})$ along the orthonormal basis $\{e_k\}_{k \geq 0}$ of Lemma 2.2 and using the Taylor expansion $K_w(z) = \sum_{k \geq 0} \frac{\pi^k \bar{w}^k}{k!} z^k = \sum_{k \geq 0} \overline{e_k(w)} e_k(z)$, one obtains the reproducing identity

$$\langle F, K_w \rangle_{\mathcal{F}^2} = \sum_{k \geq 0} a_k e_k(w) = F(w), \quad F(z) = \sum_{k \geq 0} a_k e_k(z).$$

Point evaluation is therefore continuous, and F is entire as a uniform limit on compact sets of polynomials. This is the standard description of the Bargmann–Fock space, see [4, 18, 22, 51].

(ii) Applying Cauchy–Schwarz to the reproducing identity from (i),

$$|F(z)| = |\langle F, K_z \rangle_{\mathcal{F}^2}| \leq \|F\|_{\mathcal{F}^2} \|K_z\|_{\mathcal{F}^2} = \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}, \quad (2.10)$$

which immediately gives $u_F(z) = |F(z)|^2 e^{-\pi|z|^2} \leq \|F\|_{\mathcal{F}^2}^2$. Equality at a point $z = w$ corresponds to equality in Cauchy–Schwarz, that is, $F = cK_w$ for some $c \in \mathbb{C}$; conversely, for $F = cK_w$ one computes $u_F(w) = |c|^2 e^{2\pi|w|^2} e^{-\pi|w|^2} = |c|^2 e^{\pi|w|^2} = |c|^2 \|K_w\|_{\mathcal{F}^2}^2 = \|F\|_{\mathcal{F}^2}^2$, proving the rigidity statement. The sharpness of this pointwise bound goes back to [34].

(iii) Fix $\varepsilon > 0$, and choose a polynomial P such that

$$\|F - P\|_{\mathcal{F}^2} < \varepsilon$$

using the density of polynomials in $\mathcal{F}^2(\mathbb{C})$. By the pointwise bound from (ii),

$$|F(z) - P(z)| e^{-\frac{\pi}{2}|z|^2} \leq \varepsilon \quad \text{for every } z \in \mathbb{C}.$$

Since P is a polynomial,

$$P(z) e^{-\frac{\pi}{2}|z|^2} \rightarrow 0 \quad \text{as } |z| \rightarrow \infty.$$

Hence

$$\limsup_{|z| \rightarrow \infty} |F(z)| e^{-\frac{\pi}{2}|z|^2} \leq \varepsilon.$$

As $\varepsilon > 0$ is arbitrary, we conclude that

$$|F(z)| e^{-\frac{\pi}{2}|z|^2} \rightarrow 0,$$

that is, $u_F(z) \rightarrow 0$ as $|z| \rightarrow \infty$.

(iv) Writing

$$|u_F - u_G| = ||F|^2 - |G|^2| e^{-\pi|z|^2} \leq |F - G|(|F| + |G|) e^{-\pi|z|^2},$$

the claim follows by Cauchy–Schwarz. $\square$

We now note the following crucial fact about the Bargmann transform: it transforms the basis of normalized Hermite functions $\{h_k\}_{k \geq 0}$ on $L^2(\mathbb{R})$ (see [50]) onto normalized monomials,

yielding a natural orthonormal basis on $\mathcal{F}^2(\mathbb{C})$. More precisely, one has $\mathcal{B}h_k = e_k$ for every $k \geq 0$, see [4, 18, 22].

Lemma 2.2. *The functions*

$$e_k(z) := b_k z^k, \quad k \geq 0,$$

form an orthonormal basis of $\mathcal{F}^2(\mathbb{C})$, where

$$b_k = \left(\frac{\pi^k}{k!} \right)^{1/2}.$$

In particular,

$$\langle e_k, e_m \rangle_{\mathcal{F}^2} = \delta_{k,m}.$$

Proof. Writing $z = re^{i\theta}$ and using $dA(z) = r dr d\theta$ on $\mathbb{C}$, a direct computation in polar coordinates gives

$$\int_{\mathbb{C}} |z|^{2k} e^{-\pi|z|^2} dA(z) = \int_0^{2\pi} \int_0^\infty r^{2k+1} e^{-\pi r^2} dr d\theta = 2\pi \int_0^\infty r^{2k+1} e^{-\pi r^2} dr = \frac{k!}{\pi^k},$$

where the last identity follows from the substitution $s = \pi r^2$, giving $2\pi \int_0^\infty r^{2k+1} e^{-\pi r^2} dr = \pi^{-k} \int_0^\infty s^k e^{-s} ds = \pi^{-k} \Gamma(k+1)$. Thus $\|z^k\|_{\mathcal{F}^2}^2 = \frac{k!}{\pi^k}$, which yields the normalization. For orthogonality, again writing $z = re^{i\theta}$,

$$\langle z^k, z^m \rangle_{\mathcal{F}^2} = \int_0^\infty r^{k+m+1} e^{-\pi r^2} dr \int_0^{2\pi} e^{i(k-m)\theta} d\theta = 0 \quad \text{for } k \neq m.$$

For completeness, since $\mathcal{B}: L^2(\mathbb{R}) \rightarrow \mathcal{F}^2(\mathbb{C})$ is a unitary isomorphism and $\mathcal{B}h_k = e_k$ for every $k \geq 0$, where $\{h_k\}_{k \geq 0}$ is the orthonormal basis of Hermite functions on $L^2(\mathbb{R})$ (see [50]), the family $\{e_k\}_{k \geq 0}$ is the image of an orthonormal basis under a unitary, hence itself an orthonormal basis of $\mathcal{F}^2(\mathbb{C})$. Equivalently, polynomials are dense in $\mathcal{F}^2(\mathbb{C})$; see [4, 51]. $\square$

Lemma 2.3. *Let $F \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) *F admits the expansion*

$$F(z) = \sum_{k=0}^{\infty} a_k e_k(z), \quad \sum_{k=0}^{\infty} |a_k|^2 = \|F\|_{\mathcal{F}^2}^2.$$

(ii) *(Growth bound) For every $z \in \mathbb{C}$,*

$$|F(z)| \leq \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

(iii) *(Derivative bounds) For every $z \in \mathbb{C}$,*

$$|F'(z)| \leq C(1 + |z|) \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

Proof. (i) follows from the orthonormal basis in Lemma 2.2.

(ii) follows from the reproducing kernel estimate:

$$|F(z)| = |\langle F, K_z \rangle| \leq \|F\|_{\mathcal{F}^2} \|K_z\|_{\mathcal{F}^2} = \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

(iii) Starting from the reproducing identity $F(z) = \langle F, K_z \rangle_{\mathcal{F}^2} = \int_{\mathbb{C}} F(w) e^{\pi \bar{w} z} e^{-\pi |w|^2} dA(w)$ and differentiating in z under the integral sign (justified by the absolute convergence on compacta

of z , via the pointwise bound from (ii)), we obtain

$$F'(z) = \pi \int_{\mathbb{C}} F(w) \bar{w} e^{\pi \bar{w} z} e^{-\pi |w|^2} dA(w) = \pi \langle F, w K_z \rangle_{\mathcal{F}^2},$$

since $\overline{w K_z(w)} = \bar{w} e^{\pi \bar{w} z}$. Cauchy–Schwarz then gives

$$|F'(z)| \leq \pi \|F\|_{\mathcal{F}^2} \|w K_z\|_{\mathcal{F}^2}.$$

Now, by completing the square as in (i),

$$\|w K_z\|_{\mathcal{F}^2}^2 = \int_{\mathbb{C}} |w|^2 e^{2\pi \Re(w \bar{z}) - \pi |w|^2} dA(w) = e^{\pi |z|^2} \int_{\mathbb{C}} |w + z|^2 e^{-\pi |w|^2} dA(w),$$

after the change of variables $w \mapsto w + z$. Expanding $|w + z|^2 = |w|^2 + 2\Re(w \bar{z}) + |z|^2$, and using that $\int_{\mathbb{C}} e^{-\pi |w|^2} dA(w) = 1$, $\int_{\mathbb{C}} |w|^2 e^{-\pi |w|^2} dA(w) = \frac{1}{\pi}$, and $\int_{\mathbb{C}} w e^{-\pi |w|^2} dA(w) = 0$ (by rotational invariance), we conclude

$$\int_{\mathbb{C}} |w + z|^2 e^{-\pi |w|^2} dA(w) = \frac{1}{\pi} + |z|^2 \leq C(1 + |z|^2),$$

for some absolute constant $C > 0$. Therefore

$$|F'(z)| \leq C(1 + |z|) \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2} |z|^2}.$$

□

Lemma 2.4. *Let $F \in \mathcal{F}^2(\mathbb{C})$ and set*

$$T := \sup_{z \in \mathbb{C}} u_F(z).$$

Then $0 \leq T \leq \|F\|_{\mathcal{F}^2}^2$, and $T = \|F\|_{\mathcal{F}^2}^2$ is attained if and only if F is proportional to the reproducing kernel K_w for some $w \in \mathbb{C}$.

Proof. The bound and the characterization of the equality case are immediate from Lemma 2.1(ii). □

We proceed with a few elementary facts.

Lemma 2.5. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) $I_F(s)$ is attained by a superlevel set of u_F , and

$$0 \leq I_F(s) \leq \|F\|_{\mathcal{F}^2}^2.$$

(ii) For every $c \in \mathbb{C}$,

$$I_{cF}(s) = |c|^2 I_F(s).$$

(iii) One has the Lipschitz bound

$$|I_F(s) - I_G(s)| \leq \|u_F - u_G\|_{L^1(\mathbb{C})} \leq (\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}) \|F - G\|_{\mathcal{F}^2}.$$

Proof. The identity

$$\int_{\mathbb{C}} u_F dA = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi |z|^2} dA(z) = \|F\|_{\mathcal{F}^2}^2$$

is immediate from the definition of the Fock norm. By Lemma 2.1(ii)–(iii), u_F is bounded, continuous, and satisfies $u_F(z) \rightarrow 0$ as $|z| \rightarrow \infty$; thus $u_F \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$.

The existence of a maximizing set for $I_F(s)$, of the form $\{u_F > t\}$ (possibly together with part of $\{u_F = t\}$) with $t \geq 0$ chosen so that the resulting set has measure s , is the classical bathtub principle, which applies since $u_F \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$. The bound

$$0 \leq I_F(s) \leq \int_{\mathbb{C}} u_F dA = \|F\|_{\mathcal{F}^2}^2$$

is immediate. The scaling identity in (ii) is obvious.

For (iii), for every measurable $\Omega \subset \mathbb{C}$ with $|\Omega| = s$,

$$\left| \int_{\Omega} u_F dA - \int_{\Omega} u_G dA \right| \leq \int_{\mathbb{C}} |u_F - u_G| dA.$$

Taking suprema over $|\Omega| = s$ gives

$$|I_F(s) - I_G(s)| \leq \|u_F - u_G\|_{L^1(\mathbb{C})}.$$

Finally,

$$\begin{aligned} \|u_F - u_G\|_{L^1(\mathbb{C})} &= \int_{\mathbb{C}} ||F|^2 - |G|^2| e^{-\pi|z|^2} dA \\ &\leq \int_{\mathbb{C}} |F - G|(|F| + |G|) e^{-\pi|z|^2} dA \\ &\leq \|F - G\|_{\mathcal{F}^2} (\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}) \end{aligned}$$

by Cauchy–Schwarz. □

2.3. From time-frequency analysis to free boundary problems. Having translated, in the previous subsection, the time–frequency localization problem into a question about the vanishing of a Fourier transform along the two complex lines $\eta = \pm i\xi$, we now bridge the gap between this Fourier-analytic statement and the free boundary formulation that will dominate the remainder of the paper. The bridge is provided by a divisibility lemma of Ehrenpreis–Malgrange flavour: any compactly supported distribution whose Fourier transform vanishes on the characteristic variety $\{\eta = \pm i\xi\}$ of the Laplacian must be the Laplacian of a compactly supported L^2 function which, in our setting, is moreover continuous off a finite set of points.

In order to state the lemma cleanly, we make precise the class of distributions with which we work. We say that a compactly supported distribution ω is a *singular-continuous free measure* if it defines a (finite, signed) Radon measure on $\mathbb{R}^2$ whose Lebesgue decomposition with respect to planar Lebesgue measure $dx dy$ takes the form

$$\omega = \omega_{ac} + \omega_{pp} = f dx dy + \sum_{j=1}^N c_j \delta_{p_j},$$

with $f \in L^\infty(\mathbb{R}^2)$ compactly supported, $c_j \in \mathbb{R} \setminus \{0\}$, and $p_1, \dots, p_N \in \mathbb{R}^2$ pairwise distinct. In other words, the continuous singular part ω_{cs} in the Lebesgue–Radon–Nikodym decomposition $\omega = \omega_{ac} + \omega_{cs} + \omega_{pp}$ is required to vanish identically, and the pure point part is finite. The

terminology *free measure* is suggested by the role of ω as the right-hand side of a free boundary problem arising in the next subsection.

Lemma 2.6. *Let $\omega \in \mathcal{S}'(\mathbb{R}^2)$ be a compactly supported singular-continuous free distribution such that*

$$\widehat{\omega}(w, \pm iw) = 0, \quad \forall w \in \mathbb{C}.$$

Then there exist a finite set $P \subset \mathbb{R}^2$ and a compactly supported function $u \in L^2(\mathbb{R}^2)$ such that

$$\Delta u = \omega$$

in the sense of distributions, and

$$u \in C(\mathbb{R}^2 \setminus P).$$

Proof. By the Paley–Wiener–Schwartz theorem (see, e.g., [25, 18]), the Fourier transform $\widehat{\omega}(\xi, \eta)$ extends to an entire function on $\mathbb{C}^2$ of exponential type, satisfying the bound

$$|\widehat{\omega}(\xi, \eta)| \leq C(1 + |\xi| + |\eta|)^N e^{R(|\Im \xi| + |\Im \eta|)}$$

for some $C, R > 0$ and some integer $N \geq 0$ depending on the order of ω , where R is the radius of any disk containing $\text{supp } \omega$. In particular, $\widehat{\omega}$ admits an everywhere convergent power series expansion

$$\widehat{\omega}(\xi, \eta) = \sum_{k, l \geq 0} a_{k, l} \xi^k \eta^l, \quad (\xi, \eta) \in \mathbb{C}^2.$$

Using the vanishing hypothesis on the complex lines $\eta = \pm i\xi$, we may group terms by total homogeneity: setting $\eta = \pm iw$, $\xi = w$, and collecting powers of w ,

$$0 = \widehat{\omega}(w, \pm iw) = \sum_{n \geq 0} \left(\sum_{k=0}^n a_{k, n-k} (\pm i)^{n-k} \right) w^n, \quad \forall w \in \mathbb{C}.$$

Since this holds for all $w \in \mathbb{C}$, comparing coefficients yields

$$\sum_{k=0}^n a_{k, n-k} (\pm i)^{n-k} = 0, \quad \forall n \geq 0.$$

For each $n \geq 0$, define the auxiliary one-variable polynomial

$$p_n(t) := \sum_{k=0}^n a_{k, n-k} t^{n-k}.$$

The two scalar equations above precisely say that $p_n(\pm i) = 0$, so $(t - i)(t + i) = t^2 + 1$ divides $p_n(t)$. Hence we may write

$$p_n(t) = (t^2 + 1) q_n(t),$$

with q_n a polynomial of degree at most $n - 2$ (and $q_n \equiv 0$ when $n < 2$).

We now reassemble the homogeneous pieces. Let

$$H_n(\xi, \eta) := \sum_{k=0}^n a_{k, n-k} \xi^k \eta^{n-k}, \quad n \geq 0,$$

so that $\widehat{\omega} = \sum_{n \geq 0} H_n$ is the decomposition of $\widehat{\omega}$ into homogeneous components (of degree n). The argument above shows that each H_n vanishes on the two complex lines $\eta = \pm i\xi$, hence is

divisible, as a homogeneous polynomial in two variables, by $\xi^2 + \eta^2$. Concretely,

$$H_n(\xi, \eta) = (\xi^2 + \eta^2) G_{n-2}(\xi, \eta)$$

for a (uniquely determined) homogeneous polynomial G_{n-2} of degree $n - 2$, with $G_{n-2} \equiv 0$ for $n < 2$. Summing the homogeneous pieces, the formal series

$$F(\xi, \eta) := \sum_{n \geq 2} G_{n-2}(\xi, \eta)$$

converges absolutely on every polydisc on which $\widehat{\omega}$ does, and defines an entire function on $\mathbb{C}^2$ satisfying

$$\widehat{\omega}(\xi, \eta) = (\xi^2 + \eta^2) F(\xi, \eta).$$

Since $\widehat{\omega}$ has a zero of order at least two at the origin (coming from the vanishing of H_0 and H_1 , which is forced by $p_0(\pm i) = p_1(\pm i) = 0$), this factorization is consistent at 0 and F is entire there as well.

We next quantify the size of F on $\mathbb{R}^2$. Since ω is a finite (signed) Radon measure with compact support, its Fourier transform is in fact bounded on $\mathbb{R}^2$:

$$|\widehat{\omega}(\xi, \eta)| \leq \|\omega\|_{\text{TV}}, \quad (\xi, \eta) \in \mathbb{R}^2,$$

where $\|\omega\|_{\text{TV}}$ is the total variation; equivalently, the Paley–Wiener bound holds with $N = 0$. On $\mathbb{R}^2$, $\xi^2 + \eta^2 = |\xi|^2 + |\eta|^2$, so for $|\xi|^2 + |\eta|^2 \geq 1$ we have $\xi^2 + \eta^2 \geq \frac{1}{2}(1 + |\xi|^2 + |\eta|^2)$, whence

$$|F(\xi, \eta)| = \frac{|\widehat{\omega}(\xi, \eta)|}{\xi^2 + \eta^2} \leq \frac{2\|\omega\|_{\text{TV}}}{1 + |\xi|^2 + |\eta|^2}, \quad |\xi|^2 + |\eta|^2 \geq 1.$$

Near the origin, F is entire (as established above) and hence bounded; absorbing this region into the constants we obtain

$$|F(\xi, \eta)| \leq \frac{C''}{1 + |\xi|^2 + |\eta|^2}, \quad (\xi, \eta) \in \mathbb{R}^2,$$

for some $C'' > 0$. Therefore $F|_{\mathbb{R}^2} \in L^2(\mathbb{R}^2) \cap L^1(\mathbb{R}^2)$. On $\mathbb{C}^2$, F is entire (by the formal-series construction above) and satisfies an exponential-type bound of the same rate R as $\widehat{\omega}$. Indeed, on the open set $\Omega := \{(\xi, \eta) \in \mathbb{C}^2 : |\xi^2 + \eta^2| \geq 1\}$ one has the pointwise estimate

$$|F(\xi, \eta)| = \frac{|\widehat{\omega}(\xi, \eta)|}{|\xi^2 + \eta^2|} \leq C(1 + |\xi| + |\eta|)^N e^{R(|\Im \xi| + |\Im \eta|)}, \quad (\xi, \eta) \in \Omega,$$

by the Paley–Wiener bound on $\widehat{\omega}$. On the complement $\Omega^c = \{|\xi^2 + \eta^2| < 1\}$, the function F is entire and the maximum modulus principle, applied on each complex line through a given point in Ω^c in a direction transverse to the variety $\{\xi^2 + \eta^2 = 0\}$, bounds $|F|$ at points of Ω^c by a uniform constant times the supremum of $|F|$ on the boundary of a fixed thin tubular neighbourhood of that variety, where the previous estimate applies. Combining these two regimes and absorbing the resulting bookkeeping constant yields, after enlarging C' if necessary,

$$|F(\xi, \eta)| \leq C' (1 + |\xi| + |\eta|)^N e^{R(|\Im \xi| + |\Im \eta|)}, \quad (\xi, \eta) \in \mathbb{C}^2,$$

which is the Paley–Wiener–Schwartz bound for a compactly supported distribution of order $\leq N$ in $\overline{B(0, R)}$.

By the Paley–Wiener–Schwartz theorem applied in the reverse direction (see again [25]), the entire function F satisfies an exponential-type bound on $\mathbb{C}^2$ with the same exponential rate R as $\widehat{\omega}$, and integrable decay along $\mathbb{R}^2$, so it is the Fourier–Laplace transform of a distribution of order $\leq N$ compactly supported in the closed disk $\overline{B(0, R)}$. The L^2 bound on the real locus shows that this distribution is, in fact, an $L^2(\mathbb{R}^2)$ function. Let $u \in L^2(\mathbb{R}^2)$ denote this compactly supported representative, characterised by $\widehat{u} = F$ on $\mathbb{R}^2$ and $\text{supp } u \subset \overline{B(0, R)}$.

Computing distributional Laplacians on the Fourier side, with the convention $\widehat{g}(\xi, \eta) = \int g(x, y) e^{-2\pi i(x\xi + y\eta)} dx$ we have

$$\widehat{\Delta u}(\xi, \eta) = -4\pi^2(\xi^2 + \eta^2) \widehat{u}(\xi, \eta) = -4\pi^2(\xi^2 + \eta^2) F(\xi, \eta) = -4\pi^2 \widehat{\omega}(\xi, \eta);$$

with any other standard normalisation, the same identity holds up to a nonzero real multiplicative factor. In all cases, Δu is a constant nonzero real multiple of ω . Replacing u by λu for the appropriate (nonzero, real) λ – this is the harmless rescaling referred to above – we may, and do, assume that

$$\Delta u = \omega$$

in the sense of distributions. The rescaling preserves all the previously established properties of u : compact support, membership in $L^2(\mathbb{R}^2)$, and the Fourier identity $\widehat{u} = \lambda F$.

To pin down u pointwise, observe that since ω is a compactly supported Radon measure on $\mathbb{R}^2$, the convolution

$$v(x, y) := \frac{1}{2\pi} (\log |\cdot| * \omega)(x, y)$$

is a well-defined locally integrable function on $\mathbb{R}^2$ which solves $\Delta v = \omega$ in the sense of distributions; here $G(z) = \log |z|$ is (up to the standard normalisation $\frac{1}{2\pi}$) the fundamental solution of the Laplacian in $\mathbb{R}^2$, in the sense that $\Delta(\frac{1}{2\pi} \log |\cdot|) = \delta_0$, see, e.g., [25] for the underlying distribution theory and [3] for related representations in the planar quadrature-domain context. The difference $u - v$ is then a tempered distribution which is harmonic on $\mathbb{R}^2$ (since $\Delta(u - v) = \omega - \omega = 0$), and hence, being harmonic, real-analytic. Outside the union of the supports of u and ω , the function v is in fact real-analytic and grows at most logarithmically at infinity, with $v(x) = \frac{M}{2\pi} \log |x| + O(|x|^{-1})$ as $|x| \rightarrow \infty$ where $M = \omega(\mathbb{R}^2)$, while u vanishes there because of its compact support. Hence $u - v$ is harmonic on $\mathbb{R}^2$ and grows at most logarithmically at infinity; a standard refinement of the harmonic Liouville theorem (any harmonic function on $\mathbb{R}^2$ growing slower than a polynomial of degree 1 is constant) then forces $u - v$ to be a constant, and this constant must vanish to be consistent with $u \in L^2(\mathbb{R}^2)$ and the local integrability of v . Equivalently, and more transparently: by the compact support of ω together with $\widehat{u} = F$ on $\mathbb{R}^2$, the function u is uniquely determined among compactly supported L^2 solutions of $\Delta u = \omega$, since any two such would differ by an entire L^2 function which is harmonic, hence vanishes. Thus

$$u(x, y) = \frac{1}{2\pi} \int_{\mathbb{R}^2} \log |(x, y) - (x', y')| d\omega(x', y'),$$

the logarithmic potential of ω .

We finally analyse the regularity of u . Decomposing $\omega = f \, dx \, dy + \sum_{j=1}^N c_j \delta_{p_j}$ as in the definition of singular-continuous free measure, we obtain correspondingly

$$u(x, y) = \frac{1}{2\pi} \int_{\mathbb{R}^2} \log |(x, y) - (x', y')| f(x', y') \, dx' \, dy' + \frac{1}{2\pi} \sum_{j=1}^N c_j \log |(x, y) - p_j|.$$

The first summand is the logarithmic potential of the bounded, compactly supported density $f \in L^\infty(\mathbb{R}^2)$ and is therefore not merely continuous but in fact $C^{1,\alpha}(\mathbb{R}^2)$ for every $\alpha \in (0, 1)$ (by the standard Calderón–Zygmund / Schauder estimates applied to $\Delta = \partial_x^2 + \partial_y^2$ with right-hand side in L^∞ of compact support, see, e.g., the Newtonian-potential discussion in [25]). The continuity used here is in fact elementary: by dominated convergence, $z \mapsto \int \log |z - w| f(w) \, dA(w)$ is continuous since $\log |\cdot - w|$ is locally integrable in w uniformly on compacta in z , and f is bounded with compact support. It is here that the absence of a continuous singular part in ω is essential: a generic continuous singular measure (e.g. the Hausdorff measure on a Cantor-type set of positive 1-capacity) gives rise to a logarithmic potential which need not be continuous, only quasi-continuous, and the conclusion of the lemma would fail in that generality. Each term $\log |(x, y) - p_j|$ is continuous on $\mathbb{R}^2 \setminus \{p_j\}$, with a logarithmic singularity at p_j . Setting $P := \{p_1, \dots, p_N\}$, we conclude that $u \in C(\mathbb{R}^2 \setminus P)$, as claimed. The fact that u is compactly supported has already been established in the construction above and is consistent with the representation as a logarithmic potential, which differs from a compactly supported function only by a globally harmonic correction that, as explained, must vanish identically. $\square$

We postpone the proof of Theorem 2 until after the perturbative inverse theorem. The same inverse/free-boundary mechanism used to construct domains for prescribed eigenfunctions also recovers the classical Abreu–Dörfler disk theorem in the simply connected setting; see Section 4 below.

3. PROOF OF THEOREM 1

3.1. The disk inverse step. The proof rests on a local inverse theorem at the Daubechies disk. The result below is self-contained, but its point of view is inspired by perturbative inverse-source and quadrature-domain methods: Isakov’s local free-boundary scheme for inverse source problems, Cherednichenko’s local inverse logarithmic potential theorem, and the quadrature-domain framework developed by Gustafsson, Shahgholian, Karp, and Aharonov–Schiffer–Zalcman. In the present disk-based setting the required inverse step can be proved directly by diagonalizing the full moment map in Fourier modes.

Fix $0 < \lambda < 1$, and let $r_\lambda > 0$ be determined by

$$1 - e^{-\pi r_\lambda^2} = \lambda.$$

For a real-valued 2π -periodic function φ with $\|\varphi\|_{C^0} < r_\lambda/2$, set

$$U_\varphi := \{re^{i\theta} : 0 \leq r < r_\lambda + \varphi(\theta)\}.$$

For $\tau > 0$, let $\mathcal{A}_\tau^\mathbb{R}$ be the Banach space of real-valued periodic functions

$$\varphi(\theta) = \sum_{k \in \mathbb{Z}} \varphi_k e^{ik\theta}, \quad \varphi_{-k} = \overline{\varphi_k}, \quad (3.1)$$

with norm

$$\|\varphi\|_{\mathcal{A}_\tau} := \sum_{k \in \mathbb{Z}} |\varphi_k| e^{\tau|k|} < \infty. \quad (3.2)$$

This is the real subspace of the Wiener-type Banach algebra of functions on $\mathbb{T} := \mathbb{R}/2\pi\mathbb{Z}$ admitting a holomorphic extension to the strip $|\Im \theta| < \tau$ with absolutely summable Fourier expansion. It is a Banach algebra, continuously embedded into $C^m(\mathbb{T})$ for every $m \geq 0$, and in particular into the Hölder space $C^{1,\alpha}(\mathbb{T})$ for every $\alpha \in (0, 1)$. Small elements of $\mathcal{A}_\tau^\mathbb{R}$ therefore produce real-analytic C^2 Jordan domains, given as radial graphs over the disk D_{r_λ} .

For sequences indexed by $j \geq 0$, the natural target Banach space dual to $\mathcal{A}_\tau^\mathbb{R}$ under the moment map of the proposition below is

$$\mathcal{Y}_\tau := \left\{ y = (y_j)_{j \geq 0} : y_0 \in \mathbb{R}, \|y\|_{\mathcal{Y}_\tau} := \frac{|y_0|}{a_0} + 2 \sum_{j=1}^{\infty} e^{\tau j} \frac{|y_j|}{a_j} < \infty \right\}, \quad (3.3)$$

with the weight sequence $\{a_j\}_{j \geq 0}$ given by (3.6) below. The factor of two in (3.3) accounts for the conjugate pairing $\varphi_{-j} = \overline{\varphi_j}$ used in (3.1).

Proposition 3.1 (Disk symmetrized inverse theorem). *Let $N \geq 1$, $0 < \lambda < 1$, and r_λ be as above. There are $\tau > 0$, $\varepsilon > 0$, and a C^1 map*

$$b = (b_1, \dots, b_N) \mapsto \varphi_b \in \mathcal{A}_\tau^\mathbb{R}, \quad \varphi_0 = 0,$$

defined for $\max_j |b_j| < \varepsilon$, such that, for

$$P_b(z) := 1 + \sum_{j=1}^N b_j z^j, \quad P_b^\#(z) := \overline{P_b(\bar{z})}, \quad Q_b(w) := P_b^\#(\partial_w/\pi) P_b(w),$$

and

$$U_b := U_{\varphi_b},$$

one has

$$\int_{U_b} |P_b(z)|^2 e^{-\pi|z|^2} e^{\pi \bar{z} w} dA(z) = \lambda Q_b(w), \quad w \in \mathbb{C}. \quad (3.4)$$

Moreover,

$$\|\varphi_b\|_{C^2(\mathbb{S}^1)} \leq C_{\lambda, N} \max_{1 \leq j \leq N} |b_j|,$$

and the branch is locally unique among sufficiently small analytic radial graphs in $\mathcal{A}_\tau^\mathbb{R}$.

Proof. We solve the coefficient equations obtained by expanding the entire identity (3.4). Write

$$Q_b(w) = \sum_{j=0}^N q_j(b) w^j,$$

and $q_j(b) = 0$ for $j > N$. The constant coefficient is real:

$$q_0(b) = 1 + \sum_{k=1}^N \frac{k!}{\pi^k} |b_k|^2.$$

Since

$$e^{\pi \bar{z} w} = \sum_{j=0}^{\infty} \frac{\pi^j}{j!} \bar{z}^j w^j,$$

the desired identity is equivalent to the infinite system

$$\frac{\pi^j}{j!} \int_{U_\varphi} |P_b(z)|^2 \bar{z}^j e^{-\pi|z|^2} dA(z) = \lambda q_j(b), \quad j \geq 0. \quad (3.5)$$

We introduce the weighted sequence space adapted to the linearization at the circle. Put

$$a_j := 2\pi e^{-\pi r_\lambda^2} r_\lambda^{j+1} \frac{\pi^j}{j!}, \quad j \geq 0, \quad (3.6)$$

so that $\{a_j\}$ decays super-exponentially in j , and $\mathcal{Y}_\tau$ is the Banach space defined in (3.3). Define

$$\mathcal{F}_j(b, \varphi) := \frac{\pi^j}{j!} \int_{U_\varphi} |P_b(z)|^2 \bar{z}^j e^{-\pi|z|^2} dA(z) - \lambda q_j(b), \quad j \geq 0.$$

Thus $\mathcal{F}(b, \varphi) = 0$ is precisely (3.5). At $b = 0$, $\varphi = 0$, the domain is the disk D_{r_λ} , and radial symmetry gives

$$\frac{\pi^j}{j!} \int_{D_{r_\lambda}} \bar{z}^j e^{-\pi|z|^2} dA(z) = \begin{cases} \lambda, & j = 0, \\ 0, & j \geq 1. \end{cases}$$

Hence $\mathcal{F}(0, 0) = 0$.

We next compute the Fréchet derivative of the moment map with respect to the boundary graph. Let $\psi(\theta) = \sum_{k \in \mathbb{Z}} \psi_k e^{ik\theta} \in \mathcal{A}_\tau^\mathbb{R}$, and recall the polar-coordinate formula

$$\int_{U_\varphi} H(z) dA(z) = \int_0^{2\pi} \int_0^{r_\lambda + \varphi(\theta)} H(re^{i\theta}) r dr d\theta,$$

valid for any continuous $H: \mathbb{C} \rightarrow \mathbb{C}$. Differentiating the right-hand side at $\varphi = 0$ in the direction of ψ by the fundamental theorem of calculus gives

$$D_\varphi \left(\int_{U_\varphi} H(z) dA(z) \right) \Big|_{\varphi=0} [\psi] = r_\lambda \int_0^{2\pi} \psi(\theta) H(r_\lambda e^{i\theta}) d\theta. \quad (3.7)$$

Applied to $H(z) = \frac{\pi^j}{j!} \bar{z}^j e^{-\pi|z|^2}$ and combined with the orthogonality relation $\int_0^{2\pi} e^{i(k-j)\theta} d\theta = 2\pi \delta_{kj}$, formula (3.7) yields the closed-form Fourier-multiplier expression

$$D_\varphi \mathcal{F}_j(0, 0)[\psi] = \frac{\pi^j}{j!} r_\lambda^{j+1} e^{-\pi r_\lambda^2} \int_0^{2\pi} \psi(\theta) e^{-ij\theta} d\theta = a_j \psi_j, \quad j \geq 0, \quad (3.8)$$

where a_j is the weight in (3.6). Consequently the Fréchet derivative $D_\varphi \mathcal{F}(0, 0)$ is the Fourier multiplier

$$D_\varphi \mathcal{F}(0, 0): \mathcal{A}_\tau^\mathbb{R} \longrightarrow \mathcal{Y}_\tau, \quad \psi = \sum_{k \in \mathbb{Z}} \psi_k e^{ik\theta} \longmapsto (a_j \psi_j)_{j \geq 0}, \quad (3.9)$$

acting diagonally on the non-negative Fourier modes of ψ .

The reality constraint $\psi_{-j} = \overline{\psi_j}$ shows that the kernel of the unrestricted multiplier on the complexified space consists exactly of the negative-frequency modes; on $\mathcal{A}_\tau^\mathbb{R}$ the reality constraint determines those negative modes from the non-negative ones, so $D_\varphi \mathcal{F}(0, 0)$ has trivial kernel as a real-linear map. We verify directly that (3.9) is bounded and invertible.

Boundedness. Using (3.2) and (3.3),

$$\|D_\varphi \mathcal{F}(0, 0)\psi\|_{\mathcal{Y}_\tau} = \frac{|a_0\psi_0|}{a_0} + 2 \sum_{j=1}^{\infty} e^{\tau j} \frac{|a_j\psi_j|}{a_j} = |\psi_0| + 2 \sum_{j=1}^{\infty} e^{\tau j} |\psi_j| = \|\psi\|_{\mathcal{A}_\tau}.$$

Invertibility. For $y \in \mathcal{Y}_\tau$ define

$$\psi_0 := \frac{y_0}{a_0} \in \mathbb{R}, \quad \psi_j := \frac{y_j}{a_j} \ (j \geq 1), \quad \psi_{-j} := \overline{\psi_j} \ (j \geq 1), \quad (3.10)$$

and $\psi(\theta) := \sum_{k \in \mathbb{Z}} \psi_k e^{ik\theta}$. The above computation gives $\|\psi\|_{\mathcal{A}_\tau} = \|y\|_{\mathcal{Y}_\tau}$, so $\psi \in \mathcal{A}_\tau^\mathbb{R}$, and by construction $D_\varphi \mathcal{F}(0, 0)\psi = y$. Hence (3.9) is an isometric isomorphism of $\mathcal{A}_\tau^\mathbb{R}$ onto $\mathcal{Y}_\tau$; in particular $\|D_\varphi \mathcal{F}(0, 0)^{-1}\|_{\mathcal{Y}_\tau \rightarrow \mathcal{A}_\tau^\mathbb{R}} = 1$, which provides the explicit quantitative invertibility needed for the Banach-space implicit function theorem.

It remains to justify the use of the Banach-space implicit function theorem. For $\|\varphi\|_{\mathcal{A}_\tau}$ small, the function $\theta \mapsto r_\lambda + \varphi(\theta)$ extends holomorphically to the strip $|\Im \theta| < \tau$ and stays in a fixed annulus $r_\lambda/2 < |r| < 3r_\lambda/2$. In polar coordinates,

$$\mathcal{F}_j(b, \varphi) + \lambda q_j(b) = \frac{\pi^j}{j!} \int_0^{2\pi} e^{-ij\theta} \int_0^{r_\lambda + \varphi(\theta)} P_b(re^{i\theta}) P_b^\#(re^{-i\theta}) r^{j+1} e^{-\pi r^2} dr d\theta.$$

Expanding the inner integral at r_λ gives an absolutely convergent power series in φ , whose coefficients are analytic functions of θ and polynomial functions of b . Since $\mathcal{A}_\tau$ is a Banach algebra and $P_b(re^{i\theta}) P_b^\#(re^{-i\theta})$ is a finite trigonometric polynomial of degree at most N , the resulting series and its first derivatives in (b, φ) converge in $\mathcal{Y}_\tau$ after possibly decreasing τ and restricting to $\|b\| + \|\varphi\|_{\mathcal{A}_\tau} < \eta$. More explicitly, Cauchy's estimate on a slightly wider strip gives, for $0 < \tau' < \tau$,

$$\sum_{j \geq 0} e^{\tau' j} \frac{1}{a_j} \left| \frac{\pi^j}{j!} \widehat{R_j}(j) \right| \leq C_{\lambda, N, \tau, \tau'} \|R\|_{\mathcal{A}_\tau},$$

where $\widehat{R_j}(j)$ denotes the j -th Fourier coefficient of the analytic coefficient appearing in the j -th moment equation. Applying this estimate term by term to the Taylor expansion in φ yields

$$\|\mathcal{F}(b, \varphi) - \mathcal{F}(b, \psi) - D_\varphi \mathcal{F}(b, \psi)[\varphi - \psi]\|_{\mathcal{Y}_{\tau'}} \leq C(\|b\| + \|\varphi\|_{\mathcal{A}_\tau} + \|\psi\|_{\mathcal{A}_\tau}) \|\varphi - \psi\|_{\mathcal{A}_\tau}^2,$$

and analogous estimates for differentiation in b . Shrinking the strip once and then relabelling τ' as τ , this proves that

$$\mathcal{F} : \{(b, \varphi) : \|b\| + \|\varphi\|_{\mathcal{A}_\tau} < \eta\} \longrightarrow \mathcal{Y}_\tau$$

is C^1 .

The Banach implicit function theorem now applies at $(0, 0)$, because $D_\varphi \mathcal{F}(0, 0)$ is an isomorphism. Thus, for $\|b\|$ small, there is a unique $\varphi_b \in \mathcal{A}_\tau^\mathbb{R}$, small in $\mathcal{A}_\tau$, such that $\mathcal{F}(b, \varphi_b) = 0$, and $b \mapsto \varphi_b$ is C^1 . The embedding $\mathcal{A}_\tau \hookrightarrow C^2$ gives the displayed C^2 bound.

It remains to upgrade the coefficient identities (3.5) to the global identity (3.4). Define the entire functions

$$\mathcal{L}(w) := \int_{U_b} |P_b(z)|^2 e^{-\pi|z|^2} e^{\pi\bar{z}w} dA(z), \quad \mathcal{R}(w) := \lambda Q_b(w), \quad w \in \mathbb{C}. \quad (3.11)$$

The function $\mathcal{R}$ is a polynomial of degree $\leq N$, and $\mathcal{L}$ is entire of exponential type πM , with $M = \sup_{z \in U_b} |z|$, since $|e^{\pi\bar{z}w}| \leq e^{\pi M|w|}$ on U_b . Both $\mathcal{L}$ and $\mathcal{R}$ admit absolutely convergent Taylor expansions at the origin,

$$\mathcal{L}(w) = \sum_{j \geq 0} \frac{\pi^j}{j!} \left(\int_{U_b} |P_b(z)|^2 \bar{z}^j e^{-\pi|z|^2} dA(z) \right) w^j, \quad \mathcal{R}(w) = \sum_{j=0}^N \lambda q_j(b) w^j,$$

the first by interchange of summation and integration (justified by $\sum_j \frac{(\pi M|w|)^j}{j!} < \infty$ and the boundedness of U_b). The system (3.5), namely $\mathcal{F}_j(b, \varphi_b) = 0$ for every $j \geq 0$, is exactly the identification of the Taylor coefficients of $\mathcal{L}$ and $\mathcal{R}$ at the origin, including the moments of degree $j > N$ for which $q_j(b) = 0$ and the integral on the left-hand side accordingly vanishes. Hence $\mathcal{L} = \mathcal{R}$ as formal power series, and by the identity theorem for entire functions $\mathcal{L} \equiv \mathcal{R}$ on $\mathbb{C}$, which is (3.4).

Equivalently, the same identity may be read off from the exterior Cauchy-transform picture: restricting (3.4) to real $w = t > 0$, multiplying by $e^{-\pi\zeta t}$ and integrating in $t \in (0, \infty)$ for real $\zeta > M$ gives, by Fubini's theorem (applicable since $\Re(\bar{z} - \zeta) \leq |z| - \zeta < 0$ on U_b),

$$\frac{1}{\pi} \int_{U_b} \frac{|P_b(z)|^2 e^{-\pi|z|^2}}{\zeta - \bar{z}} dA(z) = \lambda \sum_{j=0}^N q_j(b) \frac{j!}{\pi^{j+1} \zeta^{j+1}}, \quad \zeta \in (M, \infty). \quad (3.12)$$

Both sides of (3.12) are holomorphic in ζ on the connected open set $\mathbb{C} \setminus \overline{U_b^*}$ (where $U_b^* = \{\bar{z} : z \in U_b\}$), and they agree on the real ray (M, ∞) , which has an accumulation point in this set. By the identity theorem for holomorphic functions on a connected domain, (3.12) extends to all $\zeta \in \mathbb{C} \setminus \overline{U_b^*}$. Expanding the geometric series $1/(\zeta - \bar{z}) = \sum_{j \geq 0} \bar{z}^j / \zeta^{j+1}$ at $\zeta = \infty$ and matching the resulting Laurent coefficients gives the moment identities $\frac{\pi^j}{j!} \int_{U_b} |P_b|^2 \bar{z}^j e^{-\pi|z|^2} dA = \lambda q_j(b)$ of (3.5), including the vanishing of the higher moments $\int_{U_b} |P_b|^2 \bar{z}^j e^{-\pi|z|^2} dA = 0$ for every $j > N$ (equivalently $q_j(b) = 0$ for $j > N$). The two formulations are therefore equivalent, and either yields (3.4). This proves the proposition. $\square$

Remark 3.1. The proof above is the only place where analyticity of the boundary parametrization is used. The constructed domains are therefore C^∞ , hence C^2 , which is the regularity needed later. The argument does not assert a general local quadrature-domain theorem near an arbitrary reference domain; it proves exactly the disk-based inverse statement used below.

We isolate the ODE-theoretic ingredient needed for the unsymmetrization step of the proof of Theorem 1, namely the claim that the only entire function annihilated by $P^\#(\partial_w/\pi)$ and growing no faster than the Cauchy-transform of a bounded weighted measure on U is the zero function, provided the zeros of the symbol $P^\#$ escape to infinity.

Lemma 3.1 (Step 3: rigidity of exponential-polynomial solutions). *Let $N \geq 1$ and let $P(z) = 1 + \sum_{k=1}^N b_k z^k$ be a polynomial with $\max_k |b_k| \leq 1$. Let $P^\#(z) = \overline{P(\bar{z})}$ be its coefficientwise*

conjugate, and consider the constant-coefficient differential operator $P^\#(\partial_w/\pi)$ defined as in (3.21). Suppose $H: \mathbb{C} \rightarrow \mathbb{C}$ is an entire function satisfying

$$P^\#(\partial_w/\pi)H(w) = 0, \quad w \in \mathbb{C}, \quad (3.13)$$

together with the growth bound

$$|H(w)| \leq C(1 + |w|)^N e^{\pi M|w|}, \quad w \in \mathbb{C}, \quad (3.14)$$

for some $C > 0$ and some $M > 0$. If, in addition, the zeros $\zeta_1, \dots, \zeta_\ell$ of the characteristic polynomial $\tilde{P}^\#(\zeta) := P^\#(\zeta/\pi)$ on $\mathbb{C}$ all satisfy

$$|\zeta_j| > \pi M, \quad 1 \leq j \leq \ell, \quad (3.15)$$

then $H \equiv 0$.

Proof. Let $m_1, \dots, m_\ell$ be the multiplicities of the zeros $\zeta_1, \dots, \zeta_\ell$ of $\tilde{P}^\#$, and write $\tilde{P}^\#(\zeta) = \overline{b_N} \pi^{-N} \prod_{j=1}^\ell (\zeta - \zeta_j)^{m_j}$ when $b_N \neq 0$ (otherwise the leading coefficient is the highest non-zero $\overline{b_k}$ divided by π^k , and the same argument applies). The elementary structure theorem for entire solutions of constant-coefficient linear ODEs of finite exponential type [27, Ch. XII] states that every entire solution of (3.13) with finite exponential type takes the form

$$H(w) = \sum_{j=1}^\ell R_j(w) e^{\zeta_j w}, \quad (3.16)$$

where $R_j \in \mathbb{C}[w]$ is a polynomial of degree at most $m_j - 1$. The hypothesis (3.14) ensures that H has exponential type at most πM , so (3.16) indeed applies.

Suppose, for a contradiction, that some R_j does not vanish identically. Re-indexing if necessary, choose j_0 with $R_{j_0} \not\equiv 0$ and $|\zeta_{j_0}| = \max\{|\zeta_j|: R_j \not\equiv 0\}$. After a small generic rotation of the coordinate $w \mapsto e^{i\theta} w$ (which leaves both (3.13) and (3.14) invariant up to relabeling the ζ_j), we may assume that for every j with $j \neq j_0$ and $R_j \not\equiv 0$,

$$\Re(\zeta_j \overline{\zeta_{j_0}}) < |\zeta_{j_0}|^2. \quad (3.17)$$

This is automatic when $|\zeta_j| < |\zeta_{j_0}|$ from the Cauchy–Schwarz inequality $\Re(\zeta_j \overline{\zeta_{j_0}}) \leq |\zeta_j| |\zeta_{j_0}|$, and ties at maximal modulus are broken by the rotation.

Along the half-line $w_t := t \overline{\zeta_{j_0}}$, $t \geq 0$, the exponential factor of the j_0 th term has modulus $|e^{\zeta_{j_0} w_t}| = e^{t|\zeta_{j_0}|^2}$, while every other term satisfies $|R_j(w_t) e^{\zeta_j w_t}| \leq C_j (1+t)^{m_j-1} e^{t \Re(\zeta_j \overline{\zeta_{j_0}})}$, which by (3.17) grows strictly slower than $e^{t|\zeta_{j_0}|^2}$. Combining these and using $R_{j_0} \not\equiv 0$,

$$|H(w_t)| \geq c_0 e^{t|\zeta_{j_0}|^2} \quad \text{for all sufficiently large } t, \quad (3.18)$$

with $c_0 > 0$ depending only on R_{j_0} and the gap in (3.17).

On the other hand, (3.14) evaluated at w_t gives

$$|H(w_t)| \leq C(1 + t|\zeta_{j_0}|)^N e^{\pi M t |\zeta_{j_0}|}. \quad (3.19)$$

Comparing (3.18) and (3.19), and absorbing the polynomial factor into an arbitrarily small exponential, yields $|\zeta_{j_0}|^2 \leq \pi M |\zeta_{j_0}|$, i.e. $|\zeta_{j_0}| \leq \pi M$, contradicting (3.15). Hence all R_j are identically zero, and $H \equiv 0$. $\square$

Remark 3.2. Lemma 3.1 is the precise reason why no finite-dimensional obstruction is left behind by the symmetrization (3.23) $\Rightarrow$ (3.25): the kernel of $P^\#(\partial_w/\pi)$ on the full space of entire functions is the finite-dimensional exponential-polynomial space spanned by $\{w^i e^{\zeta_j w} : 0 \leq i < m_j, 1 \leq j \leq \ell\}$, but the growth constraint (3.14) together with the non-vanishing of $P^\#$ on the closure of U (which forces (3.15)) collapses this kernel to $\{0\}$. The role of $P^\#$ being non-vanishing on U is therefore not cosmetic: it is what guarantees that the zeros of $\tilde{P}^\#$ are far from the origin, hence beyond the exponential-type radius πM that any candidate solution can have.

Proof of Theorem 1. We again use the Bargmann transform. Let

$$P(z) = 1 + \sum_{k=1}^N b_k z^k,$$

where the coefficients b_k are chosen so that the Bargmann transform of

$$f_0 = h_0 + \sum_{k=1}^N a_k h_k$$

is exactly P . We also write

$$P^\#(z) := \overline{P(\bar{z})} = 1 + \sum_{k=1}^N \bar{b}_k z^k \quad (3.20)$$

for the *coefficientwise conjugate polynomial*: the operation $P \mapsto P^\#$ leaves the monomials z^k intact and conjugates only the scalar coefficients, so $P^\#$ is again a polynomial of the same degree as P , holomorphic in z , satisfying $P^\#(\bar{z}) = \overline{P(z)}$ for every $z \in \mathbb{C}$. Associated with $P^\#$ is the holomorphic constant-coefficient differential operator on the variable $w \in \mathbb{C}$ defined by

$$P^\#(\partial_w/\pi) := \sum_{k=0}^N \bar{b}_k \left(\frac{\partial_w}{\pi}\right)^k, \quad b_0 := 1, \quad (3.21)$$

i.e. by substituting $z = \partial_w/\pi$ into (3.20). Since $\partial_w = \frac{1}{2}(\partial_x - i\partial_y)$ is the standard Wirtinger derivative on entire functions of $w = u + iv$, the operator $P^\#(\partial_w/\pi)$ acts on every entire function as the finite linear combination of the iterated Wirtinger derivatives $(\partial_w/\pi)^k$, $0 \leq k \leq N$; in particular it is a finite-order, holomorphic differential operator. The key calculation is that $P^\#(\partial_w/\pi)$ acts as the multiplier $P^\#(\bar{z})$ on each plane wave $w \mapsto e^{\pi \bar{z} w}$:

$$P^\#(\partial_w/\pi)[e^{\pi \bar{z} w}] = P^\#(\bar{z}) e^{\pi \bar{z} w} = \overline{P(z)} e^{\pi \bar{z} w}, \quad z, w \in \mathbb{C}, \quad (3.22)$$

because $(\partial_w/\pi)^k e^{\pi \bar{z} w} = \bar{z}^k e^{\pi \bar{z} w}$.

Term-by-term differentiation of an integrand of the form $P(z)e^{-\pi|z|^2}e^{\pi \bar{z} w}$ against $z \in U$ is justified by dominated convergence: U is bounded, $P(z)e^{-\pi|z|^2}$ is bounded on U , and for each $k \leq N$ and each compact $K \subset \mathbb{C}$ one has the locally uniform estimate

$$\left| \left(\frac{\partial_w}{\pi}\right)^k [e^{\pi \bar{z} w}] \right| = |\bar{z}|^k e^{\pi \Re(\bar{z} w)} \leq M^k e^{\pi M|w|}, \quad z \in U, w \in K,$$

with $M := \sup_{z \in U} |z|$. Thus differentiation and integration may be exchanged finitely many times, which is all that is required to apply $P^\#(\partial_w/\pi)$ term-by-term to the entire function

defined by an integral against $z \in U$. Then the condition that f_0 be an eigenfunction of $\mathcal{L}_U$ with eigenvalue λ is equivalent to

$$\int_U P(z) e^{-\pi|z|^2} e^{\pi \bar{z} w} dA(z) = \lambda P(w), \quad w \in \mathbb{C}. \quad (3.23)$$

Set

$$Q(w) := P^\#(\partial_w/\pi)P(w). \quad (3.24)$$

By Proposition 3.1, after decreasing $\varepsilon_0(\lambda, N)$ if necessary, there exists a real-analytic radial graph $U = U_b$, C^2 -close to the Daubechies disk of radius r_λ , for which

$$\int_U |P(z)|^2 e^{-\pi|z|^2} e^{\pi \bar{z} w} dA(z) = \lambda Q(w), \quad w \in \mathbb{C}. \quad (3.25)$$

We now recover the unsymmetrized identity (3.23). Define

$$G(w) := \int_U P(z) e^{-\pi|z|^2} e^{\pi \bar{z} w} dA(z).$$

The function G is entire, and differentiating under the integral is justified by boundedness of U . Since

$$\left(\frac{\partial_w}{\pi}\right)^j e^{\pi \bar{z} w} = \bar{z}^j e^{\pi \bar{z} w},$$

we have

$$P^\#(\partial_w/\pi)G(w) = \int_U |P(z)|^2 e^{-\pi|z|^2} e^{\pi \bar{z} w} dA(z) = \lambda Q(w) = P^\#(\partial_w/\pi)(\lambda P)(w).$$

Thus $H := G - \lambda P$ is an entire solution of

$$P^\#(\partial_w/\pi)H = 0.$$

We now verify the hypotheses of Lemma 3.1. Setting $M := \sup_{z \in U} |z|$, the integral defining G yields the standard exponential-type bound

$$|G(w)| \leq \left(\sup_{z \in U} |P(z)| e^{-\pi|z|^2} \right) |U| e^{\pi M|w|} \leq C(1 + |w|)^N e^{\pi M|w|}, \quad w \in \mathbb{C}, \quad (3.26)$$

because $|e^{\pi \bar{z} w}| \leq e^{\pi|z||w|}$ for $z \in U$. Since λP grows only polynomially, $H = G - \lambda P$ inherits the same exponential-type bound, with possibly larger constant: H satisfies (3.14).

The location of the zeros of the characteristic polynomial $\tilde{P}^\#$ is controlled by b . Since $U \subset V_2$ (with V_2 defined in Proposition 3.1), and since $P^\#(z) = 1 + \sum_k \bar{b}_k z^k$ reduces to the constant 1 at $b = 0$, the zeros of $P^\#$ on $\mathbb{C}$ escape to infinity uniformly as $b \rightarrow 0$; a fortiori the zeros of $\tilde{P}^\#(\zeta) = P^\#(\zeta/\pi)$ escape to infinity. After shrinking $\varepsilon_0(\lambda, N)$ if necessary, all ℓ zeros ζ_j of $\tilde{P}^\#$ satisfy

$$|\zeta_j| > 10M \geq \pi M, \quad 1 \leq j \leq \ell,$$

which is hypothesis (3.15). Lemma 3.1 therefore implies $H \equiv 0$, which is precisely (3.23).

The Bargmann transform is unitary and intertwines the localization operator with the integral operator in (3.23). Therefore the inverse Bargmann transform f_0 of P satisfies

$$\mathcal{L}_U f_0 = \lambda f_0.$$

This proves the theorem. □

4. A CLASSICAL RIGIDITY COROLLARY

The localization operator $\mathcal{L}_U$ on the Bargmann–Fock space is diagonalized by Hermite functions whenever U is a disk centered at the origin: this is an immediate consequence of the rotational symmetry of U together with the radial structure of the Hermite weight. The converse question, raised by Abreu and Dörfler, asks whether eigenfunction status of even a *single* Hermite function already forces the underlying domain to be a disk. The next theorem records a positive answer in the simply connected C^2 regime.

For convenience we recall the statement of Theorem 2 from the introduction: *let $U \subset \mathbb{C}$ be a simply connected bounded domain with C^2 boundary, and let h_k be a Hermite function ($k \geq 0$). If h_k is an eigenfunction of the localization operator $\mathcal{L}_U$, then U is a disk.* (The original Abreu–Dörfler theorem [1] dispenses with the boundary regularity assumption; we will recover this stronger form in the second proof below.)

We will give two proofs of Theorem 2. The first, presented below, leverages the Fourier-analytic machinery developed earlier in the paper, reducing the question to a free boundary problem for a logarithmic potential. The second, in the next subsection, proceeds along more classical complex-analytic lines and, as we shall see, dispenses with the C^2 boundary assumption altogether.

Before launching into the arguments, we record, in a form convenient for later reference, the two analytic inputs underlying both proofs. The first is the precise variant of the Paley–Wiener theorem we use to interpret the moment identity (4.3) below as the vanishing of an entire function on a complex hypersurface; the second is the Ehrenpreis–Malgrange-type divisibility lemma that converts this vanishing into a Poisson equation with prescribed right-hand side. Both statements are quoted in the exact form in which they will be applied; we have stated them here, rather than referring back to Section 2.3, in order to keep the regularity hypotheses on ω visible to the reader following Section 4 in isolation.

Theorem 6 (Paley–Wiener–Schwartz, planar form). *Let $\omega \in \mathcal{D}'(\mathbb{R}^2)$ be a compactly supported distribution of order $\leq N$ with $\text{supp } \omega \subset \overline{B(0, R)}$. Then its Fourier transform $\widehat{\omega}$ extends to an entire function on $\mathbb{C}^2$ satisfying the Paley–Wiener bound*

$$|\widehat{\omega}(\xi, \eta)| \leq C (1 + |\xi| + |\eta|)^N e^{R(|\Im \xi| + |\Im \eta|)}, \quad (\xi, \eta) \in \mathbb{C}^2, \quad (4.1)$$

for some $C > 0$. Conversely, any entire function on $\mathbb{C}^2$ obeying such a bound is the Fourier–Laplace transform of a unique compactly supported distribution of order $\leq N$ supported in $\overline{B(0, R)}$. If, in addition, the bound holds with $N = 0$ and $\widehat{\omega}|_{\mathbb{R}^2} \in L^2(\mathbb{R}^2)$, then $\omega \in L^2(\mathbb{R}^2)$.

This is the classical Paley–Wiener–Schwartz theorem; we use the form stated in [26, Theorem 7.3.1], see also [18, §9.3]. The L^2 refinement in the last sentence is a direct consequence of Plancherel applied to the entire function $\widehat{\omega}|_{\mathbb{R}^2}$, combined with uniqueness of the Fourier inversion. In our application, ω is a finite signed Radon measure of compact support, so the Paley–Wiener bound (4.1) holds with $N = 0$ and $C = \|\omega\|_{\text{TV}}$.

Lemma 4.1 (Ehrenpreis–Malgrange-type divisibility along $\xi^2 + \eta^2$). *Let ω be a compactly supported singular-continuous free measure on $\mathbb{R}^2$, in the sense of Section 2.3: a finite signed Radon*

measure of the form

$$\omega = f \, dx \, dy + \sum_{j=1}^N c_j \delta_{p_j},$$

with $f \in L^\infty(\mathbb{R}^2)$ compactly supported, $c_j \in \mathbb{R} \setminus \{0\}$, and $p_j \in \mathbb{R}^2$ distinct. Suppose moreover that the Fourier–Laplace transform $\widehat{\omega}$, an entire function on $\mathbb{C}^2$ of exponential type by Theorem 6, vanishes on the complex characteristic variety of the planar Laplacian,

$$\Sigma := \{(\xi, \eta) \in \mathbb{C}^2 : \xi^2 + \eta^2 = 0\} = \{\eta = i\xi\} \cup \{\eta = -i\xi\}. \quad (4.2)$$

Then $\widehat{\omega}$ is divisible, in the ring of entire functions of exponential type on $\mathbb{C}^2$, by the polynomial $\xi^2 + \eta^2$, with quotient $F \in L^2(\mathbb{R}^2) \cap L^1(\mathbb{R}^2)$ on the real locus. Equivalently, there exists a unique compactly supported $u \in L^2(\mathbb{R}^2)$, continuous off a finite set $P \subset \mathbb{R}^2$, such that

$$\Delta u = \omega \quad \text{in } \mathcal{D}'(\mathbb{R}^2),$$

namely the logarithmic potential

$$u(z) = \frac{1}{2\pi} \int_{\mathbb{R}^2} \log |z - z'| \, d\omega(z').$$

This is exactly Lemma 2.6, restated to emphasize its character as a planar-Laplacian instance of the Ehrenpreis–Malgrange division theorem (entire functions of exponential type can be divided by a polynomial within the same Paley–Wiener class; see [26, Ch. 7]). We do *not* invoke the general Ehrenpreis–Malgrange machinery: a self-contained proof, exploiting the homogeneity decomposition of $\widehat{\omega}$ and positivity of $\xi^2 + \eta^2$ on $\mathbb{R}^2$, is given in Section 2.3. The hypotheses on ω – in particular, the absence of a continuous singular part and the finiteness of the point part – are essential for the conclusion that u is continuous off a finite set; without them the quotient F remains in L^2 , but u need only be quasi-continuous in the logarithmic-potential sense.

4.1. First proof.

First proof under the additional assumption that ∂U is C^2 . The Bargmann transform of the Hermite function h_k is a nonzero scalar multiple of the monomial z^k . Thus, arguing as in [1], the assumption that h_k is an eigenfunction of the localization operator $\mathcal{L}_U$ is equivalent to the identity

$$\int_U z^k e^{\pi w \bar{z}} e^{-\pi |z|^2} dA(z) = \lambda w^k, \quad w \in \mathbb{C}.$$

Both sides are entire functions of w . Since U is bounded and the weight $e^{-\pi |z|^2}$ is integrable, the dominated convergence theorem applied to difference quotients shows that for each multi-index α in w , the partial derivative ∂_w^α commutes with the integral on the left-hand side; the absolute integrability of $|z|^{|\alpha|+k} e^{\pi |w||z|} e^{-\pi |z|^2} \mathbf{1}_U$ for fixed w provides the required local majorant. Iterating k times, we obtain

$$\frac{1}{(\pi)^k} \partial_w^k \left[\int_U z^k e^{\pi w \bar{z}} e^{-\pi |z|^2} dA(z) \right] = \int_U z^k \bar{z}^k e^{\pi w \bar{z}} e^{-\pi |z|^2} dA(z) = \int_U |z|^{2k} e^{\pi w \bar{z}} e^{-\pi |z|^2} dA(z).$$

On the right-hand side, $\partial_w^k(\lambda w^k) = \lambda k!$. Setting $c(\lambda, k) := \lambda k!/\pi^k \in \mathbb{R}$ and using $\mathbf{1}_U$ to extend the integration to $\mathbb{C}$, we obtain

$$\int_{\mathbb{C}} \mathbf{1}_U(z) |z|^{2k} e^{-\pi|z|^2} e^{\pi\bar{z}w} dA(z) = c(\lambda, k), \quad (4.3)$$

for some real constant $c(\lambda, k)$.

Consider now the compactly supported distribution

$$\eta := |z|^{2k} e^{-\pi|z|^2} \mathbf{1}_U(z) - c(\lambda, k) \delta_0, \quad (4.4)$$

which is a finite signed Radon measure with compact support in any closed ball containing $U \cup \{0\}$. Its absolutely continuous part has bounded density $|z|^{2k} e^{-\pi|z|^2} \mathbf{1}_U \in L_c^\infty(\mathbb{R}^2)$, and its pure-point part is the single Dirac mass $-c(\lambda, k)\delta_0$. Hence η is a singular-continuous free measure in the sense of Section 2.3, and we may apply Theorem 6 and Lemma 4.1 to it.

Writing $z = x + iy$ and $w \in \mathbb{C}$, we rewrite the exponential factor in (4.3) as

$$e^{\pi\bar{z}w} = e^{\pi(x-iy)w} = e^{\pi xw} e^{-i\pi yw}.$$

The function class entering the Paley–Wiener step is now unambiguous: $\eta \in \mathcal{E}'(\mathbb{R}^2)$ is a compactly supported distribution of order zero, in fact a finite signed Radon measure. Theorem 6 therefore applies and produces an entire function $\hat{\eta}$ on $\mathbb{C}^2$ obeying the Paley–Wiener bound (4.1) with $N = 0$ and R equal to the radius of any closed disk containing $U \cup \{0\}$; in particular $\hat{\eta}|_{\mathbb{R}^2}$ is bounded by $\|\eta\|_{\text{TV}}$ and extends to $\mathbb{C}^2$ as an entire function of exponential type. With the convention $\hat{g}(\xi, \eta) := \int_{\mathbb{R}^2} g(x, y) e^{-2\pi i(x\xi + y\eta)} dx dy$ used in Section 2.3, a comparison of exponents shows that for $w \in \mathbb{C}$ the kernel $e^{\pi xw} e^{-i\pi yw}$ coincides with $e^{-2\pi i(x\xi + y\eta)}$ on the complex line $(\xi, \eta) = (\frac{iw}{2}, -\frac{w}{2})$, $w \in \mathbb{C}$, whose defining equation is $\eta = i\xi$. Reparametrizing this line in the form $(\xi, \eta) = (w, -iw)$, identity (4.3)—which says that the integral on the left-hand side, viewed as an entire function of $w \in \mathbb{C}$, is identically equal to the constant $c(\lambda, k)$, so that the Fourier–Laplace transform of the modified distribution η vanishes there—becomes

$$\hat{\eta}(w, -iw) = 0, \quad \forall w \in \mathbb{C}. \quad (4.5)$$

Symmetrically, since $c(\lambda, k) \in \mathbb{R}$ and the absolutely continuous density of η is real-valued, complex conjugation of (4.3) after the substitution $w \mapsto \bar{w}$, combined with the same Paley–Wiener identification applied to the conjugate complex line $\eta = -i\xi$, yields

$$\hat{\eta}(w, iw) = 0, \quad \forall w \in \mathbb{C}. \quad (4.6)$$

Identities (4.5) and (4.6) together assert that $\hat{\eta}$ vanishes on the complex characteristic variety $\Sigma = \{\xi^2 + \eta^2 = 0\}$ of the Laplacian, see (4.2).

Applying the divisibility lemma (Lemma 4.1, equivalently Lemma 2.6) to the distribution η , we obtain a unique compactly supported function $u \in L^2(\mathbb{R}^2)$, continuous off a finite set, such that

$$\Delta u = f(|z|) \mathbf{1}_U - c(\lambda, k) \delta_0 \quad (4.7)$$

in the sense of distributions on $\mathbb{R}^2$, where

$$f(|z|) = |z|^{2k} e^{-\pi|z|^2}.$$

Writing u explicitly as a logarithmic potential of the right-hand side of (4.7), we conclude that U satisfies

$$\int_U \log |z - z'| f(|z'|) dA(z') = c(\lambda, k) \log |z|, \quad z \in \mathbb{C} \setminus (U \cup \{0\}). \quad (4.8)$$

Reduction to $0 \notin U$. Suppose first that $0 \in U$, and pick $\rho > 0$ small enough that $\overline{D_\rho(0)} \subset U$. The radial measure $f(|z|) \mathbf{1}_{D_\rho(0)} dA$ generates a logarithmic potential

$$\Phi_\rho(z) := \int_{D_\rho(0)} \log |z - z'| f(|z'|) dA(z'),$$

which, by the planar Newton shell theorem for logarithmic potentials of radial densities (see, e.g., [28, Ch. 2]), is radial; outside $D_\rho(0)$ it equals

$$\Phi_\rho(z) = \left(\int_{D_\rho(0)} f(|z'|) dA(z') \right) \log |z| + \text{const.}$$

Subtracting the closed disk $\overline{D_\rho(0)}$ from U thus preserves identity (4.8) on the complement of the new set, at the cost of replacing $c(\lambda, k)$ by $c(\lambda, k) - M_\rho$, where $M_\rho := \int_{D_\rho(0)} f(|z'|) dA(z') > 0$. The new domain $U' := U \setminus \overline{D_\rho(0)}$ satisfies $0 \notin \overline{U'}$, so we may assume from now on that $0 \notin \overline{U}$ in (4.8), with $c(\lambda, k)$ replaced (if needed) by the real constant $c(\lambda, k) - M_\rho$, which we continue to denote by $c(\lambda, k)$. Furthermore, this constant must be nonzero: comparing the two sides of (4.8) as $|z| \rightarrow \infty$ shows that the left-hand side behaves like $(\int_U f dA) \log |z| + O(|z|^{-1})$, while the right-hand side is $c(\lambda, k) \log |z|$, forcing $c(\lambda, k) = \int_U f(|z'|) dA(z') > 0$.

Define the logarithmic potential

$$\Psi_U(z) := \int_U \log |z - z'| f(|z'|) dA(z').$$

We claim that Ψ_U is *locally radial*: for each $z_0 \in U$, there exists a ball $B(z_0) \subset U$ such that Ψ_U agrees on $B(z_0)$ with a radial function of $|z|$.

To prove this, let $B \supset U$ be a ball centered at 0, and define

$$\Phi(z) := \int_B \log |z - z'| f(|z'|) dA(z').$$

Since both B and the weight $f(|z'|)$ are radial, Φ is radial; in particular, $\nabla \Phi(z)$ is parallel to z for every $z \in \mathbb{R}^2$. Furthermore, on the interior of B ,

$$\Delta \Phi(z) = 2\pi f(|z|).$$

Now Ψ_U satisfies (4.8) on the unbounded component of $\mathbb{C} \setminus (U \cup \{0\})$. Differentiating that identity in z for $z \notin \overline{U} \cup \{0\}$ gives

$$\nabla \Psi_U(z) = c(\lambda, k) \frac{z}{|z|^2}, \quad z \in \mathbb{C} \setminus (\overline{U} \cup \{0\}).$$

The classical jump relations for the single-layer / volume logarithmic potential of an L^∞ density on a C^2 domain (see, e.g., [23]) imply that $\nabla \Psi_U$ extends continuously across ∂U from *both* sides,

and the boundary values agree. Hence on ∂U ,

$$\nabla \Psi_U(z) = c(\lambda, k) \frac{z}{|z|^2}, \quad z \in \partial U,$$

which is, in particular, parallel to z . Therefore the difference

$$v := \Psi_U - \Phi$$

satisfies $\Delta v = 2\pi f(|z|)\mathbf{1}_U - 2\pi f(|z|)\mathbf{1}_B = -2\pi f(|z|)\mathbf{1}_{B \setminus U}$ in the sense of distributions on B , so v is harmonic in U and in $B \setminus \overline{U}$, and on ∂U satisfies

$$\nabla v(x) \text{ is parallel to } x.$$

The next lemma converts this gradient condition into local radially of v .

Lemma 4.2. *Let $V \subset \mathbb{R}^2$ be a domain, and suppose there exists a harmonic function $v \in C^1(\overline{V})$ such that $\nabla v(x)$ is parallel to x for every $x \in \partial V$. Then $\nabla v(x)$ is parallel to x for every $x \in V$. In particular, on every circle $S = \{|x| = r\} \subset \mathbb{R}^2$, the function v is constant on every connected component of $S \cap V$.*

Proof. Consider the angular derivative

$$g(x) := x_1 \partial_2 v(x) - x_2 \partial_1 v(x) = \partial_\theta v(x),$$

which equals the tangential derivative of v along the circle through x centered at 0. A direct computation gives

$$\Delta g = x_1 \Delta(\partial_2 v) - x_2 \Delta(\partial_1 v) + 2(\partial_1 \partial_2 v - \partial_2 \partial_1 v) = 0,$$

since v is harmonic and partial derivatives of harmonic functions are themselves harmonic. By the boundary hypothesis, $\nabla v(x)$ is parallel to x on ∂V , so $g \equiv 0$ on ∂V . Together with $g \in C(\overline{V})$, the maximum principle for harmonic functions yields $g \equiv 0$ in V . Hence $\nabla v(x)$ is parallel to x at every interior point as well, and on any circle $S = \{|x| = r\}$ the tangential derivative $\partial_\theta v$ vanishes on $S \cap V$. Therefore v is constant on each connected component of $S \cap V$. $\square$

Applying Lemma 4.2 on small subdomains $V \subset U$ where v is harmonic and C^1 up to the boundary, we conclude that v is locally radial on U . Since Φ is globally radial, so is $\Psi_U = \Phi + v$, locally on U .

Lemma 4.3. *Let $W \subset \mathbb{R}^2 \setminus \{0\}$ be a connected open set, and let $u \in C(W)$. Suppose that for every $x \in W$ there exist $r_x > 0$ and a scalar function ϕ_x such that*

$$u(y) = \phi_x(|y|) \quad \text{for all } y \in B(x, r_x) \subset W.$$

Then there exists a single scalar function ϕ , defined on the interval

$$I_W := \{|y| : y \in W\},$$

such that

$$u(y) = \phi(|y|) \quad \text{for all } y \in W.$$

Proof. For each $x \in W$, choose a ball $B_x := B(x, r_x) \subset W$ on which u is a radial function of $|y|$, with profile ϕ_x . We first note that whenever $B_x \cap B_{x'} \neq \emptyset$,

$$\phi_x(|y|) = u(y) = \phi_{x'}(|y|) \quad \text{for every } y \in B_x \cap B_{x'}.$$

Since $B_x \cap B_{x'}$ is open and avoids the origin, the map $y \mapsto |y|$ is a submersion on it, so the set of radii it attains contains a nontrivial open interval on which ϕ_x and $\phi_{x'}$ agree.

Fix $x_* \in W$ and define a candidate function ϕ on I_W as follows. Given $y \in W$, connect x_* to y by a polygonal path in W , which exists because W is open and connected. Its image is compact, hence covered by finitely many balls $B_{x_1}, \dots, B_{x_m}$ with successive nonempty overlaps and $x_1 = x_*$, $y \in B_{x_m}$. The previous paragraph implies ϕ_{x_j} and $\phi_{x_{j+1}}$ agree on a nontrivial interval of radii, so by analytic-continuation along the chain we may unambiguously set $\phi(|y|) := \phi_{x_m}(|y|)$. Independence of the chain follows from the same overlap argument: any two chains from x_* to y can be interleaved into a single chain, and successive overlaps force the final value to coincide. Hence ϕ is well-defined on I_W and satisfies $u(y) = \phi(|y|)$ for every $y \in W$. $\square$

We now conclude the proof. Since U is simply connected and bounded, it suffices to rule out the possibility that U fails to be a disk centered at the origin. Suppose therefore, for contradiction, that U is not such a disk. Then ∂U is not a single circle centered at 0, so there exist radii $0 < r_1 < r_2$ such that the annulus

$$A := \{z \in \mathbb{R}^2 : r_1 < |z| < r_2\}$$

meets both U and ∂U in a non-trivial way; in particular, we may pick a point

$$x_0 \in \partial U \cap A.$$

Let

$$\tilde{U}_0 := U \cap B_r(x_0),$$

for $r > 0$ sufficiently small that $\tilde{U}_0$ is connected and $0 \notin \overline{\tilde{U}_0}$. Since Ψ_U is locally radial in U , Lemma 4.3 applied to $W = \tilde{U}_0$ yields a scalar function ψ_U on $I_{\tilde{U}_0}$ such that

$$\Psi_U(x) = \psi_U(|x|), \quad x \in \tilde{U}_0.$$

On the other hand, by (4.8) and the continuity of the gradient of a volume logarithmic potential with bounded density across a C^2 boundary (see, e.g., [23, 28]), the gradient $\nabla \Psi_U$ extends continuously across ∂U , and for $z \in \partial U$ we have

$$\nabla \Psi_U(z) = c \frac{z}{|z|^2}$$

for the same constant $c = c(\lambda, k) \in \mathbb{R}$ as in (4.8). Combining this with $\Psi_U(x) = \psi_U(|x|)$ on $\tilde{U}_0$ and passing to $z \in \partial \tilde{U}_0 \cap \partial U$ from the inside,

$$\psi'_U(|z|) \frac{z}{|z|} = c \frac{z}{|z|^2}, \quad z \in \partial U \cap \overline{B_r(x_0)},$$

hence

$$\psi'_U(r) = \frac{c}{r}$$

for every r in the set of radii attained by points of $\partial U \cap \overline{B_r(x_0)}$. Since ∂U is a C^2 curve, we may, after shrinking r once more, parametrize $\partial U \cap B_r(x_0)$ by a C^2 embedding $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^2$ with $\gamma(0) = x_0$. Because ∂U is not a circle centered at 0, the smooth function $t \mapsto |\gamma(t)|^2$ is non-constant on the connected C^2 arc $\partial U \cap A$. Hence its derivative does not vanish identically on any subinterval, and we may choose $x_0 \in \partial U \cap A$ so that

$$\left. \frac{d}{dt} |\gamma(t)|^2 \right|_{t=0} \neq 0,$$

which is equivalent to $\left. \frac{d}{dt} |\gamma(t)| \right|_{t=0} \neq 0$ since $|\gamma(0)| = |x_0| > 0$. Hence by the open-mapping property for C^1 functions of one variable with nonzero derivative, the image of $|\gamma|$ contains a nontrivial open interval $I \subset (r_1, r_2)$. Therefore

$$\psi'_U(r) = \frac{c}{r} \quad \text{for every } r \in I,$$

whence

$$\psi_U(r) = c \log r + C_0, \quad r \in I,$$

for some real constant C_0 .

Let

$$W_I := \{x \in \tilde{U}_0 : |x| \in I\}.$$

Then W_I is a nonempty open subset of U , and on W_I ,

$$\Psi_U(x) = c \log |x| + C_0.$$

Hence

$$\Delta \Psi_U \equiv 0 \quad \text{in } W_I,$$

since the origin is excluded from $\tilde{U}_0$. On the other hand, since $0 \notin \overline{U}$, the logarithmic potential Ψ_U has distributional Laplacian

$$\Delta \Psi_U = 2\pi f(|z|) \mathbf{1}_U$$

in $\mathbb{C}$, with no contribution at the origin. Hence on the open set $W_I \subset U \setminus \{0\}$ we have

$$\Delta \Psi_U = 2\pi f(|z|) = 2\pi |z|^{2k} e^{-\pi|z|^2}.$$

The right-hand side vanishes only at the origin, which lies outside W_I , while the left-hand side just shown to be 0 on W_I — a contradiction. This rules out the case in which U is not a disk centered at 0, completing the proof. $\square$

Remark 4.1. The structure of the argument above is in the spirit of [3], where logarithmic-potential rigidity statements were derived from Schiffer-style overdetermined boundary conditions, and of the quadrature-domain literature [23, 48, 47, 29], in which the regularity of free boundaries and the matching of an interior potential with the Newtonian potential of a radially symmetric density play a central role. Concretely, identity (4.8) can be read as saying that U is a *generalized quadrature domain* for the radial weight $f(|z|)$, with quadrature data concentrated at the origin. From this perspective, the conclusion that U is a disk is a (sharp) instance of the classical principle that radial quadrature data forces a radial domain. The role of the

C^2 boundary regularity is concentrated in the boundary extension of $\nabla\Psi_U$ and in the open-interval-of-radii argument that ends the proof; both can be relaxed at the cost of additional technicalities, which we will revisit in the second proof.

4.2. Second proof. The first proof established Theorem 2 under the auxiliary hypothesis that ∂U is of class C^2 . The argument we now give does not use any boundary regularity at all; it therefore strengthens Theorem 2 to a form which agrees with the original Abreu–Dörfler statement [1].

Theorem 7 (Strengthened rigidity, no boundary regularity). *Let $U \subset \mathbb{C}$ be an open, bounded, simply connected set, with no hypothesis on the regularity of ∂U beyond what is implicit in “simply connected”. Let h_k ($k \geq 0$) be a Hermite function. If h_k is an eigenfunction of the localization operator $\mathcal{L}_U$, then U is a disk.*

The hypotheses on U in Theorem 7 reduce to: U is open, bounded, connected, and its complement in $\overline{\mathbb{C}}$ is connected (i.e. U is simply connected). In particular, ∂U is allowed to be merely a closed set with $|U| < \infty$; no assumption of positive Lebesgue density, finite perimeter, or differentiability is imposed. The proof below shows that the boundary identity $\nabla u \equiv 0$ on ∂U , which in the first proof was extracted via boundary regularity of the logarithmic potential and jump relations on a C^2 curve, now follows directly from the explicit formula for u as a convolution with $\log|\cdot|$, together with the local integrability of the kernel $z/|z|^2$. Since u vanishes identically on the open exterior $\mathbb{R}^2 \setminus \overline{U}$, continuity of u and of ∇u on $\mathbb{R}^2 \setminus \{0\}$ forces both to vanish on ∂U , regardless of how rough the boundary is. The remainder of the proof proceeds via a holomorphic auxiliary function on U whose boundary values are forced to be real, and the maximum principle on U (which is well-defined even for rough boundaries) then yields the conclusion. We summarize this in Remark 4.2 below; the precise place where the C^2 assumption is bypassed is at the derivation of (4.11).

Proof of Theorem 7 (second proof of Theorem 2). We keep the notation

$$f(|z|) := |z|^{2k} e^{-\pi|z|^2}, \quad c := c(\lambda, k),$$

from the first proof. The argument below is in the spirit of the holomorphic moving-plane/quadrature-domain methods of [3, 23, 29], in that we exploit the existence of a holomorphic function on U with prescribed real boundary values to recover full radial symmetry. Starting again from (4.3) and applying Lemma 2.6, we obtain a compactly supported distributional solution of

$$\Delta u = f(|z|)\mathbf{1}_U - c\delta_0 \quad \text{in } \mathcal{D}'(\mathbb{R}^2). \quad (4.9)$$

Using the planar fundamental solution $(2\pi)^{-1} \log|z|$, we may write

$$u(z) := \frac{1}{2\pi} \int_U \log|z - z'| f(|z'|) dA(z') - \frac{c}{2\pi} \log|z|. \quad (4.10)$$

By (4.8), this function satisfies

$$u \equiv 0 \quad \text{on } \mathbb{R}^2 \setminus (U \cup \{0\}).$$

We first show that $0 \in U$. Since the singular term in (4.9) is $-c\delta_0$, one must have $0 \in \overline{U}$. If $0 \in \partial U$, then there exist exterior points $z_j \rightarrow 0$ with

$$u(z_j) = 0.$$

On the other hand, the first term in (4.10) remains finite as $z \rightarrow 0$, because $f \in L^\infty(U)$ and $\log|z - z'|$ is locally integrable, whereas the second term equals $-\frac{c}{2\pi} \log|z| \rightarrow +\infty$. This contradicts the exterior identity $u(z_j) = 0$. Hence $0 \notin \partial U$, and therefore

$$0 \in U.$$

We next show directly from (4.10) that

$$u \in C^1(\mathbb{R}^2 \setminus \{0\}).$$

Indeed, if we write

$$g(z) := f(|z|)\mathbf{1}_U(z) \in L_c^\infty(\mathbb{R}^2),$$

then

$$u(z) = \frac{1}{2\pi} \int_{\mathbb{R}^2} \log|z - z'| g(z') dA(z') - \frac{c}{2\pi} \log|z|.$$

Since $g \in L_c^\infty(\mathbb{R}^2)$, the only contribution to the integral defining the convolution $(K * g)(z)$ below comes from a fixed bounded set, so we may replace K by its restriction $K\mathbf{1}_{B_R}$ for any sufficiently large ball B_R . The kernel $K(\zeta) = \zeta/|\zeta|^2$ satisfies $|K(\zeta)| = |\zeta|^{-1}$, which is integrable on bounded subsets of $\mathbb{R}^2$; hence $K\mathbf{1}_{B_R} \in L^1(\mathbb{R}^2)$. It is a standard fact that the convolution of an L^1 function with a bounded function is continuous (this follows from the density of $C_c(\mathbb{R}^2)$ in $L^1(\mathbb{R}^2)$ and continuity of translations in L^1), so

$$(K * g)(z) := \int_{\mathbb{R}^2} \frac{z - z'}{|z - z'|^2} g(z') dA(z')$$

is a continuous function of $z \in \mathbb{R}^2$. On the other hand, a direct distributional computation shows

$$\nabla \left(\int_{\mathbb{R}^2} \log|z - z'| g(z') dA(z') \right) = K * g \quad \text{in } \mathcal{D}'(\mathbb{R}^2),$$

so the gradient of the convolution $\log|\cdot| * g$ coincides almost everywhere with the continuous function $K * g$. It follows that $\log|\cdot| * g$ is of class C^1 on $\mathbb{R}^2$, and hence $u \in C^1(\mathbb{R}^2 \setminus \{0\})$ (the only obstruction to global C^1 regularity being the explicit logarithmic singularity at the origin coming from the $-\frac{c}{2\pi} \log|z|$ term).

Now let $z_* \in \partial U$. Since $0 \in U$, the boundary point z_* is bounded away from 0; fix a small ball $B_\rho(z_*) \Subset \mathbb{R}^2 \setminus \{0\}$. On the exterior open set $\mathbb{R}^2 \setminus \overline{U}$ we have $u \equiv 0$, so $\nabla u \equiv 0$ there. As ∂U is contained in the boundary of the exterior, every $z_* \in \partial U$ is a limit of exterior points $z_j \rightarrow z_*$ along which $\nabla u(z_j) = 0$. By the C^1 regularity of u on $B_\rho(z_*)$, the gradient ∇u is continuous at z_* , and we conclude $\nabla u(z_*) = 0$. In complex notation $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$, this yields

$$\partial_z u = 0 \quad \text{on } \partial U. \tag{4.11}$$

We emphasize that *no* boundary regularity of ∂U has been used at this point, in contrast with the first proof: the vanishing of ∇u on ∂U is forced solely by exterior continuity, which in turn is supplied by the explicit formula (4.10).

Define

$$\Phi(t) := \int_0^t s^k e^{-\pi s} ds \quad \text{and} \quad \Psi(z) := \frac{\Phi(|z|^2)}{z}, \quad z \neq 0.$$

A direct computation, using $|z|^2 = z\bar{z}$ and $\partial_{\bar{z}}(z\bar{z}) = z$, gives, for $z \neq 0$,

$$\partial_{\bar{z}}\Psi(z) = \frac{\Phi'(|z|^2) \partial_{\bar{z}}(|z|^2)}{z} = \frac{|z|^{2k} e^{-\pi|z|^2} \cdot z}{z} = |z|^{2k} e^{-\pi|z|^2} = f(|z|).$$

At the origin, the apparent singularity $1/z$ is integrable against the vanishing factor $\Phi(|z|^2)$: since $k \geq 0$ and $s^k e^{-\pi s} \leq s^k$ for $s \geq 0$, we have $\Phi(t) = \int_0^t s^k e^{-\pi s} ds \leq \int_0^t s^k ds = t^{k+1}/(k+1) = O(t^{k+1})$ as $t \rightarrow 0$. Hence $\Psi(z) = O(|z|^{2k+1})$ near $z = 0$, so $\Psi \in L_{\text{loc}}^\infty$ (in fact continuous, with $\Psi(0) = 0$). Consequently, the identity $\partial_{\bar{z}}\Psi = f(|z|)$ persists in the distributional sense on all of U ; no extra δ_0 contribution arises at the origin.

Recall also the standard identity

$$\partial_{\bar{z}}\left(\frac{1}{z}\right) = \pi\delta_0 \quad \text{in } \mathcal{D}'(\mathbb{R}^2),$$

which is equivalent to the fact that $\frac{1}{\pi z}$ is a fundamental solution of $\partial_{\bar{z}}$; equivalently, $\Delta \log |z| = 4\partial_z \partial_{\bar{z}} \log |z| = 2\pi\delta_0$ combined with $\partial_z \log |z| = \frac{1}{2z}$. Now set

$$F(z) := 4\partial_z u(z) - \Psi(z) + \frac{c}{\pi z}, \quad z \in U \setminus \{0\}.$$

We first check that F defines a distribution on all of U (a priori it is given only on $U \setminus \{0\}$). From (4.10) we have, near $z = 0$,

$$4\partial_z u(z) = \frac{2}{\pi} \partial_z \left(\int_{\mathbb{R}^2} \log |z - z'| g(z') dA(z') \right) - \frac{c}{\pi z},$$

where the first term is locally bounded near $z = 0$ (the Cauchy-type integral $\int (z - z')^{-1} g(z') dA(z')$ of a bounded compactly supported function is locally integrable, by the same L_{loc}^1 -kernel argument as above). Adding $-\Psi$ (continuous with $\Psi(0) = 0$) and $c/(\pi z)$ cancels the explicit $1/z$ -singularity, so F is locally bounded near the origin and hence defines a distribution on U .

Using $\Delta = 4\partial_z \partial_{\bar{z}}$, the relation $\partial_{\bar{z}}(1/z) = \pi\delta_0$, the computation of $\partial_{\bar{z}}\Psi$ above, and (4.9), we compute, in $\mathcal{D}'(U)$,

$$\partial_{\bar{z}}F = 4\partial_z \partial_{\bar{z}}u - \partial_{\bar{z}}\Psi + \frac{c}{\pi} \partial_{\bar{z}}\left(\frac{1}{z}\right) = \Delta u - f(|z|) + c\delta_0 = (f(|z|) - c\delta_0) - f(|z|) + c\delta_0 = 0.$$

The singular term $-c\delta_0$ in (4.9), arriving through $4\partial_z \partial_{\bar{z}}u = \Delta u$, is thus exactly cancelled by $(c/\pi)\partial_{\bar{z}}(1/z) = c\delta_0$. Hence F is a distributional solution of the Cauchy–Riemann equation $\partial_{\bar{z}}F = 0$ on U , and by Weyl’s lemma for $\partial_{\bar{z}}$ (equivalently, hypoellipticity of the Cauchy–Riemann operator) it is represented by a genuine holomorphic function on U .

Define

$$W(z) := \pi z F(z).$$

Then W is holomorphic on U and $W(0) = 0$. Since $\partial_z u$ is continuous up to ∂U , so is W , and by (4.11),

$$W|_{\partial U} = \pi z \left(4\partial_z u - \frac{\Phi(|z|^2)}{z} + \frac{c}{\pi z} \right) \Big|_{\partial U} = c - \pi\Phi(|z|^2).$$

In particular, $W|_{\partial U}$ is real-valued. The imaginary part of W is therefore harmonic in U , continuous up to ∂U , and vanishes on ∂U . By the maximum principle,

$$\Im W \equiv 0 \quad \text{in } U.$$

Since a holomorphic function with zero imaginary part is constant, there exists $\kappa \in \mathbb{R}$ such that

$$W \equiv \kappa \quad \text{in } U.$$

Restricting to the boundary, and using that W is continuous up to ∂U with boundary value $c - \pi\Phi(|z|^2)$, we obtain

$$c - \pi\Phi(|z|^2) = \kappa \quad \text{for all } z \in \partial U.$$

The function $t \mapsto \Phi(t) = \int_0^t s^k e^{-\pi s} ds$ has derivative $\Phi'(t) = t^k e^{-\pi t} > 0$ for $t > 0$, hence is strictly increasing on $[0, \infty)$. The identity above therefore forces $|z|^2$ to take a single value on ∂U , i.e. $|z|$ is constant on ∂U . Consequently ∂U is contained in a circle centered at the origin. Let $R > 0$ be such that $\partial U \subset \{|z| = R\}$. Since U is open and $U \cap \partial U = \emptyset$, we have $U \subset \{|z| < R\} \cup \{|z| > R\}$; since U is connected and the two pieces are disjoint open sets, U must lie entirely in one of them, and boundedness of U forces $U \subset \{|z| < R\}$. If this inclusion were strict, pick $z_1 \in \{|z| < R\} \setminus U$ and $z_0 \in U$. The straight segment from z_0 to z_1 is contained in the convex set $\{|z| < R\}$, and as it joins a point of U to a point of its complement it must cross ∂U at some z_* ; but then $z_* \in \partial U \cap \{|z| < R\}$, contradicting $\partial U \subset \{|z| = R\}$. Therefore

$$U = \{|z| < R\}$$

that is, U is a disk. □

Remark 4.2. The two proofs use the boundary assumptions in rather different ways, and it is instructive to keep careful track of the regularity bookkeeping.

In the first proof, once the logarithmic potential Ψ_U is written explicitly, no elliptic boundary regularity is needed to conclude that $\Psi_U \in C^1(\mathbb{R}^2 \setminus \{0\})$: this follows directly from convolution with the locally integrable kernel $\nabla \log |z|$, exactly as in the L^1_{loc} -translation argument used here for (4.10). Thus the local radiality argument itself only uses continuity of the gradient up to the boundary. The genuine extra input appears only in the final geometric step, where one needs that, near a suitable point $x_0 \in \partial U$, the set $\partial U \cap B_r(x_0)$ contains a nontrivial boundary arc, so that the radii attained there form an interval. Accordingly, the first proof does not really need C^2 regularity; it works under any hypothesis guaranteeing such a local arc structure, for instance when U is a Jordan domain. In particular, any Lipschitz, C^1 , or C^2 boundary is more than sufficient.

By contrast, the second proof does not use any boundary regularity at all. The explicit formula (4.10) already yields $u \in C^1(\mathbb{R}^2 \setminus \{0\})$, and since $u \equiv 0$ on the exterior open set $\mathbb{R}^2 \setminus \overline{U}$, continuity implies that both u and ∇u vanish on ∂U – precisely the boundary information packaged in (4.11). This is exactly the input needed to construct the holomorphic function

$W = \pi z F$ and conclude that $|z|$ is constant on ∂U . Hence the second proof recovers the theorem under the original assumptions of Abreu and Dörfler [1], namely for arbitrary bounded simply connected domains, with no boundary smoothness hypothesis – this is the content of Theorem 7. The mechanism is also conceptually closer to the classical quadrature-domain rigidity results of [3, 23, 29], where the existence of a holomorphic function on U with real boundary values forces the boundary to lie on a level set of a real-analytic function, here the elementary radial function $z \mapsto |z|^2$.

To make the comparison explicit: the only step at which the first proof genuinely uses $\partial U \in C^2$ is in the open-mapping argument that extracts a nontrivial open interval of radii from a boundary parametrization $\gamma : (-\varepsilon, \varepsilon) \rightarrow \partial U$. In the second proof, this geometric step is replaced by the harmonic-function-with-real-boundary-values mechanism, which is purely topological (continuity of $W = \pi z F$ up to ∂U plus the maximum principle on the open set U) and is therefore insensitive to the regularity of ∂U . It is in this precise sense that Theorem 7 drops the C^2 hypothesis present in Theorem 2 while delivering the same conclusion.

5. PROOF OF THEOREM 3

We prove that the quantitative stability estimate in Corollary 1.4 of [20], which builds on the Faber–Krahn rigidity result of [40] characterizing the disk as the unique optimizer of the first localization eigenvalue (see also the foundational work of [15] on the spectrum of the Daubechies localization operator on the disk), is sharp. The proof proceeds by constructing a family of domains which are perturbations of the disk, whose first localization eigenvalue has the same first-order behaviour as the disk, but whose distance to the class of disks is of order ε .

Before turning to the construction, we collect four supporting lemmas. They furnish, respectively, (i) the identification of the constructed eigenvalue with the *principal* eigenvalue of $\mathcal{L}_{U_\varepsilon}$ via an analytic-perturbation argument from the disk, (ii) the order- ε^2 effect of the area-normalizing dilation on the principal eigenvalue, (iii) the Reynolds transport / Hadamard shape-derivative formula in the regularity used here, and (iv) the explicit symmetric-difference lower bound which witnesses that second-order harmonics cannot be eliminated by translations of the disk.

5.1. Spectral, dilational and shape-derivative preliminaries. Throughout this subsection we let $D_1 \subset \mathbb{C}$ denote the closed unit disk centered at the origin. Recall that the localization operator $\mathcal{L}_U$ on $\mathcal{F}^2(\mathbb{C})$, with $U \subset \mathbb{C}$ a bounded measurable set, is the self-adjoint compact Toeplitz-type operator

$$(\mathcal{L}_U F)(z) := \int_U F(\zeta) e^{\pi z \bar{\zeta}} e^{-\pi |\zeta|^2} dA(\zeta),$$

whose eigenvalues form a non-increasing sequence $\lambda_1(U) \geq \lambda_2(U) \geq \dots \rightarrow 0$; its principal (largest) eigenvalue is $\lambda_1(U)$.

For the unit disk $U = D_1$, the operator $\mathcal{L}_{D_1}$ is diagonalized by the orthonormal monomial basis $\{e_k(z) = (\pi^k/k!)^{1/2} z^k\}_{k \geq 0}$ of $\mathcal{F}^2(\mathbb{C})$ (cf. [15]), with eigenvalues

$$\mu_k := \frac{\gamma(k+1, \pi)}{k!} = 1 - e^{-\pi} \sum_{j=0}^k \frac{\pi^j}{j!}, \quad k \geq 0, \quad (5.1)$$

where $\gamma(s, x) = \int_0^x t^{s-1} e^{-t} dt$ is the lower incomplete gamma function. The sequence $(\mu_k)_{k \geq 0}$ is strictly decreasing: $\mu_k - \mu_{k+1} = e^{-\pi} \pi^{k+1} / (k+1)! > 0$. In particular,

$$\mu_0 = 1 - e^{-\pi} = \lambda_0, \quad \mu_1 = 1 - (1 + \pi)e^{-\pi}, \quad \mu_0 - \mu_1 = \pi e^{-\pi} > 0, \quad (5.2)$$

so the disk has a strict spectral gap above its principal eigenvalue μ_0 .

Lemma 5.1 (Spectral continuity and principal-eigenvalue identification). *Let $(U_\varepsilon)_{|\varepsilon| < \varepsilon_0}$ be a family of bounded measurable sets with $U_0 = D_1$ and $|U_\varepsilon \Delta D_1| \rightarrow 0$ as $\varepsilon \rightarrow 0$. Then for every $k \geq 1$,*

$$\lambda_k(U_\varepsilon) \xrightarrow{\varepsilon \rightarrow 0} \mu_{k-1}, \quad (5.3)$$

with μ_{k-1} given by (5.1). Hence there exists $\varepsilon_* > 0$ such that, for all $|\varepsilon| < \varepsilon_*$,

$$\lambda_2(U_\varepsilon) < \frac{\mu_0 + \mu_1}{2} < \lambda_1(U_\varepsilon), \quad (5.4)$$

and the principal eigenvalue is the unique eigenvalue of $\mathcal{L}_{U_\varepsilon}$ lying in $[\mu_0 - \frac{\pi e^{-\pi}}{4}, \mu_0 + \frac{\pi e^{-\pi}}{4}]$.

Consequently, for the family U_ε of Theorem 1 applied with $\lambda = \lambda_0 = \mu_0$ and $P_\varepsilon(w) = 1 + \varepsilon w^2$, the eigenvalue λ_0 produced by Theorem 1 is the principal eigenvalue of $\mathcal{L}_{U_\varepsilon}$, and P_ε is a corresponding first eigenfunction.

Proof. For any bounded measurable $U \subset \mathbb{C}$, the operator $\mathcal{L}_U$ is positive semi-definite, compact, and depends Lipschitz-continuously on $\mathbf{1}_U$ in operator norm: for any $F \in \mathcal{F}^2(\mathbb{C})$,

$$\langle (\mathcal{L}_{U_\varepsilon} - \mathcal{L}_{D_1})F, F \rangle = \int_{\mathbb{C}} (\mathbf{1}_{U_\varepsilon}(\zeta) - \mathbf{1}_{D_1}(\zeta)) |F(\zeta)|^2 e^{-\pi|\zeta|^2} dA(\zeta),$$

and by the standard Fock-space pointwise bound $|F(\zeta)|^2 e^{-\pi|\zeta|^2} \leq \|F\|_{\mathcal{F}^2}^2$ (cf. Lemma 2.1) one obtains

$$\|\mathcal{L}_{U_\varepsilon} - \mathcal{L}_{D_1}\|_{\text{op}} \leq |U_\varepsilon \Delta D_1|.$$

By Weyl's inequality (the min-max characterization of eigenvalues of compact self-adjoint operators),

$$|\lambda_k(U_\varepsilon) - \lambda_k(D_1)| \leq \|\mathcal{L}_{U_\varepsilon} - \mathcal{L}_{D_1}\|_{\text{op}} \leq |U_\varepsilon \Delta D_1| \xrightarrow{\varepsilon \rightarrow 0} 0,$$

which proves (5.3). The gap inequality (5.4) then follows by combining this convergence with the strict gap $\mu_0 - \mu_1 = \pi e^{-\pi} > 0$ of (5.2): choose ε_* so small that $|\lambda_1(U_\varepsilon) - \mu_0| < \pi e^{-\pi}/4$ and $|\lambda_2(U_\varepsilon) - \mu_1| < \pi e^{-\pi}/4$ for $|\varepsilon| < \varepsilon_*$; then any eigenvalue λ of $\mathcal{L}_{U_\varepsilon}$ lying in $[\mu_0 - \pi e^{-\pi}/4, \mu_0 + \pi e^{-\pi}/4]$ is greater than $\mu_1 + \pi e^{-\pi}/4 \geq \lambda_2(U_\varepsilon)$, hence equal to $\lambda_1(U_\varepsilon)$ by the ordering of the spectrum.

The same conclusion can be obtained, in a more structural form, from analytic perturbation theory in the sense of Kato [30]: by Theorem 1 the family U_ε depends in a C^1 fashion on ε (in fact analytically, since the Banach-space implicit-function theorem applied to a real-analytic moment map produces a real-analytic branch), so $\varepsilon \mapsto \mathcal{L}_{U_\varepsilon}$ is a real-analytic family of compact self-adjoint operators in $\mathcal{F}^2(\mathbb{C})$. The principal eigenvalue μ_0 of $\mathcal{L}_{D_1}$ is simple (its eigenspace is the one-dimensional space $\mathbb{C} \cdot e_0$) and isolated by (5.2). The Kato selection theorem ([30, Ch. VII, Thm. 1.8]) thus produces a real-analytic branch $\varepsilon \mapsto (\tilde{\lambda}(\varepsilon), \tilde{F}(\varepsilon))$ of eigenpairs with $\tilde{\lambda}(0) = \mu_0$, $\tilde{F}(0) = e_0$, and $\tilde{\lambda}(\varepsilon) = \lambda_1(U_\varepsilon)$ for all sufficiently small ε . The eigenpair $(\lambda_0, P_\varepsilon)$

constructed in Theorem 1 satisfies $P_0 \equiv 1 = e_0$ and depends continuously on ε ; by uniqueness of the analytic branch within the gap interval, it agrees with the Kato branch. In particular, $\lambda_0 = \tilde{\lambda}(\varepsilon) = \lambda_1(U_\varepsilon)$ for ε small. $\square$

Remark 5.1. The numbers μ_k of (5.1) are the classical Daubechies disk eigenvalues; the explicit values $\mu_0 = 1 - e^{-\pi}$ and $\mu_1 = 1 - (1 + \pi)e^{-\pi}$ appear ubiquitously in the time–frequency literature.

Lemma 5.2 (Effect of the area-normalizing dilation). *For any bounded measurable set $U \subset \mathbb{C}$ with $|U| > 0$ and any $t > 0$, the eigenvalues of $\mathcal{L}_{tU}$ are real-analytic functions of t on $(0, \infty)$. In the special case $U = D_1$, the principal eigenvalue is given by the explicit closed form*

$$\lambda_1(tD_1) = 1 - e^{-\pi t^2}, \quad t > 0, \quad (5.5)$$

attained on the constant eigenfunction $F \equiv 1$. In particular, for $r_\varepsilon = 1 + O(\varepsilon^2)$,

$$\lambda_1(r_\varepsilon D_1) = 1 - e^{-\pi r_\varepsilon^2} = \lambda_0 + 2\pi e^{-\pi}(r_\varepsilon - 1) + O((r_\varepsilon - 1)^2) = \lambda_0 + O(\varepsilon^2). \quad (5.6)$$

For the family U_ε of Theorem 1 applied with $P_\varepsilon(w) = 1 + \varepsilon w^2$, one has $|U_\varepsilon| = \pi + O(\varepsilon^2)$ (see Step 3 of the proof of Theorem 3 below), and the area-normalizing dilation factor $r_\varepsilon = \sqrt{\pi/|U_\varepsilon|} = 1 + O(\varepsilon^2)$ satisfies

$$|\lambda_1(r_\varepsilon U_\varepsilon) - \lambda_1(U_\varepsilon)| = O(\varepsilon^2). \quad (5.7)$$

Proof. The change of variable $\zeta = t\zeta'$ in $(\mathcal{L}_{tU}F)(z) = \int_{tU} F(\zeta) e^{\pi z \bar{\zeta}} e^{-\pi|\zeta|^2} dA(\zeta)$ gives

$$(\mathcal{L}_{tU}F)(z) = t^2 \int_U F(t\zeta') e^{\pi z t \bar{\zeta}'} e^{-\pi t^2 |\zeta'|^2} dA(\zeta'),$$

which depends real-analytically on t ; hence $\mathcal{L}_{tU}$ is a real-analytic family of compact self-adjoint operators (in fact in trace class), and by Kato [30, Ch. VII] its eigenvalues vary analytically in t .

For $U = D_1$, the constant function $F \equiv 1$ satisfies, by direct integration, $(\mathcal{L}_{tD_1} 1)(0) = \int_{tD_1} e^{-\pi|\zeta|^2} dA(\zeta) = 1 - e^{-\pi t^2}$. Equality of $(\mathcal{L}_{tD_1} 1)(z)$ with $(1 - e^{-\pi t^2}) \cdot 1$ for all $z \in \mathbb{C}$ follows from the fact that $F \equiv 1 = e_0$ is the principal eigenfunction of the rotationally symmetric operator $\mathcal{L}_{tD_1}$. This gives (5.5); Taylor expansion at $t = 1$ yields (5.6).

For (5.7), the dilation $F \mapsto F(r_\varepsilon \cdot)$ intertwines $\mathcal{L}_{r_\varepsilon U_\varepsilon}$ with an analytic perturbation of $\mathcal{L}_{U_\varepsilon}$ of operator-norm size $O(|r_\varepsilon - 1|) = O(\varepsilon^2)$; Weyl's inequality then yields (5.7). $\square$

Remark 5.2. For the proof of Theorem 3 only (5.6) is strictly required, since one combines it with the exact identity $\lambda_1(U_\varepsilon) = \lambda_0$ furnished by Theorem 1 and Lemma 5.1. The full Lemma 5.2 is recorded for completeness, as it isolates the (otherwise implicit) claim that area normalization perturbs the eigenvalue only at second order.

Lemma 5.3 (Reynolds transport / Hadamard shape derivative for normal-graph perturbations of the disk). *Let $(\Omega_\varepsilon)_{|\varepsilon| < \varepsilon_0}$ be a family of bounded C^1 domains in $\mathbb{C}$, with Ω_0 of class C^1 , outward unit normal ν , and arc-length measure $d\mathcal{H}^1$, and represented as the image of a one-parameter family of C^1 -diffeomorphisms V_ε of $\mathbb{C}$, $V_0 = \text{id}$, with deformation field*

$$W(z) := \partial_\varepsilon V_\varepsilon(z) \Big|_{\varepsilon=0} \in C^1(\overline{\Omega_0}; \mathbb{C}),$$

and normal velocity $V_\nu(z) := W(z) \cdot \nu(z) \in C^1(\partial\Omega_0)$. Then for any H admitting a C^1 extension to a neighborhood of $\overline{\Omega_0}$,

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_{\Omega_\varepsilon} H(z) dA(z) = \int_{\partial\Omega_0} V_\nu(z) H(z) d\mathcal{H}^1(z). \quad (5.8)$$

In the special case of a normal-graph perturbation of the unit disk, $\partial\Omega_\varepsilon = \{(1+\varphi_\varepsilon(y))y : y \in \mathbb{S}^1\}$ with $\varepsilon \mapsto \varphi_\varepsilon$ of class C^1 into $C^1(\mathbb{S}^1; \mathbb{R})$ and $\varphi_0 \equiv 0$, the boundary deformation field equals $W(y) = \dot{\varphi}_0(y)y$, the normal velocity is $V_\nu(y) = \dot{\varphi}_0(y)$, and (5.8) specializes to

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_{\Omega_\varepsilon} H(z) dA(z) = \int_{\mathbb{S}^1} \dot{\varphi}_0(y) H(y) d\mathcal{H}^1(y). \quad (5.9)$$

Applied to $H(z) = |F_0(z)|^2 e^{-\pi|z|^2}$ where $F_0 \equiv 1$ is the disk's principal eigenfunction, (5.9) together with the Hadamard shape-derivative formula for the principal eigenvalue of a self-adjoint Toeplitz-type operator (cf. Step 3 of the proof of the local-maximizer theorem in Section 6, and [24, Thm. 5.7.1] or [49] for the general formalism) yields the first variation of the principal eigenvalue along the family U_ε ,

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \lambda_1(U_\varepsilon) = \int_{\partial D_1} |F_0(y)|^2 e^{-\pi|y|^2} V_\nu(y) d\mathcal{H}^1(y) = e^{-\pi} \int_{\mathbb{S}^1} \dot{\varphi}_0(y) d\mathcal{H}^1(y). \quad (5.10)$$

Proof. The Reynolds transport formula (5.8) for a C^1 deformation of a bounded C^1 domain is classical; see, for example, [24, §5.9] or [49, Ch. 2]. We sketch the polar-coordinate derivation in the case of a normal-graph perturbation of the unit disk, which is the form actually used here. In polar coordinates $z = re^{it}$,

$$\int_{\Omega_\varepsilon} H(z) dA(z) = \int_0^{2\pi} \int_0^{1+\varphi_\varepsilon(e^{it})} H(re^{it}) r dr dt.$$

By the C^1 dependence of $\varepsilon \mapsto \varphi_\varepsilon$ into $C^1(\mathbb{S}^1)$ and the fundamental theorem of calculus,

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_{\Omega_\varepsilon} H dA = \int_0^{2\pi} \dot{\varphi}_0(e^{it}) H(e^{it})(r) \Big|_{r=1} dt = \int_{\mathbb{S}^1} \dot{\varphi}_0(y) H(y) d\mathcal{H}^1(y),$$

which is (5.9).

For the eigenvalue identity (5.10), the principal eigenfunction F_ε of $\mathcal{L}_{U_\varepsilon}$ depends real-analytically on ε by Lemma 5.1 and the Kato selection theorem; we may take it $L^2(\mathbb{C}, e^{-\pi|z|^2} dA)$ -normalized. Differentiating $\lambda_1(U_\varepsilon) \|F_\varepsilon\|^2 = \int_{U_\varepsilon} |F_\varepsilon|^2 e^{-\pi|z|^2} dA$ in ε at $\varepsilon = 0$, using $\partial_\varepsilon \|F_\varepsilon\|^2|_{\varepsilon=0} = 0$ and the self-adjointness identity $\langle (\mathcal{L}_{D_1} - \mu_0 \text{Id}) F_0, \partial_\varepsilon F_\varepsilon|_0 \rangle = 0$, the only surviving contribution is the boundary term, namely

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \lambda_1(U_\varepsilon) = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_{U_\varepsilon} |F_0|^2 e^{-\pi|z|^2} dA(z),$$

to which (5.8) applies with $H = |F_0|^2 e^{-\pi|z|^2}$. Since $F_0 \equiv 1$ and $|y| = 1$ on $\mathbb{S}^1$, this gives (5.10). The corresponding parallel computation in Step 3 of the proof of the local-maximizer theorem (Section 6) spells out the divergence-theorem manipulation in greater detail. $\square$

Lemma 5.4 (Symmetric-difference lower bound: second-order harmonics cannot be eliminated by translations). *Let $\dot{\varphi}_0 \in C^1(\mathbb{S}^1; \mathbb{R})$ be of pure second-harmonic form,*

$$\dot{\varphi}_0(e^{it}) = a \cos(2t) + b \sin(2t), \quad (a, b) \in \mathbb{R}^2 \setminus \{(0, 0)\}, \quad (5.11)$$

and fix $K > 0$. Then there exists a constant $\delta_K > 0$, depending only on (a, b) and K , with the explicit lower bound

$$\delta_K \geq \frac{\pi(a^2 + b^2)}{4(K+1) + 2\sqrt{a^2 + b^2}}, \quad (5.12)$$

such that, for every $\theta \in [0, 2\pi)$ and every $\eta \in [-K, K]$,

$$\int_{\mathbb{S}^1} |\dot{\varphi}_0(y) - \eta \Re(e^{-i\theta} y)| d\mathcal{H}^1(y) \geq \delta_K. \quad (5.13)$$

Consequently, for the area-normalized family $\tilde{U}_\varepsilon$ of Theorem 1 with second-harmonic $\dot{\varphi}_0$ of the form (5.11) and $(a, b) \neq (0, 0)$, there exist constants $c = c(a, b, K) > 0$ and $c' = c'(a, b) > 0$ and a threshold $\varepsilon_1 > 0$ such that, for all $0 < |\varepsilon| < \varepsilon_1$,

$$|\tilde{U}_\varepsilon \Delta (D_1 + a)| \geq c|\varepsilon| \quad \text{whenever } |a| \leq K|\varepsilon|, \quad (5.14)$$

and, combining with the comparison for distant disks,

$$\inf_{\substack{D \subset \mathbb{C} \\ |D| = \pi}} |\tilde{U}_\varepsilon \Delta D| \geq c'|\varepsilon|. \quad (5.15)$$

Proof. Define $g^{\theta, \eta}(t) := a \cos 2t + b \sin 2t - \eta \cos(t - \theta)$. Then $g^{\theta, \eta}$ lies in the four-dimensional subspace $V := \text{span}\{\cos t, \sin t, \cos 2t, \sin 2t\} \subset L^1(\mathbb{S}^1)$. Expanding in the orthogonal basis $\{\cos t, \sin t, \cos 2t, \sin 2t\}$ of $L^2(\mathbb{S}^1)$ (each of squared L^2 -norm π),

$$g^{\theta, \eta}(t) = -\eta \cos \theta \cdot \cos t - \eta \sin \theta \cdot \sin t + a \cos 2t + b \sin 2t,$$

hence by Parseval

$$\|g^{\theta, \eta}\|_{L^2(\mathbb{S}^1)}^2 = \pi(\eta^2 + a^2 + b^2) \geq \pi(a^2 + b^2),$$

the second-harmonic coefficients (a, b) being independent of (θ, η) . The crucial structural feature is that the orthogonal projection of $g^{\theta, \eta}$ onto $\text{span}\{\cos 2t, \sin 2t\}$ equals $a \cos 2t + b \sin 2t$ for every (θ, η) ; translations of the disk shift only the first-harmonic coefficients of the boundary perturbation.

For the L^1 lower bound, recall the elementary inequality $\|g\|_{L^1(\mathbb{S}^1)} \geq \|g\|_{L^2(\mathbb{S}^1)}^2 / \|g\|_{L^\infty(\mathbb{S}^1)}$, valid for any $g \in L^\infty(\mathbb{S}^1) \setminus \{0\}$. For $g = g^{\theta, \eta}$,

$$\|g^{\theta, \eta}\|_{L^\infty(\mathbb{S}^1)} \leq |a| + |b| + |\eta| \leq \sqrt{2(a^2 + b^2)} + K \leq 2\sqrt{a^2 + b^2} + 2(K+1).$$

Hence

$$\|g^{\theta, \eta}\|_{L^1(\mathbb{S}^1)} \geq \frac{\pi(a^2 + b^2)}{4(K+1) + 2\sqrt{a^2 + b^2}},$$

which is (5.12), and (5.13) follows.

For (5.14), recall that for a disk $D = D_1 + a$ of area π with $a = \eta e^{i\theta}$ and $|\eta| \leq K$, the translated set $\tilde{U}_\varepsilon - a$ is a normal graph over $\mathbb{S}^1$ with radial perturbation $\psi_\varepsilon^{\theta, \eta}(y) = \varepsilon g^{\theta, \eta}(y) + O(\varepsilon^2)$ in $C^1(\mathbb{S}^1)$, uniformly for $|\eta| \leq K$ (see Step 5 of the proof of Theorem 3 below for the detailed expansion).

The polar-coordinate representation of the symmetric difference yields

$$|\tilde{U}_\varepsilon \triangle D| = \frac{1}{2} \int_{\mathbb{S}^1} |2\varepsilon g^{\theta, \eta}(y) + O(\varepsilon^2)| d\mathcal{H}^1(y) \geq |\varepsilon| \delta_K - O(\varepsilon^2),$$

which is at least $\frac{1}{2}|\varepsilon| \delta_K$ for $|\varepsilon| < \varepsilon_1$ sufficiently small. This proves (5.14) with $c = \frac{1}{2} \delta_K$.

For (5.15), we use the explicit symmetric-difference formula for two unit disks: for $a \in \mathbb{C}$ with $|a| \leq 2$,

$$|D_1 \triangle (D_1 + a)| = 2\pi - 4 \arccos(|a|/2) + 2|a| \sqrt{1 - |a|^2/4}, \quad (5.16)$$

which satisfies $|D_1 \triangle (D_1 + a)| \geq \sqrt{3}|a|$ for $|a| \leq 1$; hence $|D_1 \triangle (D_1 + a)| \geq c_0 \min(|a|, 1)$ with absolute constant $c_0 = \sqrt{3}$. Choose K so large that $c_0 K \geq 1 + C$, where C is the implicit constant in $|\tilde{U}_\varepsilon \triangle D_1| \leq C|\varepsilon|$ of Step 4 of the proof of Theorem 3. Then for $|a| \geq K|\varepsilon|$,

$$|\tilde{U}_\varepsilon \triangle D| \geq |D_1 \triangle D| - |\tilde{U}_\varepsilon \triangle D_1| \geq c_0 K |\varepsilon| - C |\varepsilon| \geq |\varepsilon|.$$

Combining with (5.14) for $|a| \leq K|\varepsilon|$, (5.15) follows with $c' = \min(\frac{1}{2}\delta_K, 1)$ and the explicit δ_K of (5.12). $\square$

Remark 5.3 (Translations cannot eliminate the second harmonic). The structural fact made quantitative by Lemma 5.4 is that, in the polar-graph representation of a small perturbation of D_1 , translations of the disk contribute *only* to the first Fourier harmonic $\cos t, \sin t$ of the radial perturbation, and have zero second-harmonic component. Hence any non-trivial second-harmonic mode $(a, b) \neq (0, 0)$ in $\dot{\varphi}_0$ is *intrinsic to the shape*, and provides a lower bound on the symmetric-difference distance to the entire family of disks of the same area, of order proportional to $|\varepsilon| \sqrt{a^2 + b^2}$ (with K -dependence given in (5.12)).

5.2. Proof of Theorem 3.

Proof of Theorem 3. Fix

$$\lambda_0 := 1 - e^{-\pi},$$

which is the eigenvalue of the localization operator of the unit disk D_1 associated with the ground state h_0 . For ε small, define

$$P_\varepsilon(w) := 1 + \varepsilon w^2.$$

By Theorem 1, applied with $\lambda = \lambda_0$ and $P_\varepsilon(w) = 1 + \varepsilon w^2$, for all sufficiently small $|\varepsilon|$ there exists a domain U_ε such that

$$\int_{U_\varepsilon} P_\varepsilon(z) e^{\pi \bar{z} w} e^{-\pi |z|^2} dA(z) = \lambda_0 P_\varepsilon(w), \quad w \in \mathbb{C}. \quad (5.17)$$

Moreover, by construction, U_ε is a C^2 perturbation of the unit disk. Hence, after possibly shrinking the range of admissible ε , its boundary can be written as

$$\partial U_\varepsilon = \{(1 + \varphi_\varepsilon(y))y : y \in \mathbb{S}^1\}, \quad (5.18)$$

where

$$\varphi_\varepsilon \in C^2(\mathbb{S}^1), \quad \|\varphi_\varepsilon\|_{C^2(\mathbb{S}^1)} \lesssim |\varepsilon|.$$

The map

$$\varepsilon \mapsto \varphi_\varepsilon$$

may in fact be chosen C^1 as a map from a neighborhood of 0 into $C^1(\mathbb{S}^1)$. To see this, we briefly retrace the proof of Theorem 1. The construction is the disk moment-map construction of Proposition 3.1. Thus φ_ε is obtained from an equation of the form $F(\varepsilon, \varphi) = 0$, where

$$F : (-\varepsilon_0, \varepsilon_0) \times \mathcal{U} \rightarrow \mathcal{V}$$

is the C^1 moment map between the analytic boundary spaces used there, with $F(0, 0) = 0$ and $\partial_\varphi F(0, 0)$ invertible by the explicit Fourier diagonalization at the unit disk. Since the polynomial $P_\varepsilon(z) = 1 + \varepsilon z^2$ depends polynomially on ε , the dependence of F on ε is C^∞ . The C^1 version of the implicit function theorem in Banach spaces then yields a C^1 branch $\varepsilon \mapsto \varphi_\varepsilon$ into $C^1(\mathbb{S}^1)$. We therefore write

$$\dot{\varphi}_0(y) := \partial_\varepsilon|_{\varepsilon=0} \varphi_\varepsilon(y).$$

Step 1: first variation of the eigenvalue identity. We differentiate (5.17) at $\varepsilon = 0$. Since $P_\varepsilon \rightarrow 1$ in C^∞ and $\varepsilon \mapsto \varphi_\varepsilon$ is C^1 into $C^1(\mathbb{S}^1)$, all terms in (5.17) are differentiable at $\varepsilon = 0$. Since ∂U_ε is C^2 and admits the explicit normal graph parametrization (5.18) over $\mathbb{S}^1$ with φ_ε depending C^1 -smoothly on ε , the hypotheses of Lemma 5.3 are satisfied with normal velocity $V_\nu(y) = \dot{\varphi}_0(y)$; the Reynolds transport identity (5.9) applies, and reduces here to a direct application of the fundamental theorem of calculus in polar coordinates. Concretely, with $z = re^{it}$, for any smooth test function H we have

$$\int_{U_\varepsilon} H(z) dA(z) = \int_0^{2\pi} \int_0^{1+\varphi_\varepsilon(e^{it})} H(re^{it}) r dr dt.$$

Differentiating in ε at $\varepsilon = 0$ and using the fundamental theorem of calculus,

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_{U_\varepsilon} H(z) dA(z) = \int_0^{2\pi} \dot{\varphi}_0(e^{it}) H(e^{it}) dt = \int_{\mathbb{S}^1} \dot{\varphi}_0(z) H(z) d\mathcal{H}^1(z),$$

where the factor $r = 1$ on the unit circle has been absorbed into the arc-length measure. This is precisely Reynolds' transport identity:

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_{U_\varepsilon} H(z) dA(z) = \int_{\mathbb{S}^1} \dot{\varphi}_0(z) H(z) d\mathcal{H}^1(z).$$

Applying this to

$$H(z) = e^{\pi \bar{z} w} e^{-\pi |z|^2},$$

and differentiating the factor $P_\varepsilon(z) = 1 + \varepsilon z^2$, we obtain

$$\int_{\mathbb{S}^1} \dot{\varphi}_0(z) e^{\pi \bar{z} w} e^{-\pi |z|^2} d\mathcal{H}^1(z) + \int_{D_1} z^2 e^{\pi \bar{z} w} e^{-\pi |z|^2} dA(z) = \lambda_0 w^2. \quad (5.19)$$

Since D_1 is radial, the monomial z^2 is an eigenfunction of the localization operator of D_1 ; therefore there exists $\lambda_2 > 0$ such that

$$\int_{D_1} z^2 e^{\pi \bar{z} w} e^{-\pi |z|^2} dA(z) = \lambda_2 w^2.$$

Hence (5.19) becomes

$$\int_{\mathbb{S}^1} \dot{\varphi}_0(z) e^{\pi \bar{z} w} e^{-\pi |z|^2} d\mathcal{H}^1(z) = (\lambda_0 - \lambda_2) w^2. \quad (5.20)$$

In fact, λ_2 can be computed explicitly. Expanding $e^{\pi \bar{z} w} = \sum_{n \geq 0} \frac{\pi^n}{n!} \bar{z}^n w^n$ and using orthogonality of the monomials $\bar{z}^n$ over the disk, the only term contributing to w^2 is the one with $n = 2$:

$$\int_{D_1} z^2 e^{\pi \bar{z} w} e^{-\pi |z|^2} dA(z) = \frac{\pi^2}{2} w^2 \int_{D_1} z^2 \bar{z}^2 e^{-\pi |z|^2} dA(z) = \frac{\pi^2}{2} w^2 \int_{D_1} |z|^4 e^{-\pi |z|^2} dA(z).$$

Passing to polar coordinates and integrating by parts twice,

$$\int_{D_1} |z|^4 e^{-\pi |z|^2} dA(z) = 2\pi \int_0^1 r^4 e^{-\pi r^2} r dr = \pi \int_0^1 s^2 e^{-\pi s} ds$$

where we substituted $s = r^2$. A direct computation, integrating by parts twice, yields

$$\pi \int_0^1 s^2 e^{-\pi s} ds = \frac{2}{\pi^2} \left(1 - \left(1 + \pi + \frac{\pi^2}{2} \right) e^{-\pi} \right).$$

Multiplying by $\frac{\pi^2}{2}$, we obtain

$$\lambda_2 = \frac{\pi^2}{2} \int_{D_1} |z|^4 e^{-\pi |z|^2} dA(z) = 1 - \left(1 + \pi + \frac{\pi^2}{2} \right) e^{-\pi}.$$

Therefore

$$\lambda_0 - \lambda_2 = (1 - e^{-\pi}) - \left(1 - \left(1 + \pi + \frac{\pi^2}{2} \right) e^{-\pi} \right) = \pi \left(1 + \frac{\pi}{2} \right) e^{-\pi} > 0,$$

so the coefficient of w^2 on the right-hand side is indeed nonzero. In particular, since $|z| = 1$ on $\mathbb{S}^1$, we may rewrite this as

$$e^{-\pi} \int_{\mathbb{S}^1} \dot{\varphi}_0(z) e^{\pi \bar{z} w} d\mathcal{H}^1(z) = (\lambda_0 - \lambda_2) w^2. \quad (5.21)$$

Setting $w = 0$ gives

$$\int_{\mathbb{S}^1} \dot{\varphi}_0 d\mathcal{H}^1 = 0. \quad (5.22)$$

Step 2: identification of the leading mode. Expanding

$$e^{\pi \bar{z} w} = \sum_{n \geq 0} \frac{\pi^n}{n!} \bar{z}^n w^n,$$

and matching the coefficient of w^n on both sides of (5.21), we find

$$\int_{\mathbb{S}^1} \dot{\varphi}_0(z) \bar{z}^n d\mathcal{H}^1(z) = 0 \quad \text{for all } n \neq 2,$$

whereas the coefficient corresponding to $n = 2$ equals $\frac{2}{\pi^2} (\lambda_0 - \lambda_2) e^\pi \neq 0$. Since on $\mathbb{S}^1$ we have $\bar{z} = e^{-it}$ when $z = e^{it}$, this says that the Fourier coefficient $\widehat{\dot{\varphi}_0}(n)$ vanishes for every $n \neq 2$. Since $\dot{\varphi}_0$ is real-valued, $\widehat{\dot{\varphi}_0}(-n) = \overline{\widehat{\dot{\varphi}_0}(n)}$, so the $n = -2$ mode is also nonzero and conjugate to the $n = 2$ mode, while all other modes vanish. We conclude that

$$\dot{\varphi}_0(e^{it}) = a \cos(2t) + b \sin(2t) \quad (5.23)$$

for some pair $(a, b) \neq (0, 0)$. In particular,

$$\int_{\mathbb{S}^1} |\dot{\varphi}_0| d\mathcal{H}^1 > 0. \quad (5.24)$$

Step 3: normalization of the area. The area of U_ε is given by

$$|U_\varepsilon| = \frac{1}{2} \int_{\mathbb{S}^1} (1 + \varphi_\varepsilon(y))^2 d\mathcal{H}^1(y). \quad (5.25)$$

Since

$$\varphi_\varepsilon = \varepsilon \dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1),$$

and (5.22) gives the vanishing of the first-order term, we obtain from (5.25)

$$|U_\varepsilon| = \pi + O(\varepsilon^2). \quad (5.26)$$

Let

$$r_\varepsilon := \sqrt{\frac{\pi}{|U_\varepsilon|}}, \quad \tilde{U}_\varepsilon := r_\varepsilon U_\varepsilon.$$

Then $|\tilde{U}_\varepsilon| = \pi$, and by (5.26),

$$r_\varepsilon = 1 + O(\varepsilon^2).$$

Hence $\tilde{U}_\varepsilon$ still admits a normal graph representation

$$\partial \tilde{U}_\varepsilon = \{(1 + \tilde{\varphi}_\varepsilon(y))y : y \in \mathbb{S}^1\},$$

with

$$\tilde{\varphi}_\varepsilon = \varepsilon \dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1).$$

Thus the area normalization does not alter the first-order deformation.

Step 4: distance from the centered disk. Since both $\tilde{U}_\varepsilon$ and D_1 are star-shaped with respect to the origin, the symmetric difference may be computed in polar coordinates:

$$|\tilde{U}_\varepsilon \triangle D_1| = \frac{1}{2} \int_{\mathbb{S}^1} |(1 + \tilde{\varphi}_\varepsilon(y))^2 - 1| d\mathcal{H}^1(y).$$

Since $\tilde{\varphi}_\varepsilon = \varepsilon \dot{\varphi}_0 + O(\varepsilon^2)$ in $C^1(\mathbb{S}^1)$, we have in particular $\|\tilde{\varphi}_\varepsilon\|_{L^\infty(\mathbb{S}^1)} \lesssim |\varepsilon|$, whence $\tilde{\varphi}_\varepsilon^2 = O(\varepsilon^2)$ in $L^\infty(\mathbb{S}^1)$. Hence

$$(1 + \tilde{\varphi}_\varepsilon)^2 - 1 = 2\tilde{\varphi}_\varepsilon + \tilde{\varphi}_\varepsilon^2 = 2\varepsilon \dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } L^\infty(\mathbb{S}^1),$$

where the implied constant depends only on the C^1 -norm bound on $\tilde{\varphi}_\varepsilon$. Therefore

$$|\tilde{U}_\varepsilon \triangle D_1| = \frac{1}{2} \int_{\mathbb{S}^1} |2\varepsilon \dot{\varphi}_0 + O(\varepsilon^2)| d\mathcal{H}^1.$$

Set $\alpha := \int_{\mathbb{S}^1} |\dot{\varphi}_0| d\mathcal{H}^1$. By (5.24), $\alpha > 0$. The reverse triangle inequality in L^1 and the $O(\varepsilon^2)$ remainder give

$$\int_{\mathbb{S}^1} |2\varepsilon \dot{\varphi}_0 + O(\varepsilon^2)| d\mathcal{H}^1 \geq 2|\varepsilon|\alpha - O(\varepsilon^2)$$

and similarly the upper bound

$$\int_{\mathbb{S}^1} |2\varepsilon \dot{\varphi}_0 + O(\varepsilon^2)| d\mathcal{H}^1 \leq 2|\varepsilon|\alpha + O(\varepsilon^2).$$

Consequently there exist constants $c, C > 0$ such that

$$c|\varepsilon| \leq |\tilde{U}_\varepsilon \triangle D_1| \leq C|\varepsilon| \quad (5.27)$$

for all sufficiently small ε .

Step 5: comparison with arbitrary disks. Let D be any disk of area π . Then $D = D_1 + a$ for some $a \in \mathbb{C}$. We first consider disks whose centers are not too close to the origin. Since

$$|D_1 \triangle (D_1 + a)| \gtrsim \min\{|a|, 1\},$$

the triangle inequality and (5.27) imply that there exists $K \gg 1$ such that, if $|a| \geq K|\varepsilon|$, then

$$|\tilde{U}_\varepsilon \triangle D| \geq |D_1 \triangle D| - |\tilde{U}_\varepsilon \triangle D_1| \gtrsim |a| - C|\varepsilon| \gtrsim |\varepsilon|. \quad (5.28)$$

It remains to consider disks whose centers satisfy $|a| \leq K|\varepsilon|$. Write

$$a = \eta \varepsilon e^{i\theta}, \quad |\eta| \leq K, \quad \theta \in [0, 2\pi).$$

We claim that, for ε sufficiently small, the translated domain

$$\tilde{U}_\varepsilon - a$$

is still a normal graph over $\mathbb{S}^1$:

$$\partial(\tilde{U}_\varepsilon - a) = \{(1 + \psi_\varepsilon^{\theta, \eta}(y))y : y \in \mathbb{S}^1\},$$

with

$$\psi_\varepsilon^{\theta, \eta} \in C^1(\mathbb{S}^1), \quad \|\psi_\varepsilon^{\theta, \eta}\|_{C^1} \lesssim |\varepsilon|,$$

uniformly in (θ, η) with $|\eta| \leq K$. This follows because $\tilde{U}_\varepsilon$ is already an $O(\varepsilon)$ - C^1 perturbation of D_1 , and the translation vector a is also of order ε . We now compute the first-order term of $\psi_\varepsilon^{\theta, \eta}$. A translation by a contributes, to first order in ε , the normal displacement

$$-\eta \Re(e^{-i\theta} y).$$

Indeed, for $y \in \mathbb{S}^1$,

$$|y - a| = (1 - 2\Re(\bar{y}a) + |a|^2)^{1/2} = 1 - \Re(\bar{y}a) + O(|a|^2),$$

uniformly in y , and here $a = \eta \varepsilon e^{i\theta}$. Therefore

$$\psi_\varepsilon^{\theta, \eta}(y) = \varepsilon \left(\dot{\varphi}_0(y) - \eta \Re(e^{-i\theta} y) \right) + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1), \quad (5.29)$$

uniformly for $|\eta| \leq K$.

Define

$$g^{\theta, \eta}(y) := \dot{\varphi}_0(y) - \eta \Re(e^{-i\theta} y).$$

Using (5.23), we see that $g^{\theta, \eta}$ belongs to the finite-dimensional space spanned by the first and second harmonics. More precisely,

$$g^{\theta, \eta}(e^{it}) = a \cos(2t) + b \sin(2t) - \eta \cos(t - \theta).$$

On the Fourier side this is the key cancellation point: the translation correction $-\eta \Re(e^{-i\theta} y)$ contributes only to the first harmonics (its Fourier modes are ± 1), while the perturbation $\dot{\varphi}_0$ lives entirely in the second harmonics (± 2). Hence the second-harmonic component is invariant under the translation correction, and in particular the second-harmonic part of $g^{\theta, \eta}$ is independent of (θ, η) and nonzero, because $(a, b) \neq (0, 0)$. To extract a uniform L^1 lower bound, note that $g^{\theta, \eta}$

lies in the 4-dimensional space

$$V := \text{span}\{\cos t, \sin t, \cos 2t, \sin 2t\} \subset L^1(\mathbb{S}^1).$$

On this finite-dimensional space, all norms are equivalent; in particular, the $L^1(\mathbb{S}^1)$ and $L^\infty(\mathbb{S}^1)$ norms are equivalent, and convergence in coefficients is equivalent to convergence in any of these norms. The map

$$(\theta, \eta) \in [0, 2\pi) \times [-K, K] \mapsto g^{\theta, \eta} \in V$$

is continuous, and its image

$$\mathcal{G}_K := \{g^{\theta, \eta} : \theta \in [0, 2\pi), |\eta| \leq K\}$$

is the continuous image of a compact set, hence compact in V . Moreover, the $(\cos 2t, \sin 2t)$ -coefficients of every $g^{\theta, \eta}$ are exactly $(a, b) \neq (0, 0)$, independent of (θ, η) ; thus $0 \notin \mathcal{G}_K$. The function $g \mapsto \|g\|_{L^1(\mathbb{S}^1)}$ is continuous and strictly positive on $V \setminus \{0\}$, so it attains a positive minimum on the compact set $\mathcal{G}_K$. It follows that there exists $\delta_K > 0$ such that

$$\int_{\mathbb{S}^1} |g^{\theta, \eta}(y)| d\mathcal{H}^1(y) \geq \delta_K \quad \text{for all } \theta \in [0, 2\pi), |\eta| \leq K. \quad (5.30)$$

The constant δ_K admits the explicit lower bound $\delta_K \geq \pi(a^2 + b^2)/(4(K+1) + 2\sqrt{a^2 + b^2})$ provided by Lemma 5.4, see (5.12); this quantitative estimate sharpens the qualitative compactness argument above.

Now,

$$|\tilde{U}_\varepsilon \Delta D| = |(\tilde{U}_\varepsilon - a) \Delta D_1|.$$

Since $\tilde{U}_\varepsilon - a$ is star-shaped with radial graph $1 + \psi_\varepsilon^{\theta, \eta}$, the same polar-coordinate computation as in Step 4 gives

$$|(\tilde{U}_\varepsilon - a) \Delta D_1| = \frac{1}{2} \int_{\mathbb{S}^1} \left| (1 + \psi_\varepsilon^{\theta, \eta}(y))^2 - 1 \right| d\mathcal{H}^1(y).$$

Using (5.29),

$$(1 + \psi_\varepsilon^{\theta, \eta})^2 - 1 = 2\varepsilon g^{\theta, \eta} + O(\varepsilon^2) \quad \text{in } L^\infty(\mathbb{S}^1),$$

uniformly for $|\eta| \leq K$. Thus, by (5.30), there exists $c_K > 0$ such that

$$|\tilde{U}_\varepsilon \Delta D| = |(\tilde{U}_\varepsilon - a) \Delta D_1| \geq c_K |\varepsilon| \quad (5.31)$$

whenever $|a| \leq K|\varepsilon|$. Combining (5.28) and (5.31), we conclude that

$$\inf_{\substack{D \text{ disk} \\ |D|=\pi}} |\tilde{U}_\varepsilon \Delta D| \gtrsim |\varepsilon|.$$

Since $r_\varepsilon = 1 + O(\varepsilon^2)$, the dilation from U_ε to $\tilde{U}_\varepsilon$ changes the set only by

$$|\tilde{U}_\varepsilon \Delta U_\varepsilon| = O(\varepsilon^2),$$

and therefore does not affect the first-order distance estimate. Thus the area-normalized family $\tilde{U}_\varepsilon$ remains at symmetric-difference distance $\gtrsim |\varepsilon|$ from the class of disks.

Step 6: square-root sharpness of the deficit estimate. It remains to translate the distance estimate into the corresponding deficit estimate. Recall that, by construction via Theorem 1, the

identity (5.17) states that P_ε is an eigenfunction with eigenvalue λ_0 for the localization operator on U_ε . By Lemma 5.1, the eigenvalue λ_0 is in fact the *principal* (largest) eigenvalue of $\mathcal{L}_{U_\varepsilon}$ for all sufficiently small ε , and P_ε is a corresponding first eigenfunction; the gap (5.4) ensures that λ_0 cannot be confused with any lower-order eigenvalue. Hence

$$\lambda_1(U_\varepsilon) = \lambda_0 \quad \text{exactly, for } |\varepsilon| < \varepsilon_*. \quad (5.32)$$

By (5.26), $|U_\varepsilon| = \pi + O(\varepsilon^2)$, so the area-normalizing dilation factor $r_\varepsilon = \sqrt{\pi/|U_\varepsilon|} = 1 + O(\varepsilon^2)$. By Lemma 5.2 (specifically (5.7)), the dilation changes the principal eigenvalue by

$$|\lambda_1(\tilde{U}_\varepsilon) - \lambda_1(U_\varepsilon)| = |\lambda_1(r_\varepsilon U_\varepsilon) - \lambda_0| = O(\varepsilon^2).$$

Combining with (5.32), the eigenvalue deficit of $\tilde{U}_\varepsilon$ with respect to the disk satisfies

$$\lambda_0 - \lambda(\tilde{U}_\varepsilon) = O(\varepsilon^2),$$

where $\lambda(\tilde{U}_\varepsilon)$ denotes the first localization eigenvalue of $\tilde{U}_\varepsilon$. On the other hand, by Step 5, the symmetric-difference asymmetry satisfies

$$\inf_{\substack{D \text{ disk} \\ |D|=\pi}} |\tilde{U}_\varepsilon \Delta D| \gtrsim |\varepsilon|.$$

Comparing the two,

$$\inf_{\substack{D \text{ disk} \\ |D|=\pi}} |\tilde{U}_\varepsilon \Delta D| \gtrsim |\varepsilon| \simeq \sqrt{\lambda_0 - \lambda(\tilde{U}_\varepsilon)}.$$

This shows that the inequality

$$\inf_{\substack{D \text{ disk} \\ |D|=\pi}} |\Omega \Delta D| \lesssim \sqrt{\lambda_0 - \lambda(\Omega)}$$

of Corollary 1.4 in [20] is square-root-tight along the family $\tilde{U}_\varepsilon$: the deficit is of order ε^2 , the asymmetry is of order ε , and so the square root of the deficit and the asymmetry are of comparable size. This exhibits the optimality of the exponent 1/2 and concludes the proof of the sharpness of the estimate in Corollary 1.4 of [20], complementing the rigidity result of [40]. $\square$

6. LOCAL MAXIMIZERS FOR THE FABER–KRAHN PROBLEM IN THE GAUSSIAN STFT SETTING

The purpose of this section is to establish a local rigidity statement for the Faber–Krahn problem associated with the Gaussian STFT. The global problem, solved in [40], asserts that, among measurable subsets of $\mathbb{C}$ of fixed measure, the centered disk uniquely maximizes the first eigenvalue of the Gaussian STFT localization operator. Here we show that this rigidity is already visible at the *local* level: any set whose first eigenvalue cannot be increased by small symmetric-difference perturbations must, up to a translation, be a disk. This complements the quantitative stability result of [20] and provides the variational counterpart of the inverse-spectral analysis carried out in the previous sections.

We work throughout in the Bargmann–Fock representation; we refer the reader to [4, 18, 22] for the relevant background. Thus, if $F \in \mathcal{F}^2(\mathbb{C})$ is the Bargmann transform of $f \in L^2(\mathbb{R})$, then

$$|V_\varphi f(z)|^2 = |F(z)|^2 e^{-\pi|z|^2},$$

up to the normalization already fixed in the previous sections. Accordingly, for $F \in \mathcal{F}^2(\mathbb{C})$ we set

$$u_F(z) := |F(z)|^2 e^{-\pi|z|^2}.$$

For a measurable set $\Omega \subset \mathbb{C}$, let

$$\lambda_{1,\varphi}(\Omega) := \sup_{\|f\|_2=1} \int_{\Omega} |V_\varphi f(z)|^2 dz. \quad (6.1)$$

6.1. The weighted symmetric-difference topology. The locality of a maximizer must be measured in a topology compatible with the Lebesgue-measure constraint and with the natural symmetries of the localization operator. The right choice in our setting is the weighted symmetric-difference metric obtained by integrating against the Gaussian density $e^{-\pi|z|^2} dA(z)$, which is the same density that appears both in the Fock-space norm and in the eigenfunction u_F .

Definition 6.1 (Weighted symmetric-difference metric). For measurable sets $\Omega_1, \Omega_2 \subset \mathbb{C}$ of finite Lebesgue measure, set

$$d_w(\Omega_1, \Omega_2) := \int_{\Omega_1 \Delta \Omega_2} w(z) dA(z), \quad w(z) := e^{-\pi|z|^2}. \quad (6.2)$$

We call d_w the *weighted symmetric-difference distance*, and we denote by $\mathcal{M}_a$ the Borel-measurable subsets of $\mathbb{C}$ of Lebesgue measure $a > 0$, modulo Lebesgue-null modifications. The map $d_w: \mathcal{M}_a \times \mathcal{M}_a \rightarrow [0, \infty)$ is a genuine metric on $\mathcal{M}_a$: symmetry, the triangle inequality, and vanishing only on null modifications all follow from $w(z) = e^{-\pi|z|^2} > 0$ and standard properties of the symmetric difference.

Remark 6.1 (Naturality of d_w for the Gaussian Faber–Krahn problem). Three observations clarify why d_w is the natural topology for our problem.

- (i) (*Compatibility with the measure constraint.*) Since w is bounded above by 1 and bounded below by $e^{-\pi R^2} > 0$ on every disk $\{|z| \leq R\}$, the metric d_w is, on every fixed bounded region of $\mathbb{C}$, equivalent to the unweighted symmetric-difference distance $d_0(\Omega_1, \Omega_2) := |\Omega_1 \Delta \Omega_2|$. Moreover, every d_w -perturbation E of Ω used in this section is constructed to satisfy the equimeasurability constraint $|E| = |\Omega|$; hence the d_w -balls intersected with $\mathcal{M}_a$ form a basis of neighborhoods on the measure-constraint manifold.
- (ii) (*Translation- and rotation-invariance via Weyl.*) The Gaussian density $w(z) = e^{-\pi|z|^2}$ is the symbol density associated with the metaplectic representation: under the Weyl translations $T_a: F \mapsto e^{\pi a z - \frac{\pi}{2}|a|^2} F(z - a)$ (cf. (2.5)) and rotations $F(z) \mapsto F(e^{-i\theta} z)$, one has the covariance

$$u_{T_a F}(z) = u_F(z - a), \quad u_{F(e^{-i\theta} \cdot)}(z) = u_F(e^{-i\theta} z),$$

so that the spectral data $(\lambda_{1,\varphi}, u_F)$ of the localization operator transform unitarily under these symmetries. Any choice of weight other than a Gaussian would break this covariance with the operator algebra of the Fock space; see, e.g., [18, 22].

- (iii) (*Control of Fock-space tails.*) For any $F \in \mathcal{F}^2(\mathbb{C})$ one has $u_F(z) \leq \|F\|_{\mathcal{F}^2}^2$ for every $z \in \mathbb{C}$, as a consequence of the reproducing property (6.11) below and Cauchy–Schwarz. Hence, for first eigenfunctions F (which are bounded; see Lemma 6.4),

$$\left| \int_{\Omega_1} u_F dA - \int_{\Omega_2} u_F dA \right| \leq \int_{\Omega_1 \triangle \Omega_2} u_F dA \leq \|F\|_{\mathcal{F}^2}^2 e^{\pi R^2} d_w(\Omega_1, \Omega_2)$$

on any bounded region $\{|z| \leq R\}$, so that the localization functional $\Omega \mapsto \int_{\Omega} u_F dA$ is Lipschitz in d_w on bounded sets. The Gaussian weight thus tames the lack of compactness at infinity by ensuring that d_w -small perturbations cannot transport mass to large $|z|$ without paying a proportional cost.

In summary, d_w is essentially the unique Gaussian-weighted L^1 -metric on indicator functions that is compatible with the equimeasurable constraint $|\Omega| = a$, respects the Weyl-translation and rotation symmetries of the Fock space, and renders $\lambda_{1,\varphi}$ and the localization energy continuous on bounded sets.

Definition 6.2 (Local maximizer). A set $\Omega \in \mathcal{M}_a$ is a *local maximizer* for $\lambda_{1,\varphi}$ under the Lebesgue-measure constraint $|\cdot| = a$ if there exists $\delta_0 > 0$ such that, whenever $E \in \mathcal{M}_a$ satisfies $d_w(\Omega, E) < \delta_0$, one has

$$\lambda_{1,\varphi}(\Omega) \geq \lambda_{1,\varphi}(E). \quad (6.3)$$

We can now state the main theorem of this section.

Theorem 8 (Local maximizers are disks). *Let $\Sigma \in \mathcal{M}_a$ be a local maximizer for $\lambda_{1,\varphi}$ under the Lebesgue-measure constraint, in the sense of Definition 6.2. Then there exists $a_0 \in \mathbb{C}$ such that Σ coincides, up to a Lebesgue-null modification, with the centered disk*

$$\Sigma = D_R + a_0, \quad R = \sqrt{a/\pi}.$$

The remainder of this section is devoted to the proof of Theorem 8. The proof proceeds in six steps. Steps 1–3 are essentially the analogues, in our setting, of standard arguments in the Faber–Krahn theory: Σ is forced to be a level set of u_F , the corresponding eigenvalue is simple, and a first-variation identity ties F to a divergence-free vector field built from holomorphic data. The crucial new ingredient lies in Step 4, where we show that the *derivative* F' is again a first eigenfunction. Combined with the simplicity from Step 2, this forces F to satisfy a first-order linear ODE, and hence to be a Gaussian. Once this is in place, Steps 5–6 use the Weyl invariance of the Bargmann–Fock setting and a one-dimensional rearrangement argument to identify Σ with a centered disk.

Proof of the local-maximizer theorem. Let $\Sigma \subset \mathbb{C}$ be a local maximizer for $\lambda_{1,\varphi}$ under the measure constraint, and let $F \in \mathcal{F}^2(\mathbb{C})$ be a normalized first eigenfunction for the localization operator associated with Σ . Equivalently,

$$\int_{\Sigma} H(z) \overline{F(z)} d\mu(z) = \lambda_{1,\varphi}(\Sigma) \int_{\mathbb{C}} H(z) \overline{F(z)} d\mu(z) \quad (6.4)$$

for every $H \in \mathcal{F}^2(\mathbb{C})$, where

$$d\mu(z) := e^{-\pi|z|^2} dA(z).$$

The proof of the superlevel-set characterization in Step 1 below uses the following classical rearrangement inequality, in the formulation of Lieb–Loss [35, Theorem 1.14] (the so-called *bathtub principle*). We record it here for ease of reference and to fix notation.

Lemma 6.1 (Bathtub principle). *Let (X, Σ_X, μ_X) be a measure space and let $g: X \rightarrow [0, \infty)$ be a measurable function such that $\mu_X(\{g > t\}) < \infty$ for every $t > 0$. Fix $0 < a < \mu_X(X)$. Then*

$$\sup \left\{ \int_X g \mathbf{1}_E d\mu_X \mid E \in \Sigma_X, \mu_X(E) = a \right\} \quad (6.5)$$

is attained, and any maximizer E^ coincides, up to a μ_X -null set, with a superlevel set $\{g > t_*\}$ of g (possibly with a measure-zero correction at the level set $\{g = t_*\}$ needed to fit the constraint $\mu_X(E^*) = a$). The threshold $t_* \geq 0$ is uniquely determined by the condition*

$$\mu_X(\{g > t_*\}) \leq a \leq \mu_X(\{g \geq t_*\}),$$

and the maximum is strict in the sense that, if g has no plateaus of positive measure on the level $\{g = t_\}$, then $E^* = \{g > t_*\}$ μ_X -a.e.*

In our application, $(X, \mu_X) = (\mathbb{C}, \text{Leb}_{\mathbb{C}})$ and $g = u_F$. Since u_F is real-analytic on the (open) complement of the zero set of F , its level sets $\{u_F = t\}$ are real-analytic varieties of strictly lower dimension and hence Lebesgue-null for every $t > 0$; the no-plateau hypothesis of Lemma 6.1 is therefore automatic, and the maximizer is unique up to null sets.

Step 1: Σ is a superlevel set of u_F .

We claim that there exists $t_0 \geq 0$ such that, up to sets of measure zero,

$$\Sigma = \{u_F > t_0\}. \quad (6.6)$$

The strategy is a bathtub-style perturbation argument: if Σ were not a superlevel set of u_F , one could move a small piece of Σ where u_F is small to a region where u_F is larger, increasing the localization integral while preserving Lebesgue measure and remaining within the local-maximizer neighborhood of Σ (cf. Lemma 6.1). The contradiction we extract is local, but as we shall see in Lemma 6.2, the same argument actually proves the slightly stronger statement that Σ is the unique μ -essential maximizer of the functional $E \mapsto \int_E u_F dA$ under the equimeasurable constraint, in agreement with the bathtub principle.

Recall that F is admissible in the variational problem defining $\lambda_{1,\varphi}(\Sigma')$ for any measurable set Σ' of finite measure. In particular,

$$\lambda_{1,\varphi}(\Sigma') \geq \int_{\Sigma'} u_F dA, \quad \lambda_{1,\varphi}(\Sigma) = \int_{\Sigma} u_F dA,$$

the latter equality holding because F is the actual eigenfunction for Σ . It therefore suffices to find a competitor Σ' with $|\Sigma'| = |\Sigma|$, close to Σ in the sense of weighted symmetric difference, and with $\int_{\Sigma'} u_F dA > \int_{\Sigma} u_F dA$; that would violate local maximality.

Suppose, for the sake of contradiction, that (6.6) fails. The function u_F is real-analytic on $\mathbb{C} \setminus \{F = 0\}$, so for almost every $t \geq 0$ the level set $\{u_F = t\}$ has Lebesgue measure zero;

consequently the distribution function $t \mapsto |\{u_F > t\}|$ is continuous and strictly decreasing on its support. Hence there exists a unique $t_0 \geq 0$ such that

$$A := \{u_F > t_0\} \quad \text{satisfies} \quad |A| = |\Sigma|;$$

since (6.6) fails by assumption, A and Σ do not coincide up to a null set, so both $A \setminus \Sigma$ and $\Sigma \setminus A$ have positive Lebesgue measure (and in fact positive weighted measure, since $e^{-\pi|z|^2} > 0$ everywhere).

We now perform the bathtub construction explicitly. The set $A = \{u_F > t_0\}$ is open since u_F is continuous, while $\Sigma \setminus A$ is only known to be measurable. By inner regularity of the Lebesgue measure, choose a compact set $L \subset A \setminus \Sigma$ with $|L| > 0$ and a compact set $K \subset \Sigma \setminus A$ with $|K| > 0$ (both nonempty by the previous paragraph). Note that $L \subset A$ is then contained in the open set A , and we can shrink L further if needed so that $\int_L d\mu, \int_K d\mu < \delta_0/4$. Since the Lebesgue measure restricted to a compact set has the intermediate-value property, we may choose subsets $S \subset L$ and $T \subset K$ with the common Lebesgue measure

$$|S| = |T| = \frac{1}{2} \min(|L|, |K|) > 0.$$

Then $S \subset A \setminus \Sigma$, $T \subset \Sigma \setminus A$, and

$$\int_{\mathbb{C}} (\mathbf{1}_S + \mathbf{1}_T) e^{-\pi|z|^2} dz < \frac{\delta_0}{2} < \delta_0.$$

Define

$$\Sigma' := (\Sigma \setminus T) \cup S.$$

Then $|\Sigma'| = |\Sigma|$ and

$$\int_{\mathbb{C}} |\mathbf{1}_{\Sigma} - \mathbf{1}_{\Sigma'}| e^{-\pi|z|^2} dz = \int_{\mathbb{C}} (\mathbf{1}_S + \mathbf{1}_T) e^{-\pi|z|^2} dz < \delta_0.$$

On the other hand, by construction,

$$u_F(z) > t_0 \quad \text{on } S \subset A, \quad u_F(z) \leq t_0 \quad \text{on } T \subset \Sigma \setminus A.$$

Moreover, since u_F is continuous and $L \subset A = \{u_F > t_0\}$ is compact, the continuous function u_F attains its minimum on L ; hence there exists $\eta > 0$ with $u_F \geq t_0 + \eta$ on L , and a fortiori on S . It then follows that

$$\int_S u_F dA \geq (t_0 + \eta) |S|, \quad \int_T u_F dA \leq t_0 |T| = t_0 |S|,$$

and hence

$$\int_{\Sigma'} u_F dA = \int_{\Sigma} u_F dA + \int_S u_F dA - \int_T u_F dA \geq \int_{\Sigma} u_F dA + \eta |S| > \int_{\Sigma} u_F dA.$$

Since F is admissible for the variational problem defining $\lambda_{1,\varphi}(\Sigma')$,

$$\lambda_{1,\varphi}(\Sigma') \geq \int_{\Sigma'} u_F dA > \int_{\Sigma} u_F dA = \lambda_{1,\varphi}(\Sigma),$$

contradicting the local maximality of Σ . This proves (6.6).

In particular, since u_F is real-analytic outside the (discrete) zero set of F , u_F is constant on $\partial\Sigma$ in the sense that there exists $c_\Sigma \geq 0$ with

$$u_F(z) = c_\Sigma \quad \text{for } z \in \partial\Sigma. \quad (6.7)$$

We will see in Step 2 that F does not vanish on $\bar{\Sigma}$, and in particular $c_\Sigma > 0$.

We collect the conclusions of this step in a self-contained statement that we will reuse below.

Lemma 6.2 (Superlevel-set characterization). *Let $\Sigma \in \mathcal{M}_a$ be a local maximizer for $\lambda_{1,\varphi}$ in the sense of Definition 6.2, and let $F \in \mathcal{F}^2(\mathbb{C})$ be a normalized first eigenfunction associated with Σ . Then there exists $t_0 > 0$ such that*

$$\Sigma = \{z \in \mathbb{C} : u_F(z) > t_0\} \quad \text{up to a Lebesgue-null set}, \quad (6.8)$$

and Σ coincides up to a null set with the unique (up to null sets) maximizer of the bathtub problem $\sup\{\int_E u_F dA : |E| = a\}$ provided by Lemma 6.1 applied to $g = u_F$.

Proof. The first part is exactly (6.6); the strict positivity $t_0 > 0$ follows from the fact, derived in Step 2 below, that F cannot vanish on $\bar{\Sigma}$, so that $c_\Sigma = t_0 > 0$. Uniqueness within bathtub maximizers follows from the no-plateau remark below Lemma 6.1, since u_F is real-analytic on the open set $\{F \neq 0\} \supset \bar{\Sigma}$ and hence has no positive-measure level sets there. $\square$

Lemma 6.3 (Boundary regularity at a local maximizer). *Under the hypotheses of Lemma 6.2, choosing the open superlevel-set representative $\Sigma = \{u_F > t_0\}$, we have:*

- (i) Σ is open and bounded (because $u_F(z) \rightarrow 0$ as $|z| \rightarrow \infty$ and $t_0 > 0$);
- (ii) F does not vanish on $\bar{\Sigma}$, and u_F is real-analytic on a neighborhood of $\bar{\Sigma}$;
- (iii) $\partial\Sigma$ is a real-analytic 1-manifold, except for a finite set $\Sigma_{\text{sing}} \subset \partial\Sigma$ consisting of points where ∇u_F vanishes; in particular, $\partial\Sigma$ is locally Lipschitz, has finite $\mathcal{H}^1$ -measure, and $\mathcal{H}^1(\Sigma_{\text{sing}}) = 0$;
- (iv) the divergence theorem $\int_\Sigma \operatorname{div} X dA = \int_{\partial\Sigma} X \cdot \nu d\mathcal{H}^1$ holds for every C^1 -vector field X defined on a neighborhood of $\bar{\Sigma}$.

Proof. Statement (i) is immediate, since $u_F(z) = |F(z)|^2 e^{-\pi|z|^2}$ tends to 0 at infinity (as F is entire and the Gaussian factor decays super-exponentially against any polynomial growth of $|F|^2$ along any sequence $|z| \rightarrow \infty$, since $F \in \mathcal{F}^2$ is in particular bounded along any horizontal line by [18, Ch. 1]). The non-vanishing in (ii) is established in Step 2 (more precisely, in Lemma 6.4 below; the argument in Step 2 only uses (i) of the present lemma plus (6.8)). Once F is non-vanishing on $\bar{\Sigma}$, u_F is real-analytic on a neighborhood of $\bar{\Sigma}$. Statement (iii) is the standard local structure theorem for level sets of non-constant real-analytic functions (see, e.g., [32, 36]), which gives that $\{u_F = t_0\}$ decomposes locally into finitely many real-analytic arcs meeting at finitely many singular points (the zeros of ∇u_F on $\{u_F = t_0\}$, of which there are finitely many on the compact set $\bar{\Sigma}$ by analyticity). Lipschitz regularity of the boundary then follows by parametrizing each arc by arc length, and the divergence-theorem statement (iv) follows from the Gauss–Green theorem on Lipschitz domains (cf. [17]), using that the Lipschitz regular set $\partial\Sigma \setminus \Sigma_{\text{sing}}$ carries $\mathcal{H}^1$ -almost every point of $\partial\Sigma$. $\square$

Remark 6.2. No boundary regularity is hypothesized on the local maximizer: the only input is the measurability of Σ , and the real-analyticity of $\partial\Sigma$ (away from finitely many points) is *derived* from the superlevel-set characterization combined with the entirety of F and the Gaussian weight. Thus the divergence-theorem identity used in Step 3 is fully justified by Lemma 6.3(iv), even though Σ is a priori only a measurable set.

Step 2: simplicity of the first eigenvalue.

In this step we prove the following lemma, which lies at the heart of the remainder of the argument: it is what allows us, in Step 4, to conclude from the fact that F' is a first eigenfunction that F' is a scalar multiple of F and hence that F satisfies a linear ODE.

Lemma 6.4 (Simplicity at a local maximizer). *Let Σ be a local maximizer for $\lambda_{1,\varphi}$ in the sense of Definition 6.2 and let $F \in \mathcal{F}^2(\mathbb{C})$ be a normalized first eigenfunction associated with Σ . Then the eigenspace $\ker(\mathcal{L}_\Sigma - \lambda_{1,\varphi}(\Sigma)\mathbf{I})$ is one-dimensional; that is, $\lambda_{1,\varphi}(\Sigma)$ is simple, and F is uniquely determined up to a unimodular constant. In particular, F does not vanish on $\bar{\Sigma}$, and F is bounded on $\mathbb{C}$.*

The proof uses a modular-deformation principle: any two first eigenfunctions have the same modulus on $\partial\Sigma$ up to a constant, and a holomorphic function with constant modulus on a connected open set must itself be constant. This argument plays a role also in the stability analysis of [20].

Proof of Lemma 6.4. Suppose, for contradiction, that $G \in \mathcal{F}^2(\mathbb{C})$ is another first eigenfunction, linearly independent of F . Step 1 applied to G shows that Σ is also a superlevel set of u_G ; hence there exists $c'_\Sigma \geq 0$ such that

$$u_G(z) = c'_\Sigma \quad \text{for } z \in \partial\Sigma.$$

Equivalently,

$$|F(z)|^2 e^{-\pi|z|^2} = c_\Sigma, \quad |G(z)|^2 e^{-\pi|z|^2} = c'_\Sigma, \quad z \in \partial\Sigma.$$

We next observe that first eigenfunctions are zero-free in the interior of Σ . Indeed, by Step 1 we have $u_F > c_\Sigma$ on the (open) interior of Σ . Since

$$u_F = |F|^2 e^{-\pi|z|^2}$$

and the Gaussian factor $e^{-\pi|z|^2}$ is strictly positive everywhere, this yields $|F| > 0$ throughout the interior of Σ ; the same applies to $|G|$. In particular, $c_\Sigma > 0$ and $c'_\Sigma > 0$. Moreover, since F, G are entire and continuous, $|F| > 0$ on a neighborhood of any compact subset of Σ° ; by Step 1, we may take a slight enlargement of Σ (still a superlevel set of u_F) on which $|F|, |G| > 0$, so we may work on a connected open set $U \ni \Sigma$ where both F and G are zero-free.

The ratio F/G is holomorphic and nowhere zero on U , and the function

$$\log \left| \frac{F}{G} \right| = \frac{1}{2} \log \frac{|F|^2}{|G|^2} = \frac{1}{2} \log \frac{u_F}{u_G}$$

is harmonic on U : it is the real part of any local branch of $\log(F/G)$, which is well defined on simply connected subsets, and harmonic in any case as the logarithm of the modulus of a zero-free holomorphic function. Note that the Gaussian factors cancel inside the logarithm,

so on $\partial\Sigma$ by (6.7) this function equals the constant $\frac{1}{2} \log(c_\Sigma/c'_\Sigma)$. By the maximum principle applied separately on each connected component of Σ , the function $\log |F/G|$ is constant on each component. Hence

$$\left| \frac{F}{G} \right| \equiv \text{constant} \quad \text{on each connected component of } \Sigma.$$

A nonconstant holomorphic function is open, so a holomorphic function with constant modulus on a connected open set must itself be constant. Therefore

$$F = \gamma_j G \quad \text{on each connected component } \Sigma_j \text{ of } \Sigma.$$

The relation $F = \gamma_j G$ is an identity between two entire functions on a set with an accumulation point; the analytic continuation principle gives $F = \gamma_j G$ on all of $\mathbb{C}$ for any single j . In particular all γ_j coincide, and F and G are linearly dependent, contradicting our assumption. This proves simplicity.

The non-vanishing of F on $\bar{\Sigma}$ was established in the second paragraph of this step, while the boundedness of F on $\mathbb{C}$ follows from the Berezin-type estimate $u_F(z) = |\langle F, K_z \rangle|^2 / \|K_z\|_{\mathcal{F}^2}^2 \leq \|F\|_{\mathcal{F}^2}^2$, so that $|F(z)| \leq \|F\|_{\mathcal{F}^2} e^{\pi|z|^2/2}$ and $\|u_F\|_\infty \leq \|F\|_{\mathcal{F}^2}^2 < \infty$. This completes the proof of Lemma 6.4. $\square$

Step 3: a first-variation identity.

By Lemma 6.4, the eigenfunction F is determined up to a unimodular constant. We now derive a first-variation identity satisfied by F , based on the fact that Σ is a level set of u_F and that holomorphic vector fields are divergence-free.

We work throughout this step with the open superlevel-set representative $\Sigma = \{u_F > t_0\}$ from Lemma 6.2 (which is permissible since local maximality only sees the equivalence class of Σ up to null sets). By Lemma 6.3, F does not vanish on $\bar{\Sigma}$, u_F is real-analytic on a neighborhood of $\bar{\Sigma}$, the boundary $\partial\Sigma$ is a finite union of real-analytic arcs joined at finitely many singular points (the zeros of ∇u_F on $\{u_F = t_0\}$), and the divergence theorem applies on Σ to every C^1 -vector field defined in a neighborhood of $\bar{\Sigma}$. All boundary integrations below are therefore justified without any a priori smoothness assumption on Σ ; cf. Remark 6.2.

Let ψ be holomorphic on an open neighborhood V of $\bar{\Sigma}$, and write

$$\psi = u + iv$$

with u, v real-valued. Consider the real vector field

$$X_\psi := (u, -v).$$

Since ψ is holomorphic, the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x,$$

imply

$$\operatorname{div} X_\psi = u_x + (-v)_y = u_x - v_y = 0 \quad \text{in } V.$$

Because $u_F \equiv c_\Sigma$ on $\partial\Sigma$, the divergence theorem on Σ gives

$$\int_{\Sigma} \langle \nabla u_F, X_\psi \rangle dA = \int_{\Sigma} \operatorname{div}(u_F X_\psi) dA = \int_{\partial\Sigma} u_F X_\psi \cdot \nu d\mathcal{H}^1 = c_\Sigma \int_{\partial\Sigma} X_\psi \cdot \nu d\mathcal{H}^1 = 0,$$

where the last equality uses $\int_{\partial\Sigma} X_\psi \cdot \nu d\mathcal{H}^1 = \int_{\Sigma} \operatorname{div} X_\psi dA = 0$ by Cauchy–Riemann. Since $u_F = |F|^2 e^{-\pi|z|^2}$ and $\nabla(e^{-\pi|z|^2}) = -2\pi e^{-\pi|z|^2}(x, y)$, this becomes

$$0 = \int_{\Sigma} (\langle \nabla |F|^2, X_\psi \rangle - 2\pi |F|^2 \langle z, X_\psi \rangle) d\mu. \quad (6.9)$$

We now express (6.9) in complex notation. A direct computation, using $\partial_z |F|^2 = F' \bar{F}$ and $\partial_{\bar{z}} |F|^2 = \bar{F}' F$, gives

$$\partial_x |F|^2 = F' \bar{F} + \bar{F}' F = 2\Re(F' \bar{F}), \quad \partial_y |F|^2 = i(F' \bar{F} - \bar{F}' F) = -2\Im(F' \bar{F}).$$

Writing $\psi = u + iv$ and recalling $X_\psi = (u, -v)$,

$$\langle \nabla |F|^2, X_\psi \rangle = 2u \Re(F' \bar{F}) + 2v \Im(F' \bar{F}) = 2\Re(\bar{\psi} F' \bar{F}) = 2\Re(\psi \bar{F}' F),$$

where in the last equality we used $\Re w = \Re \bar{w}$. Likewise, writing $z = x + iy$,

$$\langle z, X_\psi \rangle = xu + y(-v) = xu - yv = \Re((x + iy)(u + iv)) = \Re(z\psi).$$

Substituting these expressions into (6.9), we obtain

$$0 = \Re \int_{\Sigma} \psi(z) \left(\overline{F'(z)} F(z) - \pi z |F(z)|^2 \right) d\mu(z).$$

Replacing ψ by $i\psi$ (which is also holomorphic on V) gives the analogous identity for the imaginary part:

$$0 = \Im \int_{\Sigma} \psi(z) \left(\overline{F'(z)} F(z) - \pi z |F(z)|^2 \right) d\mu(z).$$

Combining the two, and using $|F|^2 = F \bar{F}$ to factor out $F(z)$ from both terms inside the integrand (the F from $\bar{F}' F$ and one copy of F from $|F|^2 = F \bar{F}$),

$$0 = \int_{\Sigma} \psi(z) F(z) \left(\overline{F'(z)} - \pi z \overline{F(z)} \right) d\mu(z), \quad (6.10)$$

which holds for every holomorphic ψ on a neighborhood of $\bar{\Sigma}$. Equivalently, expanding the product,

$$0 = \int_{\Sigma} \psi(z) \left(\overline{F'(z)} F(z) - \pi z |F(z)|^2 \right) d\mu(z).$$

The form (6.10) is the one used in Step 4.

Step 4: F' is again a first eigenfunction.

This step is the crux of the argument and is the main new ingredient compared to classical Faber–Krahn rigidity proofs. Rather than working with general test functions in the eigenvalue equation, we exploit the divergence identity (6.10) together with a clever choice of test function involving the reproducing kernel of the Fock space. The end result is that *the derivative* F' of any first eigenfunction is itself a first eigenfunction; since the eigenspace is one-dimensional by Step 2, this forces a first-order ODE on F , and hence pins down F up to translation as a Gaussian.

For every $w \in \mathbb{C}$, set

$$K_w(z) := e^{\pi \bar{w} z};$$

this is the reproducing kernel of $\mathcal{F}^2(\mathbb{C})$ at the point w , in the sense that

$$\int_{\mathbb{C}} H(z) e^{\pi \bar{w} z} d\mu(z) = H(w), \quad H \in \mathcal{F}^2(\mathbb{C}); \quad (6.11)$$

see, e.g., [18, Ch. 1] or [4]. Equivalently, $\langle H, K_w \rangle_{\mathcal{F}^2} = H(w)$.

By Step 1, after replacing Σ by the corresponding superlevel-set representative if necessary, we may suppose that

$$\Sigma = \{u_F > c_\Sigma\}.$$

In particular, $u_F \geq c_\Sigma > 0$ on $\bar{\Sigma}$. Since $u_F(z) = |F(z)|^2 e^{-\pi|z|^2}$, it follows that F has no zeros on $\bar{\Sigma}$. By continuity of F and the compactness of $\bar{\Sigma}$, there exists an open neighborhood $V \supset \bar{\Sigma}$ on which F has no zeros. Hence, for every $w \in \mathbb{C}$, the function

$$\psi_w(z) := \frac{K_w(z)}{F(z)} = \frac{e^{\pi \bar{w} z}}{F(z)}$$

is holomorphic on V . Applying (6.10) with $\psi = \psi_w$, so that $\psi_w(z)F(z) = e^{\pi \bar{w} z}$, we obtain

$$0 = \int_{\Sigma} e^{\pi \bar{w} z} \left(\overline{F'(z)} - \pi z \overline{F(z)} \right) d\mu(z),$$

or, separating the two terms,

$$\int_{\Sigma} e^{\pi \bar{w} z} \overline{F'(z)} d\mu(z) = \int_{\Sigma} \pi z e^{\pi \bar{w} z} \overline{F(z)} d\mu(z). \quad (6.12)$$

We now relate the right-hand side of (6.12) to $\overline{F'(w)}$ using the eigenvalue equation (6.4) together with the reproducing identity (6.11). Choosing

$$H(z) := z e^{\pi \bar{w} z} \in \mathcal{F}^2(\mathbb{C})$$

(the product of the kernel K_w by the polynomial z ; both factors preserve membership in $\mathcal{F}^2$) in (6.4), we obtain

$$\int_{\Sigma} z e^{\pi \bar{w} z} \overline{F(z)} d\mu(z) = \lambda_{1,\varphi}(\Sigma) \int_{\mathbb{C}} z e^{\pi \bar{w} z} \overline{F(z)} d\mu(z). \quad (6.13)$$

On the other hand, taking complex conjugates in (6.11) applied to F at the point w , one finds

$$\int_{\mathbb{C}} e^{\pi \bar{w} z} \overline{F(z)} d\mu(z) = \overline{F(w)}.$$

Both sides depend on w through the antiholomorphic variable $\bar{w}$; we differentiate with respect to $\bar{w}$, exchanging differentiation and integration on the left. The interchange is legitimate because, for w in any compact set $\{|w| \leq R\}$, the integrand $\partial_{\bar{w}}(e^{\pi \bar{w} z} \overline{F(z)}) = \pi z e^{\pi \bar{w} z} \overline{F(z)}$ is dominated, in absolute value, by $\pi|z| e^{\pi R|z|} |F(z)|$. The standard pointwise bound for Fock-space functions gives $|F(z)| \leq \|F\|_{\mathcal{F}^2} e^{\pi|z|^2/2}$, so this dominator is bounded by a constant times $|z| e^{\pi R|z|} e^{\pi|z|^2/2}$; after multiplication by the Gaussian weight $e^{-\pi|z|^2}$ of $d\mu$, the resulting integrand $|z| e^{\pi R|z|} e^{-\pi|z|^2/2}$ is integrable on $\mathbb{C}$. Dominated convergence therefore justifies the interchange. We conclude

$$\pi \int_{\mathbb{C}} z e^{\pi \bar{w} z} \overline{F(z)} d\mu(z) = \overline{F'(w)}.$$

Multiplying (6.13) by π and substituting,

$$\pi \int_{\Sigma} z e^{\pi \bar{w} z} \overline{F(z)} d\mu(z) = \lambda_{1,\varphi}(\Sigma) \overline{F'(w)}.$$

Combining this with (6.12),

$$\int_{\Sigma} e^{\pi \bar{w} z} \overline{F'(z)} d\mu(z) = \lambda_{1,\varphi}(\Sigma) \overline{F'(w)}, \quad w \in \mathbb{C}.$$

Taking complex conjugates, this is exactly the eigenvalue equation (6.4) applied with F replaced by F' (and an arbitrary K_w playing the role of the test function). Since the linear span of $\{K_w\}_{w \in \mathbb{C}}$ is dense in $\mathcal{F}^2(\mathbb{C})$, the identity extends to every $H \in \mathcal{F}^2(\mathbb{C})$, and we conclude that F' is a first eigenfunction of the localization operator on Σ , with eigenvalue $\lambda_{1,\varphi}(\Sigma)$.

By the simplicity of $\lambda_{1,\varphi}(\Sigma)$ established in Lemma 6.4,

$$F' = \theta F$$

for some constant $\theta \in \mathbb{C}$. Solving this ODE gives

$$F(z) = C e^{\theta z},$$

for some nonzero constant C .

Step 5: reduction to the constant eigenfunction.

We now use the Weyl invariance of the Bargmann–Fock setting to reduce the analysis to the case $F \equiv 1$. Recall from (2.5) that, for $a \in \mathbb{C}$, the Weyl translation

$$(T_a F)(z) = e^{\pi a z - \frac{\pi}{2} |a|^2} F(z - a)$$

is unitary on $\mathcal{F}^2(\mathbb{C})$ and satisfies $u_{T_a F}(z) = u_F(z - a)$. In particular, T_a maps eigenfunctions of the localization operator on Σ to eigenfunctions of the localization operator on the translated set $\Sigma + a$, with the same eigenvalue, and the quantity $\int_{\Sigma} u_F dA$ is invariant under the simultaneous substitutions $F \mapsto T_a F$ and $\Sigma \mapsto \Sigma + a$.

We claim that we may choose $a \in \mathbb{C}$ so that $T_{-a} F$ is a constant function. Indeed, by Step 4, $F(z) = C e^{\theta z}$ for some $C, \theta \in \mathbb{C}$ with $C \neq 0$. Compute, using the identity $u_{T_{-a} F}(z) = u_F(z + a)$:

$$u_{T_{-a} F}(z) = |F(z + a)|^2 e^{-\pi |z + a|^2} = |C|^2 e^{2\Re(\theta(z + a))} e^{-\pi |z + a|^2}.$$

Expanding,

$$2\Re(\theta(z + a)) - \pi |z + a|^2 = 2\Re(\theta z) + 2\Re(\theta a) - \pi |z|^2 - 2\pi \Re(\bar{a} z) - \pi |a|^2.$$

The linear terms in z on the right cancel if and only if $2\Re(\theta z) = 2\pi \Re(\bar{a} z)$ for every $z \in \mathbb{C}$, which is the condition $\bar{a} = \theta/\pi$, i.e.

$$a := \frac{\bar{\theta}}{\pi}.$$

With this choice,

$$u_{T_{-a} F}(z) = |C'|^2 e^{-\pi |z|^2}, \quad |C'|^2 := |C|^2 e^{2\Re(\theta a) - \pi |a|^2} = |C|^2 e^{|\theta|^2/\pi}.$$

Since $T_{-a} F$ is entire and $|T_{-a} F(z)|^2 e^{-\pi |z|^2} = |C'|^2 e^{-\pi |z|^2}$, it follows that $|T_{-a} F| \equiv |C'|$ is constant; a holomorphic function with constant modulus is constant, hence $T_{-a} F \equiv C''$ for

some $C''' \in \mathbb{C}$ with $|C'''| = |C'|$. Normalizing, we may absorb the unimodular constant into the eigenfunction and assume

$$F \equiv 1$$

after the change $\Sigma \rightsquigarrow \Sigma - a$ (and the corresponding unitary substitution $F \rightsquigarrow T_{-a}F$).

In this normalization the ground state h_0 (corresponding to $F \equiv 1$ in Bargmann–Fock coordinates) is an eigenfunction of the localization operator of Σ . By Step 1 (applied with the new normalization) we have, up to a null set,

$$\Sigma = \{z \in \mathbb{C} : |C'|^2 e^{-\pi|z|^2} > c_\Sigma\}.$$

Since $r \mapsto e^{-\pi r^2}$ is *strictly* decreasing, this superlevel set is automatically a centered open disk; it cannot be an annulus. Thus Σ is, up to a null set, the centered disk D_R with $|D_R| = |\Sigma|$, and Step 6 below records the corresponding quantitative rearrangement statement, which is an instance of the global Faber–Krahn theorem of [40].

Step 6: exclusion of annuli and conclusion.

It remains to verify that, among radial sets of prescribed measure, the only local maximizer is the centered disk. Although this is essentially automatic after Step 5, we record the standard rearrangement argument since it makes explicit *why* no annular local maximizer can exist; this provides the endpoint matching the global Faber–Krahn theorem of [40].

Suppose, more generally, that Σ is any radial set of finite measure which arises as a superlevel set of a radial real-analytic non-constant function on $\mathbb{C}$; write

$$\tilde{\Sigma} := \{r > 0 : re^{it} \in \Sigma \text{ for some } t \in [0, 2\pi)\}.$$

By Step 1, Σ admits an open representative; combined with the real-analytic structure established in Step 3, the boundary $\partial\Sigma$ is a finite union of real-analytic arcs, hence $\tilde{\Sigma} \subset (0, \infty)$ is open and is a finite union of disjoint open intervals:

$$\tilde{\Sigma} = \bigcup_{j=1}^N (a_j, b_j), \quad 0 \leq a_1 < b_1 \leq a_2 < b_2 \leq \cdots \leq a_N < b_N.$$

Then, using $\int r e^{-\pi r^2} dr = -\frac{1}{2\pi} e^{-\pi r^2}$,

$$\int_{\Sigma} e^{-\pi|z|^2} dA(z) = 2\pi \int_{\tilde{\Sigma}} r e^{-\pi r^2} dr = \sum_{j=1}^N (e^{-\pi a_j^2} - e^{-\pi b_j^2}),$$

under the constraint

$$|\Sigma| = 2\pi \int_{\tilde{\Sigma}} r dr = \pi \sum_{j=1}^N (b_j^2 - a_j^2).$$

We are therefore led to the optimization problem

$$\text{maximize} \quad \sum_{j=1}^N (e^{-\pi a_j^2} - e^{-\pi b_j^2}) \quad \text{subject to} \quad \sum_{j=1}^N (b_j^2 - a_j^2) = c, \quad (6.14)$$

where $c = |\Sigma|/\pi$ is fixed. Substituting $s_j := a_j^2$, $t_j := b_j^2$ (with $0 \leq s_1 < t_1 \leq s_2 < \dots$), and writing $\Phi(x) := e^{-\pi x}$, (6.14) becomes

$$\text{maximize} \quad \sum_{j=1}^N (\Phi(s_j) - \Phi(t_j)) \quad \text{subject to} \quad \sum_{j=1}^N (t_j - s_j) = c.$$

The function Φ is strictly convex and strictly decreasing on $[0, \infty)$; its derivative $-\Phi'(\sigma) = \pi e^{-\pi\sigma}$ is strictly decreasing in σ . For any $t > s \geq 0$,

$$\Phi(s) - \Phi(t) = \int_s^t (-\Phi'(\sigma)) d\sigma,$$

so

$$\sum_{j=1}^N (\Phi(s_j) - \Phi(t_j)) = \int_{\tilde{\Sigma}_2} (-\Phi'(\sigma)) d\sigma, \quad \tilde{\Sigma}_2 := \bigcup_{j=1}^N (s_j, t_j).$$

The set $\tilde{\Sigma}_2 \subset [0, \infty)$ has Lebesgue measure equal to c , and the integrand $-\Phi'$ is strictly decreasing in σ . The bathtub principle (Lemma 6.1 applied to $g = -\Phi'$) shows that, among all measurable subsets of $[0, \infty)$ of total Lebesgue measure c , the integral $\int_{\tilde{\Sigma}_2} (-\Phi'(\sigma)) d\sigma$ is strictly maximized by the leftmost interval $(0, c)$. This corresponds to $N = 1$, $s_1 = 0$, $t_1 = c$, i.e.

$$\tilde{\Sigma} = (0, R), \quad R := \sqrt{c},$$

and equivalently Σ is the centered disk D_R with $|D_R| = |\Sigma|$. Any annular configuration ($N \geq 2$, or $N = 1$ with $s_1 > 0$) produces a strictly smaller value of $\int_{\Sigma} u_F dA$, and is therefore not a local maximizer.

We conclude that every local maximizer of $\lambda_{1,\varphi}$ under the measure constraint is, up to translation, a disk. $\square$

7. GLOBAL MAXIMIZERS FOR THE FABER–KRAHN PROBLEM IN THE GAUSSIAN STFT SETTING

In this section we prove existence of global maximizers for the Gaussian time-frequency concentration problem under a measure constraint, and then combine that existence result with the local rigidity theorem from the previous section to recover the disk as the unique maximizer, up to translation. The strategy is by now classical in time-frequency analysis: a profile decomposition in the spirit of P.-L. Lions' concentration-compactness, transplanted to the Fock space via the Bargmann transform, dichotomizes the loss of compactness of a maximizing sequence into the action of the Weyl translation group. A vanishing remainder is then ruled out by an iterative extraction whose maximality forces all the energy to remain captured by the extracted profiles. Closely related arguments have appeared in the time-frequency setting in [39, 38, 19, 46], and the disk-rigidity result we recover is the celebrated theorem of Nicola–Tilli [40, 41].

Since the Bargmann transform is unitary, the variational problem may be written on the Fock space side as

$$M(s) := \sup_{|\Omega|=s} \lambda_{1,\varphi}(\Omega) = \sup_{\|F\|_{\mathcal{F}^2}=1} I_F(s).$$

Here, and throughout the section, we exploit the translation invariance (2.8) together with the pointwise identity (2.7) (cf. also (2.6)), the basic regularity properties recorded in Lemma 2.5, and the elementary properties of the Fock space gathered in Lemma 2.1.

We begin with a Weyl-orthogonality lemma that captures the asymptotic decoupling of two functions whose translation parameters drift apart. This plays the role of *Pythagorean orthogonality at infinity* on the Fock space.

Lemma 7.1. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$, and let $\{a_k\} \subset \mathbb{C}$ satisfy $|a_k| \rightarrow \infty$. Then*

$$T_{a_k}G \rightharpoonup 0 \quad \text{weakly in } \mathcal{F}^2(\mathbb{C}),$$

and in particular

$$\langle F, T_{a_k}G \rangle_{\mathcal{F}^2} \rightarrow 0.$$

Consequently,

$$\|F + T_{a_k}G\|_{\mathcal{F}^2}^2 = \|F\|_{\mathcal{F}^2}^2 + \|G\|_{\mathcal{F}^2}^2 + o(1).$$

More generally, if $|a_k - b_k| \rightarrow \infty$, then

$$\langle T_{a_k}F, T_{b_k}G \rangle_{\mathcal{F}^2} \rightarrow 0.$$

Proof. Using unitarity of the Weyl operators T_a on $\mathcal{F}^2(\mathbb{C})$ (cf. (2.5)),

$$\langle F, T_{a_k}G \rangle_{\mathcal{F}^2} = \langle T_{-a_k}F, G \rangle_{\mathcal{F}^2}.$$

Now $T_{-a_k}F$ is bounded in $\mathcal{F}^2$ by unitarity, and the identity (2.7) (equivalently (2.6)) yields

$$u_{T_{-a_k}F}(z) = u_F(z + a_k) \quad (z \in \mathbb{C}),$$

so that the densities $u_{T_{-a_k}F}$ drift uniformly to infinity. Equivalently, since $u_F \in C_0(\mathbb{C})$ by Lemma 2.1(iii), for every compact $K \subset \mathbb{C}$,

$$\sup_{z \in K} |T_{-a_k}F(z)| e^{-\pi|z|^2/2} = \sup_{z \in K} u_{T_{-a_k}F}(z)^{1/2} = \sup_{z \in K} u_F(z + a_k)^{1/2} \rightarrow 0.$$

Hence $T_{-a_k}F \rightarrow 0$ locally uniformly. Combining this with the boundedness of $\{T_{-a_k}F\}$ in $\mathcal{F}^2(\mathbb{C})$, and using the fact that the linear span of the reproducing kernels $\{K_w\}_{w \in \mathbb{C}}$ is dense in $\mathcal{F}^2$, we conclude that $T_{-a_k}F \rightharpoonup 0$ weakly in $\mathcal{F}^2$; in particular $\langle T_{-a_k}F, G \rangle \rightarrow 0$. This proves the first claim.

For the norm decoupling, expand the square:

$$\|F + T_{a_k}G\|_{\mathcal{F}^2}^2 = \|F\|_{\mathcal{F}^2}^2 + \|T_{a_k}G\|_{\mathcal{F}^2}^2 + 2\Re \langle F, T_{a_k}G \rangle_{\mathcal{F}^2} = \|F\|_{\mathcal{F}^2}^2 + \|G\|_{\mathcal{F}^2}^2 + o(1),$$

where we used unitarity of T_{a_k} and the cross-term decay just proved.

Finally, the general case reduces to the first by writing

$$\langle T_{a_k}F, T_{b_k}G \rangle_{\mathcal{F}^2} = \langle F, T_{-a_k}T_{b_k}G \rangle_{\mathcal{F}^2} = e^{i\theta_k} \langle F, T_{b_k - a_k}G \rangle_{\mathcal{F}^2},$$

for some real phase θ_k coming from the cocycle of the Weyl representation; since this phase factor has modulus one, the conclusion follows from $|b_k - a_k| \rightarrow \infty$. $\square$

The next lemma compares the density of a finite sum of widely separated translates with the density of each individual translate. The precise statement is what makes the profile decomposition compatible with the variational quantities $I_F(s)$, since these depend only on the densities and not on the phases of the underlying functions.

Lemma 7.2. *Fix $J \geq 1$, let $G^{(1)}, \dots, G^{(J)} \in \mathcal{F}^2(\mathbb{C})$, and let $\{a_k^{(j)}\}_{k \geq 1} \subset \mathbb{C}$ for $1 \leq j \leq J$ satisfy*

$$|a_k^{(j)} - a_k^{(\ell)}| \rightarrow \infty \quad \text{whenever } j \neq \ell.$$

For each k , set

$$S_k := \sum_{j=1}^J T_{a_k^{(j)}} G^{(j)}.$$

Then the following hold.

(i) For every $R > 0$ and every $1 \leq j \leq J$,

$$\sup_{z \in D_R(a_k^{(j)})} |u_{S_k}(z) - u_{T_{a_k^{(j)}} G^{(j)}}(z)| \rightarrow 0 \quad (k \rightarrow \infty).$$

(ii) For every $R > 0$, if

$$E_{k,R} := \mathbb{C} \setminus \bigcup_{j=1}^J D_R(a_k^{(j)}),$$

then

$$\limsup_{k \rightarrow \infty} \sup_{z \in E_{k,R}} u_{S_k}(z) \leq \left(\sum_{j=1}^J \sup_{\zeta \in \mathbb{C} \setminus D_R} u_{G^{(j)}}(\zeta)^{1/2} \right)^2.$$

In particular, given $\varepsilon > 0$, there exists $R_\varepsilon > 0$ such that for every $R \geq R_\varepsilon$,

$$\sup_{z \in E_{k,R}} u_{S_k}(z) \leq \varepsilon$$

for all sufficiently large k .

Proof. Throughout the proof we use repeatedly the identity

$$u_{T_a F}(z) = u_F(z - a), \quad z, a \in \mathbb{C},$$

which is (2.7) (equivalently (2.6)); together with the elementary subadditivity

$$u_{F+G}(z)^{1/2} \leq u_F(z)^{1/2} + u_G(z)^{1/2}, \quad z \in \mathbb{C},$$

which follows from the definition $u_F = |F|^2 e^{-\pi|z|^2}$ and the triangle inequality applied to $|F+G| \leq |F| + |G|$.

Proof of (i). Fix $R > 0$ and $j \in \{1, \dots, J\}$. Write

$$S_k = T_{a_k^{(j)}} G^{(j)} + H_{k,j}, \quad H_{k,j} := \sum_{\ell \neq j} T_{a_k^{(\ell)}} G^{(\ell)}.$$

For every $z \in \mathbb{C}$, the elementary inequality $||a+b|^2 - |a|^2| \leq 2|a||b| + |b|^2$ yields

$$|u_{S_k}(z) - u_{T_{a_k^{(j)}} G^{(j)}}(z)| = e^{-\pi|z|^2} ||T_{a_k^{(j)}} G^{(j)}(z) + H_{k,j}(z)|^2 - |T_{a_k^{(j)}} G^{(j)}(z)|^2|$$

$$\leq 2 u_{T_{a_k^{(j)}} G^{(j)}}(z)^{1/2} u_{H_{k,j}}(z)^{1/2} + u_{H_{k,j}}(z).$$

By Lemma 2.1(ii),

$$u_{T_{a_k^{(j)}} G^{(j)}}(z) \leq \|G^{(j)}\|_{\mathcal{F}^2}^2 \quad \text{for every } z \in \mathbb{C} \text{ and every } k.$$

Thus, to prove (i) it is enough to show that

$$\sup_{z \in D_R(a_k^{(j)})} u_{H_{k,j}}(z) \rightarrow 0.$$

For $z \in D_R(a_k^{(j)})$ and $\ell \neq j$,

$$z - a_k^{(\ell)} \in D_R(a_k^{(j)} - a_k^{(\ell)}).$$

Since $|a_k^{(j)} - a_k^{(\ell)}| \rightarrow \infty$,

$$\text{dist}(D_R(a_k^{(j)} - a_k^{(\ell)}), 0) \geq |a_k^{(j)} - a_k^{(\ell)}| - R \rightarrow \infty.$$

Because $u_{G^{(\ell)}} \in C_0(\mathbb{C})$ by Lemma 2.1(iii),

$$\sup_{z \in D_R(a_k^{(j)})} u_{T_{a_k^{(\ell)}} G^{(\ell)}}(z) = \sup_{\zeta \in D_R(a_k^{(j)} - a_k^{(\ell)})} u_{G^{(\ell)}}(\zeta) \xrightarrow{k \rightarrow \infty} 0.$$

The subadditivity of $u_F^{1/2}$ yields

$$u_{H_{k,j}}(z)^{1/2} \leq \sum_{\ell \neq j} u_{T_{a_k^{(\ell)}} G^{(\ell)}}(z)^{1/2},$$

so the preceding convergence implies

$$\sup_{z \in D_R(a_k^{(j)})} u_{H_{k,j}}(z)^{1/2} \rightarrow 0.$$

This proves (i).

Proof of (ii). Fix $R > 0$ and $z \in E_{k,R}$. By definition, $z \notin D_R(a_k^{(j)})$ for every j , and hence $|z - a_k^{(j)}| \geq R$. Therefore

$$u_{T_{a_k^{(j)}} G^{(j)}}(z) = u_{G^{(j)}}(z - a_k^{(j)}) \leq \sup_{\zeta \in \mathbb{C} \setminus D_R} u_{G^{(j)}}(\zeta).$$

Using again the subadditivity of $u_F^{1/2}$,

$$u_{S_k}(z)^{1/2} \leq \sum_{j=1}^J u_{T_{a_k^{(j)}} G^{(j)}}(z)^{1/2} \leq \sum_{j=1}^J \sup_{\zeta \in \mathbb{C} \setminus D_R} u_{G^{(j)}}(\zeta)^{1/2}.$$

Squaring yields

$$u_{S_k}(z) \leq \left(\sum_{j=1}^J \sup_{\zeta \in \mathbb{C} \setminus D_R} u_{G^{(j)}}(\zeta)^{1/2} \right)^2.$$

Taking the supremum over $z \in E_{k,R}$ proves the displayed estimate in (ii). Since each $u_{G^{(j)}} \in C_0(\mathbb{C})$, the right-hand side tends to 0 as $R \rightarrow \infty$; the final claim follows by choosing R_ε so that the displayed bracket is at most $\sqrt{\varepsilon}$. $\square$

The next lemma is a robust uniform-density perturbation estimate. Concretely, it says that adding to a bounded sequence in $\mathcal{F}^2(\mathbb{C})$ a residue whose density vanishes uniformly leaves the density essentially unchanged in L^∞ . This will be applied with S_k the truncated profile sum and R_k the residual of the profile decomposition.

Lemma 7.3. *Let $\{S_k\} \subset \mathcal{F}^2(\mathbb{C})$ be bounded and let $\{R_k\} \subset \mathcal{F}^2(\mathbb{C})$ satisfy*

$$\|u_{R_k}\|_{L^\infty(\mathbb{C})} \rightarrow 0.$$

Then

$$\|u_{S_k+R_k} - u_{S_k}\|_{L^\infty(\mathbb{C})} \rightarrow 0.$$

More precisely, if $\sup_k \|S_k\|_{\mathcal{F}^2} \leq M$, then

$$\|u_{S_k+R_k} - u_{S_k}\|_{L^\infty(\mathbb{C})} \leq 2M \|u_{R_k}\|_{L^\infty(\mathbb{C})}^{1/2} + \|u_{R_k}\|_{L^\infty(\mathbb{C})}.$$

Proof. For every $z \in \mathbb{C}$,

$$\begin{aligned} |u_{S_k+R_k}(z) - u_{S_k}(z)| &= e^{-\pi|z|^2} ||S_k(z) + R_k(z)|^2 - |S_k(z)|^2| \\ &\leq 2|S_k(z)|^{1/2} |R_k(z)|^{1/2} + |R_k(z)|, \end{aligned}$$

again by the elementary inequality $||a+b|^2 - |a|^2| \leq 2|a||b| + |b|^2$. By Lemma 2.1(ii),

$$u_{S_k}(z) \leq \|S_k\|_{\mathcal{F}^2}^2 \leq M^2 \quad (z \in \mathbb{C}).$$

Hence

$$|u_{S_k+R_k}(z) - u_{S_k}(z)| \leq 2M \|u_{R_k}\|_{L^\infty}^{1/2} + \|u_{R_k}\|_{L^\infty}.$$

Taking the supremum over z yields the claim. $\square$

The next lemma supplies the soft input on the function $s \mapsto I_F(s)$ that will be used at the end of the variational argument. It is essentially a consequence of the bathtub principle and the basic regularity of u_F , already noted in Lemma 2.5.

Lemma 7.4. *Let $F \in \mathcal{F}^2(\mathbb{C})$. Then the function*

$$s \mapsto I_F(s)$$

is finite, nondecreasing, and continuous on $[0, \infty)$.

Proof. By Lemma 2.1(ii)–(iii) together with the identity $\int_{\mathbb{C}} u_F dA = \|F\|_{\mathcal{F}^2}^2$, we have $u_F \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$, so $I_F(s)$ is finite for every $s \geq 0$ and bounded above by $\|F\|_{\mathcal{F}^2}^2$. Monotonicity is immediate from the definition of $I_F(s)$ as a supremum over $|\Omega| = s$: if $s_1 \leq s_2$, every admissible set for s_1 can be enlarged to a set of measure s_2 without decreasing the integral, since $u_F \geq 0$.

To prove continuity, write $u := u_F$ and let u^* denote the decreasing rearrangement of u on $[0, \infty)$. Since $u \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$, $u^* \in L^1([0, \infty))$ and is equimeasurable with u . The bathtub principle yields

$$I_F(s) = \int_0^s u^*(\tau) d\tau, \quad s \geq 0.$$

The right-hand side is the integral of an $L^1_{\text{loc}}([0, \infty))$ function over $[0, s]$ and is therefore absolutely continuous, hence continuous, in s . $\square$

The next result is the profile decomposition adapted to the present setting, in the spirit of P.-L. Lions' concentration-compactness method. Although the overall scheme is by now standard, we spell out the points used in the sequel; the version stated here is a Fock-space variant of profile decompositions used in time-frequency analysis, see e.g. [39, 38, 19, 46].

Remark 7.1 (Translation non-compactness on $\mathcal{F}^2$). Although the Weyl translations T_a act unitarily on $\mathcal{F}^2(\mathbb{C})$, the family $\{T_a\}_{a \in \mathbb{C}}$ is not relatively compact in the strong operator topology: by Lemma 7.1, for every nonzero $F \in \mathcal{F}^2$ and every sequence $\{a_k\} \subset \mathbb{C}$ with $|a_k| \rightarrow \infty$ one has $T_{a_k}F \rightharpoonup 0$ weakly, while the norm $\|T_{a_k}F\|_{\mathcal{F}^2} = \|F\|_{\mathcal{F}^2}$ is preserved. Consequently, a bounded maximizing sequence in $\mathcal{F}^2$ need not admit any strongly convergent subsequence, even after factoring out the Weyl translation group. The profile decomposition stated below is the device that quantifies this non-compactness: it identifies all directions along which mass can escape, by extracting a (possibly infinite) family of nontrivial profiles $G^{(j)}$ translated by mutually divergent centers $a_k^{(j)}$. It is essential to stress that, since the $\mathcal{F}^2$ -norm is invariant under T_a , the remainder $R_k^{(J)}$ does *not* converge to zero in $\mathcal{F}^2$: the Pythagorean identity (iii) below merely redistributes mass between the profiles and the remainder, and pure $\mathcal{F}^2$ -vanishing of the remainder is incompatible with the unitarity of T_a when several nontrivial profiles are present. The natural and sharp vanishing statement for the remainder is therefore the dispersive estimate (iv) below, expressed through the density $u_{R_k^{(J)}}$; this is exactly the quantity that controls all the localization functionals $I_F(s)$ used in the sequel, which depend only on the density u_F and not on the phase of F . The exposition below extracts what is needed in the sequel; an analogous decomposition with dispersive-style remainders in the Fock setting is discussed in [39, 38, 19, 46].

Proposition 7.1 (Profile decomposition in the Fock space). *Let $\{F_k\}_{k \geq 1} \subset \mathcal{F}^2(\mathbb{C})$ be a bounded sequence. Then, after passing to a subsequence, there exist profiles*

$$G^{(j)} \in \mathcal{F}^2(\mathbb{C}), \quad j \geq 1,$$

and sequences of points

$$a_k^{(j)} \in \mathbb{C}, \quad j \geq 1, \quad k \geq 1,$$

such that:

(i) *for each $j \neq \ell$,*

$$|a_k^{(j)} - a_k^{(\ell)}| \rightarrow \infty \quad (k \rightarrow \infty);$$

(ii) *for every $J \geq 1$, if*

$$R_k^{(J)} := F_k - \sum_{j=1}^J T_{a_k^{(j)}} G^{(j)},$$

then

$$T_{-a_k^{(j)}} R_k^{(J)} \rightarrow 0 \quad \text{locally uniformly on } \mathbb{C}$$

for every $1 \leq j \leq J$;

(iii) *for every $J \geq 1$,*

$$\|F_k\|_{\mathcal{F}^2}^2 = \sum_{j=1}^J \|G^{(j)}\|_{\mathcal{F}^2}^2 + \|R_k^{(J)}\|_{\mathcal{F}^2}^2 + o_k(1);$$

(iv)

$$\lim_{J \rightarrow \infty} \left(\limsup_{k \rightarrow \infty} \|u_{R_k^{(J)}}\|_{L^\infty(\mathbb{C})} \right) = 0.$$

Proof. We argue by the standard iterative extraction scheme, but spell out the points used in the rest of the section. Throughout, we say that a function in $\mathcal{F}^2(\mathbb{C})$ is *nontrivial* if it is not identically zero.

Extraction of the first profile. If

$$\limsup_{k \rightarrow \infty} \|u_{F_k}\|_{L^\infty(\mathbb{C})} = 0,$$

there is nothing to prove: take all profiles equal to zero, all centers arbitrary (say zero), and observe that (i)–(iii) hold trivially while (iv) holds by hypothesis. Otherwise, after passing to a subsequence, choose $a_k^{(1)} \in \mathbb{C}$ such that

$$u_{F_k}(a_k^{(1)}) \geq \frac{1}{2} \|u_{F_k}\|_{L^\infty(\mathbb{C})}.$$

By (2.8) and the density-translation identity (2.7), the sequence $T_{-a_k^{(1)}} F_k$ has the same Fock norm and the corresponding density translated; relabelling, we may assume $a_k^{(1)} = 0$. Since $\{F_k\}$ is bounded in $\mathcal{F}^2(\mathbb{C})$, the reproducing-kernel bound $|F_k(z)| \leq \|F_k\|_{\mathcal{F}^2} e^{\pi|z|^2/2}$ (Lemma 2.3(ii)) makes $\{F_k\}$ a normal family of entire functions on every compact subset of $\mathbb{C}$. Passing to a subsequence and applying Montel's theorem, $F_k \rightarrow G^{(1)}$ locally uniformly on $\mathbb{C}$, with $G^{(1)} \in \mathcal{F}^2(\mathbb{C})$ by Fatou's lemma. By the choice of $a_k^{(1)} = 0$,

$$u_{G^{(1)}}(0) = \lim_{k \rightarrow \infty} u_{F_k}(0) \geq \frac{1}{2} \limsup_{k \rightarrow \infty} \|u_{F_k}\|_{L^\infty(\mathbb{C})} > 0,$$

so $G^{(1)}$ is nontrivial. Set

$$R_k^{(1)} := F_k - G^{(1)}.$$

Iteration. Suppose profiles $G^{(1)}, \dots, G^{(J-1)}$ and centers $a_k^{(1)}, \dots, a_k^{(J-1)}$ have been extracted, with remainder $R_k^{(J-1)}$. If

$$\limsup_{k \rightarrow \infty} \|u_{R_k^{(J-1)}}\|_{L^\infty(\mathbb{C})} = 0,$$

we stop, set $G^{(j)} = 0$ and $a_k^{(j)} = 0$ for all $j \geq J$, and properties (i)–(iv) are easily seen to hold. Otherwise choose $a_k^{(J)} \in \mathbb{C}$ so that

$$u_{R_k^{(J-1)}}(a_k^{(J)}) \geq \frac{1}{2} \|u_{R_k^{(J-1)}}\|_{L^\infty(\mathbb{C})}.$$

Property (i) at this stage (proved below) forces $|a_k^{(J)} - a_k^{(m)}| \rightarrow \infty$ for every $m < J$. Granted this, the sequence $\{T_{-a_k^{(J)}} R_k^{(J-1)}\}$ is bounded in $\mathcal{F}^2$ (by unitarity of $T_{-a_k^{(J)}}$ together with (iii) at stage $J-1$), so after passing to a further subsequence and applying Montel's theorem, it converges locally uniformly to some $G^{(J)} \in \mathcal{F}^2(\mathbb{C})$. By the choice of $a_k^{(J)}$,

$$u_{G^{(J)}}(0) \geq \frac{1}{2} \limsup_{k \rightarrow \infty} \|u_{R_k^{(J-1)}}\|_{L^\infty(\mathbb{C})} > 0,$$

so $G^{(J)}$ is nontrivial. Set

$$R_k^{(J)} := R_k^{(J-1)} - T_{a_k^{(J)}} G^{(J)}.$$

Iterating either terminates at some finite step (in which case (iv) is immediate) or produces an infinite sequence of nontrivial profiles. By a standard diagonal extraction we obtain a single subsequence (still denoted $\{F_k\}$) for which all the partial decompositions are valid.

Proof of (i). Fix $j < \ell$, and suppose for contradiction that $|a_k^{(\ell)} - a_k^{(j)}|$ does not tend to infinity. After passing to a subsequence, $|a_k^{(\ell)} - a_k^{(j)}| \leq C$ for some constant C . At the ℓ -th step, by construction

$$T_{-a_k^{(\ell)}} R_k^{(\ell-1)} \rightarrow G^{(\ell)} \quad \text{locally uniformly on } \mathbb{C},$$

with $G^{(\ell)}$ nontrivial. On the other hand, property (ii) verified up to stage $\ell - 1$ (proved below) gives

$$T_{-a_k^{(j)}} R_k^{(\ell-1)} \rightarrow 0 \quad \text{locally uniformly on } \mathbb{C}.$$

The Weyl identity $T_{-a_k^{(\ell)}} = e^{i\theta_k} T_{-a_k^{(j)}} T_{a_k^{(j)} - a_k^{(\ell)}}$ (for an appropriate real phase θ_k coming from the Weyl co-cycle, of modulus one) shows that

$$T_{-a_k^{(\ell)}} R_k^{(\ell-1)}(z) = e^{i\theta_k} (T_{-a_k^{(j)}} R_k^{(\ell-1)})(z - (a_k^{(\ell)} - a_k^{(j)})).$$

Since $|a_k^{(\ell)} - a_k^{(j)}| \leq C$, the points $z - (a_k^{(\ell)} - a_k^{(j)})$ remain in a fixed compact set as z varies over a compact set; so the local uniform convergence of $T_{-a_k^{(j)}} R_k^{(\ell-1)}$ to 0 forces $T_{-a_k^{(\ell)}} R_k^{(\ell-1)} \rightarrow 0$ locally uniformly as well. This contradicts the nontriviality of $G^{(\ell)}$. Hence

$$|a_k^{(j)} - a_k^{(\ell)}| \rightarrow \infty \quad \text{whenever } j \neq \ell.$$

Proof of (ii). Fix $J \geq 1$ and $1 \leq j \leq J$. By construction, at the j -th extraction step,

$$T_{-a_k^{(j)}} R_k^{(j-1)} \rightarrow G^{(j)} \quad \text{locally uniformly on } \mathbb{C}.$$

Subtracting the j -th extracted profile,

$$T_{-a_k^{(j)}} R_k^{(j)} = T_{-a_k^{(j)}} R_k^{(j-1)} - G^{(j)} \rightarrow 0 \quad \text{locally uniformly on } \mathbb{C}.$$

For $J > j$, since

$$R_k^{(J)} = R_k^{(j)} - \sum_{\ell=j+1}^J T_{a_k^{(\ell)}} G^{(\ell)},$$

the linearity of the Weyl operators yields

$$T_{-a_k^{(j)}} R_k^{(J)} = T_{-a_k^{(j)}} R_k^{(j)} - \sum_{\ell=j+1}^J e^{i\theta_{k,\ell}} T_{a_k^{(\ell)} - a_k^{(j)}} G^{(\ell)}$$

for some real phases $\theta_{k,\ell}$ coming from the Weyl co-cycle. The first term tends to 0 locally uniformly by the previous paragraph. For each fixed $\ell > j$, part (i) gives $|a_k^{(\ell)} - a_k^{(j)}| \rightarrow \infty$, and the identity $u_{T_b G}(z) = u_G(z - b)$ (i.e. (2.7)) implies that $T_{a_k^{(\ell)} - a_k^{(j)}} G^{(\ell)} \rightarrow 0$ locally uniformly in modulus (its density is $u_{G^{(\ell)}}(z - (a_k^{(\ell)} - a_k^{(j)})) \rightarrow 0$ for z in a compact set, since $u_{G^{(\ell)}} \in C_0(\mathbb{C})$). Thus

$$T_{-a_k^{(j)}} R_k^{(J)} \rightarrow 0 \quad \text{locally uniformly on } \mathbb{C},$$

which is exactly (ii).

Proof of (iii). For fixed $J \geq 1$,

$$F_k = \sum_{j=1}^J T_{a_k^{(j)}} G^{(j)} + R_k^{(J)}.$$

Expanding the squared $\mathcal{F}^2$ -norm and using Lemma 7.1 together with the pairwise divergence of the translation parameters from (i),

$$\left\| \sum_{j=1}^J T_{a_k^{(j)}} G^{(j)} \right\|_{\mathcal{F}^2}^2 = \sum_{j=1}^J \|G^{(j)}\|_{\mathcal{F}^2}^2 + o_k(1).$$

Moreover, since

$$T_{-a_k^{(m)}} R_k^{(J)} \rightarrow 0 \quad \text{locally uniformly on } \mathbb{C}$$

for every $1 \leq m \leq J$ by (ii), and the sequence $\{T_{-a_k^{(m)}} R_k^{(J)}\}_k$ is bounded in $\mathcal{F}^2$, the same weak-convergence argument as in Lemma 7.1 yields

$$\langle T_{a_k^{(m)}} G^{(m)}, R_k^{(J)} \rangle_{\mathcal{F}^2} = \langle G^{(m)}, T_{-a_k^{(m)}} R_k^{(J)} \rangle_{\mathcal{F}^2} \rightarrow 0,$$

for every fixed m . Expanding $\|F_k\|_{\mathcal{F}^2}^2$ and collecting terms,

$$\|F_k\|_{\mathcal{F}^2}^2 = \sum_{j=1}^J \|G^{(j)}\|_{\mathcal{F}^2}^2 + \|R_k^{(J)}\|_{\mathcal{F}^2}^2 + o_k(1).$$

In particular, since $\|F_k\|_{\mathcal{F}^2}$ is bounded and the terms on the right are nonnegative, summing the profile-norm contributions over j yields the energy bound

$$\sum_{j \geq 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 \leq \limsup_{k \rightarrow \infty} \|F_k\|_{\mathcal{F}^2}^2. \quad (7.1)$$

This is the key compactness ingredient that controls the iteration and ensures (iv).

Proof of (iv). Set

$$\eta_J := \limsup_{k \rightarrow \infty} \|u_{R_k^{(J)}}\|_{L^\infty(\mathbb{C})},$$

which is monotone nonincreasing in J by construction (since $R_k^{(J+1)}$ is obtained from $R_k^{(J)}$ by extracting one further nontrivial profile). Let $\eta_\infty := \lim_{J \rightarrow \infty} \eta_J$; we must show $\eta_\infty = 0$. Suppose, for contradiction, that $\eta_\infty \geq \delta > 0$.

By the iteration, at each stage J such that $\eta_J \geq \delta$, the extracted profile $G^{(J+1)}$ satisfies $u_{G^{(J+1)}}(0) \geq \delta/2$ by construction. By Lemma 2.1(ii),

$$\|G^{(J+1)}\|_{\mathcal{F}^2}^2 \geq u_{G^{(J+1)}}(0) \geq \delta/2.$$

If the iteration produced infinitely many such stages (which is the case under the assumption $\eta_\infty \geq \delta$), then

$$\sum_{j \geq 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 = \infty,$$

contradicting (7.1). Hence $\eta_\infty = 0$, proving (iv). $\square$

The profile decomposition implies an asymptotic decoupling of the localization functional $I_{F_k}(s)$ along the profiles, which we record here for later use. This is the precise sense in which the functional $I_F(s)$ “splits” across mutually divergent translates, even though the underlying profiles themselves do not separate in $\mathcal{F}^2$.

Lemma 7.5 (Decoupling of the localization functional). *Let $\{F_k\} \subset \mathcal{F}^2(\mathbb{C})$ be bounded, and let $\{G^{(j)}\}_{j \geq 1}$ and $\{a_k^{(j)}\}$ be the profiles and centers extracted by Proposition 7.1. For every $s > 0$ one has, after passing to a subsequence,*

$$\limsup_{k \rightarrow \infty} I_{F_k}(s) \leq \lim_{J \rightarrow \infty} \sup_{\substack{s_1 + \dots + s_J \leq s \\ s_j \geq 0}} \sum_{j=1}^J I_{G^{(j)}}(s_j). \quad (7.2)$$

Conversely, for every $J \geq 1$ and every $(s_1, \dots, s_J)$ with $s_j \geq 0$ and $\sum_j s_j \leq s$,

$$\liminf_{k \rightarrow \infty} I_{F_k}(s) \geq \sum_{j=1}^J I_{G^{(j)}}(s_j). \quad (7.3)$$

Sketch of proof. The upper bound (7.2) is exactly the content of Step 1 in the proof of Theorem 9 below: by Proposition 7.1(iv) and Lemma 7.3, the density u_{F_k} agrees, up to an L^∞ error that vanishes as $J \rightarrow \infty$, with the density of the truncated profile sum $\sum_{j=1}^J T_{a_k^{(j)}} G^{(j)}$; on the latter, the pairwise divergence of the centers (Proposition 7.1(i)) and Lemma 7.2 force the contribution to $I_{F_k}(s)$ coming from any maximizing superlevel set $\Omega_k \subset \mathbb{C}$ to split into pieces $E_{k,j} \subset D_R(a_k^{(j)})$ of measures $s_{k,j} \rightarrow s_j$ with $\sum_j s_j \leq s$, plus an exterior remainder of measure $s - \sum_j s_j$ on which $u_{F_k} \rightarrow 0$ uniformly. The lower bound (7.3) is obtained by taking, for each j , a near-optimal set $E^{(j)}$ of measure s_j for the profile $G^{(j)}$ and using its translate $E^{(j)} + a_k^{(j)}$ as a competitor for F_k ; the pairwise divergence of the centers and Lemma 7.2(i) ensure that $\bigcup_j (E^{(j)} + a_k^{(j)})$ is admissible (its pieces become disjoint for k large) and that the integrals decouple to leading order. $\square$

The decoupling (7.2)–(7.3) immediately yields the variational subadditivity of the global concentration function $M(s)$, which will rule out dichotomy in the proof of existence below.

Lemma 7.6 (Subadditivity and strict concavity of M). *The function $M : [0, \infty) \rightarrow [0, \infty)$ defined by*

$$M(s) := \sup_{\|F\|_{\mathcal{F}^2}=1} I_F(s) = \sup_{|\Omega|=s} \lambda_{1,\varphi}(\Omega)$$

satisfies the following properties:

- (i) $M(0) = 0$, M is nondecreasing and continuous on $[0, \infty)$, and $M(s) < 1$ for every $s \in (0, \infty)$, with $M(s) \rightarrow 1$ as $s \rightarrow \infty$;
- (ii) (Subadditivity) for every $s_1, s_2 \geq 0$,

$$M(s_1 + s_2) \leq M(s_1) + M(s_2); \quad (7.4)$$

- (iii) (Strict subadditivity / strict concavity) for every $s_1, s_2 > 0$,

$$M(s_1 + s_2) < M(s_1) + M(s_2), \quad (7.5)$$

and M is strictly concave on $[0, \infty)$.

Proof. (i) $M(0) = 0$ is immediate, monotonicity follows from the definition (a larger admissible class yields a larger supremum), and the strict bound $M(s) < 1$ for $s \in (0, \infty)$ follows from $\|u_F\|_{L^1(\mathbb{C})} = \|F\|_{\mathcal{F}^2}^2 = 1$ together with the fact that $u_F \in C_0(\mathbb{C})$, so that no superlevel set of finite measure can saturate the full L^1 -mass; the limit $M(s) \rightarrow 1$ as $s \rightarrow \infty$ follows from dominated convergence applied to a fixed $F \in \mathcal{F}^2(\mathbb{C})$. Continuity follows from Lemma 7.4 together with the bathtub-principle representation $I_F(s) = \int_0^s u_F^*(\tau) d\tau$ and a standard equicontinuity argument.

(iii) (We prove (iii) first, since it gives (ii) as a consequence.) Set

$$m(\tau) := \sup_{\|F\|_{\mathcal{F}^2}=1} u_F^*(\tau), \quad \tau \geq 0,$$

where u_F^* denotes the decreasing rearrangement of u_F on $[0, \infty)$. For each fixed F the function u_F^* is nonincreasing, and the bathtub-principle identity used in Lemma 7.4 yields

$$I_F(s) = \int_0^s u_F^*(\tau) d\tau, \quad M(s) = \sup_{\|F\|_{\mathcal{F}^2}=1} \int_0^s u_F^*(\tau) d\tau.$$

A standard exchange-of-supremum argument (see, e.g., [40, 39]) shows that

$$M(s) = \int_0^s m(\tau) d\tau, \tag{7.6}$$

and that m is *strictly* decreasing on $[0, \infty)$: indeed, $m(\tau) > 0$ for every finite τ (test with the Gaussian $F \equiv 1$, whose density $u_F = e^{-\pi|z|^2}$ has rearrangement $u_F^*(\tau) = e^{-\tau}$), and $m(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$ since each $u_F \in C_0(\mathbb{C})$. Hence M is the integral of a strictly decreasing strictly positive function on $(0, \infty)$, which makes M strictly concave on $[0, \infty)$. Strict subadditivity (7.5) follows from $M(0) = 0$ together with the classical fact that any strictly concave function vanishing at the origin is strictly subadditive on $(0, \infty)$.

(ii) Inequality (7.4) is the non-strict version of (iii) and follows directly from (7.6) and the monotonicity of m :

$$M(s_1+s_2) = \int_0^{s_1+s_2} m(\tau) d\tau = \int_0^{s_1} m(\tau) d\tau + \int_{s_1}^{s_1+s_2} m(\tau) d\tau \leq \int_0^{s_1} m(\tau) d\tau + \int_0^{s_2} m(\tau) d\tau = M(s_1) + M(s_2).$$

□

Remark 7.2 (Ruling out dichotomy). The strict subadditivity (7.5) is the classical Lions criterion for ruling out dichotomy in the concentration-compactness alternative. Concretely, if a (subsequential) profile decomposition of a maximizing sequence $\{F_k\}$ carried mass across two or more nontrivial profiles $G^{(j)}$ of positive masses $\|G^{(j)}\|_{\mathcal{F}^2}^2 = t_j > 0$ with $\sum_j t_j \leq 1$, then by the homogeneity $I_{cG}(\sigma) = |c|^2 I_G(\sigma)$ and the decoupling (7.2), one would have

$$M(s) = \lim_{k \rightarrow \infty} I_{F_k}(s) \leq \sum_j t_j M(\sigma_j/t_j)$$

for some $\sigma_j \geq 0$ with $\sum_j \sigma_j \leq s$. Strict concavity (7.5) of M , applied to the convex combination of points σ_j/t_j with weights t_j , then gives $\sum_j t_j M(\sigma_j/t_j) < M(\sum_j \sigma_j) \leq M(s)$ unless the decomposition is trivial, i.e. a single profile carries all the $\mathcal{F}^2$ -mass and all the prescribed measure s . This is the precise mechanism by which dichotomy is excluded in Step 2 of the proof of Theorem 9 below: *every* mass splitting across two or more nontrivial profiles in Proposition 7.1

loses energy at the variational level, so only a single profile can carry the full $\mathcal{F}^2$ -mass of a maximizer.

We now turn to the global problem. The proof of existence consists of combining the profile decomposition with the elementary continuity properties of $I_F(s)$, and using the homogeneity $I_{cF}(s) = |c|^2 I_F(s)$ recorded in Lemma 2.5(ii) to redistribute mass to a single nontrivial profile.

Theorem 9. *For every $s > 0$, there exists $F_* \in \mathcal{F}^2(\mathbb{C})$ with $\|F_*\|_{\mathcal{F}^2} = 1$ such that*

$$I_{F_*}(s) = M(s).$$

Equivalently, there exists a measurable set $\Omega_ \subset \mathbb{C}$ with $|\Omega_*| = s$ such that*

$$\lambda_{1,\varphi}(\Omega_*) = M(s).$$

Proof. Let $\{F_k\} \subset \mathcal{F}^2(\mathbb{C})$ be a maximizing sequence:

$$\|F_k\|_{\mathcal{F}^2} = 1, \quad I_{F_k}(s) \rightarrow M(s).$$

By replacing F_k with a Weyl translate if necessary, and using the translation invariance (2.8) together with (2.7), we may suppose that

$$u_{F_k}(0) = \|u_{F_k}\|_{L^\infty(\mathbb{C})}.$$

Note that $M(s) > 0$: indeed, taking $F \equiv 1$, one has $F \in \mathcal{F}^2(\mathbb{C})$ with $\|F\|_{\mathcal{F}^2}^2 = 1$ and density $u_F(z) = e^{-\pi|z|^2} > 0$ everywhere, so for any $s > 0$, $I_F(s) > 0$. Consequently, $\|u_{F_k}\|_{L^\infty(\mathbb{C})}$ is bounded below by a positive constant uniformly in k ; otherwise, for any $\eta > 0$ we would have $\|u_{F_k}\|_{L^\infty} < \eta$ eventually, hence $I_{F_k}(s) \leq \eta s$, contradicting $I_{F_k}(s) \rightarrow M(s) > 0$.

Applying Proposition 7.1, after passing to a subsequence we may write

$$F_k = \sum_{j=1}^J T_{a_k^{(j)}} G^{(j)} + R_k^{(J)}$$

for every fixed $J \geq 1$, with the conclusions of that proposition.

Step 1: an upper bound for the limit. Fix $J \geq 1$ and let $\Omega_k^{(J)}$ be a measurable set of measure s such that

$$I_{F_k}(s) = \int_{\Omega_k^{(J)}} u_{F_k} dA;$$

by Lemma 2.5(i) we may take $\Omega_k^{(J)}$ to be a superlevel set of u_{F_k} . Let $\varepsilon > 0$. Since $u_{G^{(j)}} \in L^1(\mathbb{C})$ for every j (with $\int_{\mathbb{C}} u_{G^{(j)}} dA = \|G^{(j)}\|_{\mathcal{F}^2}^2$), choose $R > 0$ so large that

$$\int_{\mathbb{C} \setminus D_R} u_{G^{(j)}} dA < \varepsilon \quad \text{for all } 1 \leq j \leq J,$$

and large enough that the conclusion of Lemma 7.2(ii) holds with this choice of R . Since the translation parameters are pairwise divergent by Proposition 7.1(i), for k large enough the disks

$$D_R(a_k^{(1)}), \dots, D_R(a_k^{(J)})$$

are pairwise disjoint. Write

$$B_{k,j} := D_R(a_k^{(j)}), \quad E_{k,j} := \Omega_k^{(J)} \cap B_{k,j}, \quad E_{k,0} := \Omega_k^{(J)} \setminus \bigcup_{j=1}^J B_{k,j}.$$

Set

$$s_{k,j} := |E_{k,j}|, \quad 0 \leq j \leq J.$$

Then, since the $\{B_{k,j}\}_{j=1}^J$ are disjoint and $\Omega_k^{(J)}$ has measure s ,

$$\sum_{j=0}^J s_{k,j} = s.$$

Set

$$S_k^{(J)} := \sum_{j=1}^J T_{a_k^{(j)}} G^{(j)}.$$

By Proposition 7.1(iii), the sequence $\{S_k^{(J)}\}_k$ is bounded in $\mathcal{F}^2(\mathbb{C})$, with bound $\leq 1 + o_k(1)$. By Proposition 7.1(iv), if J is taken sufficiently large then

$$\limsup_{k \rightarrow \infty} \|u_{R_k^{(J)}}\|_{L^\infty(\mathbb{C})} < \varepsilon^2.$$

Lemma 7.3 therefore gives

$$\|u_{F_k} - u_{S_k^{(J)}}\|_{L^\infty(\mathbb{C})} \rightarrow 0 \quad (k \rightarrow \infty),$$

and in particular, after taking k large enough,

$$\|u_{F_k} - u_{S_k^{(J)}}\|_{L^\infty(\mathbb{C})} < \varepsilon. \quad (7.7)$$

Now

$$E_{k,0} = \Omega_k^{(J)} \setminus \bigcup_{j=1}^J B_{k,j} \subset \mathbb{C} \setminus \bigcup_{j=1}^J D_R(a_k^{(j)}).$$

Lemma 7.2(i) gives, for each fixed $j \in \{1, \dots, J\}$,

$$\sup_{z \in B_{k,j}} |u_{S_k^{(J)}}(z) - u_{T_{a_k^{(j)}} G^{(j)}}(z)| \rightarrow 0 \quad (k \rightarrow \infty),$$

while Lemma 7.2(ii) yields, for our choice of R and for k sufficiently large,

$$\sup_{z \in E_{k,0}} u_{S_k^{(J)}}(z) \leq \varepsilon.$$

Combining these bounds with (7.7), we conclude that for all sufficiently large k ,

(a) on each $B_{k,j}$,

$$u_{F_k}(z) \leq u_{T_{a_k^{(j)}} G^{(j)}}(z) + 2\varepsilon;$$

(b) on $E_{k,0}$,

$$u_{F_k}(z) \leq 2\varepsilon.$$

Therefore, decomposing the integral defining $I_{F_k}(s)$ according to the partition $\Omega_k^{(J)} = E_{k,0} \sqcup \bigsqcup_{j=1}^J E_{k,j}$,

$$\begin{aligned} I_{F_k}(s) &= \int_{\Omega_k^{(J)}} u_{F_k} dA \\ &\leq \sum_{j=1}^J \int_{E_{k,j}} u_{T_{a_k^{(j)}} G^{(j)}} dA + 2\varepsilon \sum_{j=0}^J |E_{k,j}| \\ &= \sum_{j=1}^J \int_{E_{k,j} - a_k^{(j)}} u_{G^{(j)}} dA + 2\varepsilon s \\ &\leq \sum_{j=1}^J I_{G^{(j)}}(s_{k,j}) + 2\varepsilon s, \end{aligned}$$

where in the next-to-last step we used the change of variables $z \mapsto z - a_k^{(j)}$ together with (2.7), and in the last step the definition of $I_{G^{(j)}}(s_{k,j})$ as the supremum of $\int_E u_{G^{(j)}} dA$ over sets of measure $s_{k,j} = |E_{k,j}| = |E_{k,j} - a_k^{(j)}|$.

Passing to a further subsequence if necessary, we may suppose

$$s_{k,j} \rightarrow \sigma_j \in [0, s] \quad (k \rightarrow \infty)$$

for each $1 \leq j \leq J$. By the continuity of $I_{G^{(j)}}$ in the measure parameter, given by Lemma 7.4,

$$M(s) = \lim_{k \rightarrow \infty} I_{F_k}(s) \leq \sum_{j=1}^J I_{G^{(j)}}(\sigma_j) + 2\varepsilon s.$$

Since $\varepsilon > 0$ was arbitrary,

$$M(s) \leq \sum_{j=1}^J I_{G^{(j)}}(\sigma_j). \quad (7.8)$$

Step 2: redistribution of mass and saturation of the bound. We now use Lemma 2.5(ii). For each j with $G^{(j)} \neq 0$, set

$$\widehat{G}^{(j)} := \frac{G^{(j)}}{\|G^{(j)}\|_{\mathcal{F}^2}}, \quad \|\widehat{G}^{(j)}\|_{\mathcal{F}^2} = 1,$$

and use the homogeneity $I_{cF}(\sigma) = |c|^2 I_F(\sigma)$ (with $c = \|G^{(j)}\|_{\mathcal{F}^2}$) to obtain

$$I_{G^{(j)}}(\sigma_j) = \|G^{(j)}\|_{\mathcal{F}^2}^2 I_{\widehat{G}^{(j)}}(\sigma_j).$$

Since $\widehat{G}^{(j)}$ has unit $\mathcal{F}^2$ -norm, by definition of M and the monotonicity of $I_{\widehat{G}^{(j)}}$ given by Lemma 7.4 (using $\sigma_j \leq s$),

$$I_{\widehat{G}^{(j)}}(\sigma_j) \leq I_{\widehat{G}^{(j)}}(s) \leq M(s).$$

Hence

$$I_{G^{(j)}}(\sigma_j) \leq \|G^{(j)}\|_{\mathcal{F}^2}^2 M(s).$$

For j with $G^{(j)} = 0$ the inequality is trivial. Summing over $1 \leq j \leq J$ and using Proposition 7.1(iii) together with the normalization $\|F_k\|_{\mathcal{F}^2}^2 = 1$,

$$\sum_{j=1}^J I_{G^{(j)}}(\sigma_j) \leq M(s) \sum_{j=1}^J \|G^{(j)}\|_{\mathcal{F}^2}^2 \leq M(s).$$

Combined with (7.8), this yields

$$M(s) \leq \sum_{j=1}^J I_{G^{(j)}}(\sigma_j) \leq M(s)$$

for every J . Therefore equality holds throughout for every J , which forces

$$\sum_{j \geq 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 = 1$$

(so the residual $R_k^{(J)}$ carries no $\mathcal{F}^2$ -mass in the limit), and moreover, for every j with $G^{(j)} \neq 0$,

$$I_{\widehat{G}^{(j)}}(\sigma_j) = M(s).$$

By monotonicity of $I_{\widehat{G}^{(j)}}$, $I_{\widehat{G}^{(j)}}(s) \geq I_{\widehat{G}^{(j)}}(\sigma_j) = M(s)$; since $M(s)$ is the supremum over unit-norm functions, we conclude

$$I_{\widehat{G}^{(j)}}(s) = M(s).$$

Thus every nonzero normalized profile is a global maximizer.

Step 3: producing a maximizing set. Choose any j_0 with $G^{(j_0)} \neq 0$ (such j_0 exists since $\sum_{j \geq 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 = 1$) and set $F_* := \widehat{G}^{(j_0)}$, so that

$$\|F_*\|_{\mathcal{F}^2} = 1, \quad I_{F_*}(s) = M(s).$$

By Lemma 2.5(i), there exists a superlevel set Ω_* of u_{F_*} with $|\Omega_*| = s$ and

$$\int_{\Omega_*} u_{F_*} dA = I_{F_*}(s) = M(s).$$

Then

$$\lambda_{1,\varphi}(\Omega_*) \geq \int_{\Omega_*} u_{F_*} dA = M(s),$$

while by definition of $M(s)$ we have $\lambda_{1,\varphi}(\Omega_*) \leq M(s)$. Hence

$$\lambda_{1,\varphi}(\Omega_*) = M(s),$$

and Ω_* is a global maximizer of measure s . □

We may now identify the global maximizers, by combining the existence theorem above with the local-rigidity theorem of the previous section.

Theorem 10. *For every $s > 0$, every global maximizer for*

$$\sup_{|\Omega|=s} \lambda_{1,\varphi}(\Omega)$$

is, up to translation, a disk.

Proof. Let Ω_* be a global maximizer of measure s (which exists by Theorem 9). Then Ω_* is, in particular, a local maximizer of $\lambda_{1,\varphi}$ under the same measure constraint: every sufficiently small smooth deformation of Ω_* preserving the measure $|\Omega| = s$ yields a competitor whose value of $\lambda_{1,\varphi}$ is at most $M(s) = \lambda_{1,\varphi}(\Omega_*)$. In particular, every competitor E with $|E| = |\Omega_*|$ and Gaussian-weighted symmetric-difference distance $\int_{\mathbb{C}} |\mathbf{1}_{\Omega_*} - \mathbf{1}_E| e^{-\pi|z|^2} dz < \delta_0$ to Ω_* (the gauge used in the local-maximizer definition of Section 6) satisfies $\lambda_{1,\varphi}(E) \leq \lambda_{1,\varphi}(\Omega_*)$, because $\lambda_{1,\varphi}(E) \leq M(s) = \lambda_{1,\varphi}(\Omega_*)$ by the defining property of $M(s)$. The local-rigidity theorem proved in the previous section therefore applies: every such local maximizer is, up to translation, a disk. Since the problem is invariant under Weyl translations, this characterization is sharp.

Uniqueness up to translation. The argument above shows the following sharp dichotomy: every global maximizer Ω_* of measure s is a translate of the centered disk D_R of area s (so that $R = \sqrt{s/\pi}$). Indeed, by the previous paragraph there exists $a \in \mathbb{C}$ and $R \geq 0$ with

$$\Omega_* = a + D_R, \quad |D_R| = \pi R^2 = s.$$

The radius R is determined by the measure constraint $|\Omega_*| = s$, and the translation parameter a ranges over all of $\mathbb{C}$. Conversely, *every* translate of D_R is also a global maximizer: by Lemma 2.5, the value $\lambda_{1,\varphi}(\Omega)$ is invariant under the action of the Weyl translation group $\{T_b\}_{b \in \mathbb{C}}$ on the underlying eigenfunction, which on the geometric side corresponds to translating the set $\Omega \mapsto a + \Omega$; hence $\lambda_{1,\varphi}(a + D_R) = \lambda_{1,\varphi}(D_R) = M(s)$. The set of global maximizers is therefore exactly the orbit of the centered disk D_R under the translation group $\mathbb{C}$ of $\mathbb{C}$, and this orbit is sharp: any two disks of the same area are translates of one another, and no further symmetry-breaking maximizers exist. In particular, the classical Nicola–Tilli theorem [40, 41] is recovered, with the precise uniqueness statement: the maximizer is unique up to a translation by an element of $\mathbb{C}$. $\square$

As an immediate consequence, we recover the classical Nicola–Tilli theorem [40, 41].

Corollary 7.1. *Among all measurable sets of prescribed measure, the disk maximizes Gaussian time-frequency concentration:*

$$\lambda_{1,\varphi}(\Omega) \leq \lambda_{1,\varphi}(D)$$

whenever $|\Omega| = |D|$ and D is a disk.

Proof. Existence of a global maximizer is given by Theorem 9, and Theorem 10 shows that every such maximizer is a disk, up to translation. Since translation by an element of $\mathbb{C}$ preserves both Lebesgue measure and the value of $\lambda_{1,\varphi}$, the claim follows. $\square$

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